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MORC2 Mediates Transcriptional Regulation Through Liquid-Liquid Phase Separation

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eLife Assessment

This **important** study has demonstrated that MORC2 undergoes phase separation in cells and established multiple interactions responsible for the phase separation. Upon revision, the data generally provide **solid** support to the claim that MORC2 condensates are functionally relevant in gene regulation and begins to demonstrate the importance of the physical properties of biological condensates. Nevertheless, there remains some weakness in the connection between condensates and function.

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Abstract

MORC2 is a chromatin-associated ATPase essential for transcriptional silencing and genome stability, yet the biophysical principles governing its regulatory activity remain elusive. Here, we demonstrate that full-length MORC2 undergoes biomolecular condensation to form dynamic nuclear assemblies, a process fundamentally required for its repressor function. Endogenous MORC2 forms discrete, dynamic condensates in neurons from *EGFP-MORC2* knock-in mice, supporting the physiological relevance of these assemblies *in vivo*. Mechanistically, a 3.1 Å crystal structure of coiled-coil 3 (CC3) identifies a dimeric scaffold that serves as a structural hub, while multivalent ‘sticker’ interactions between an intrinsically disordered region (IDR) and a newly defined IDR-binding domain (IBD) drive condensation. We show that DNA acts as a molecular scaffold that triggers MORC2 condensation, which in turn allosterically stimulates its ATPase activity. Critically, by employing a “killswitch” strategy to decouple assembly from internal fluidity, we reveal that only dynamic MORC2 condensates—not static aggregates or condensation-deficient mutants—can restore transcriptional repression in MORC2-knockout cells. Furthermore, pathogenic variants linked to CMT2Z and SMA differentially perturb these material properties and enzymatic turnover, providing a mechanistic link between condensate dysregulation and human neuropathies. Together, our findings establish a DNA-templated condensation mechanism for MORC2 and provide a molecular framework for understanding how the material state of chromatin-associated machinery dictates gene regulation and disease pathogenesis

Introduction

Chromatin dynamically regulates DNA accessibility to orchestrate essential processes such as transcription, replication, and DNA repair^{1–5}. These processes are dynamically regulated by chromatin-associated complexes, such as the human silencing hub (HUSH) complex⁶, which control the accessibility of genetic material to regulatory machinery. The Microorchidia (MORC) protein family, characterized by a conserved N-terminal GHL-ATPase domain and a structurally diverse C-terminal region, regulates gene expression and chromatin organization^{7–10}. These C-terminal domains are thought to contribute to dimerization and functional specificity^{11–14}. Notably, MORC2 has been identified as a component of the HUSH complex and is required for HUSH-mediated transcriptional repression¹².

GHL ATPases, including DNA gyrase, Hsp90, and MutL, rely on ATP-dependent dimerization to drive conformational changes essential for substrate engagement and enzymatic regulation^{15–19}. Similar mechanisms have been proposed for MORC family members. In humans, the MORC family comprises five members—MORC1, MORC2, MORC3, MORC4, and SMCHD1, and different MORC paralogs target distinct genomic loci, including transposable elements, viral DNA, and long-exon genes^{7,20,21}. The conserved N-terminal GHL-ATPase domain of MORC family proteins has been extensively characterized, whereas the evolutionarily diversified, multi-domain C-terminal regions are thought to act as key determinants of chromatin targeting and functional specificity. Despite these shared structural features, the molecular basis for MORC-mediated transcriptional regulation remains poorly understood, partly due to limited structural insight into the full-length proteins and C-terminal regions.

MORC2 has recently been identified as a key regulator of chromatin architecture and gene silencing, primarily through its recruitment to HUSH-target loci via C-terminal CC2 domain-mediated interactions with TASOR and MPP8¹². While its silencing activity was initially attributed to the N-terminal ATPase domain, genetic complementation studies have revealed that the CW domain and all three coiled-coil (CC) regions are also indispensable for its function^{12,13}. Beyond its structured domains, sequence analyses and structural predictions have uncovered an extended intrinsically disordered region (IDR) spanning the central to C-terminal regions of the protein, suggesting potential regulatory mechanisms by facilitating multivalent and dynamic interactions beyond enzymatic activity²².

Transcriptional silencing is frequently spatially organized within membrane-less nuclear compartments, often driven by the process of biomolecular condensation^{23,24}. Sequence analyses of MORC2 reveal an extended intrinsically disordered region (IDR) spanning the central to C-terminal domains, suggesting a capacity for the multivalent, weak interactions that underpin phase separation^{25,26}. While IDRs are common in chromatin-associated proteins, their presence alone does not guarantee condensation; rather, phase separation often emerges from the coordinated interplay between disordered segments and structured oligomerization hubs. How such mesoscale organization modulates MORC2's enzymatic turnover and its ability to remodel the genome remains a fundamental question in the field.

In this study, we demonstrate that MORC2 undergoes phase separation to form dynamic nuclear condensates. We define the structural basis for this behavior, identifying a dimeric CC3 scaffold that, together with multivalent interactions between the IDR and a newly identified IDR-binding domain (IBD), drives condensation. We further show that DNA binding acts as a molecular scaffold to promote MORC2 assembly, which in turn allosterically enhances its ATPase activity. Using a combination of cellular “killswitch” assays and *in vivo* models, we establish that the dynamic material properties of MORC2 condensates—rather than mere protein recruitment—are strictly required for transcriptional regulation. Finally, we provide mechanistic insights into how neuropathic disease-associated mutations in the MORC2 N-terminus perturb this biophysical balance, linking condensate dysregulation to human pathology.

Results

Coiled-coil 3 (CC3) mediates dimerization of full-length MORC2

To determine whether MORC2 has N- and C-terminal dimerization interfaces similar to those in other GHL-type ATPases²⁷, which previously identified the role of CC3 (previously designated as CC4) in MORC2 dimerization¹⁴, we established an *in vitro* system to purify full-length (FL) human MORC2 (MORC2FL, residues 1-1032) from HEK293F cells, yielding nucleic acid-free, highly pure proteins for structural and functional analyses (Figure 1a, b [↗](#)). Analytical size-exclusion chromatography coupled with static light scattering (SLS) demonstrated that MORC2FL forms dimers, with a molecular weight of approximately 239 ± 1 kDa under buffer conditions containing 1000 mM NaCl even in the absence of ATP (Figure 1c [↗](#)). Notably, the addition of 601 DNA resulted in a detectable increase in molecular weight (303 ± 2 kDa), indicative of DNA-protein complex formation, but did not disrupt the dimeric assembly of MORC2 in the same buffer condition (Figure 1c [↗](#)).

To elucidate the atomic structure of MORC2FL, we initially employed negative-stain electron microscopy to reveal that MORC2FL particles were poorly resolved and exhibited considerable heterogeneity in buffer containing 300 mM NaCl. Notably, a subset of particles displayed a morphology consistent with dimer formation (Figure S1a [↗](#)). To further investigate this, we performed structural prediction of MORC2FL using AlphaFold3 and identified a predicted structure that closely resembled the observed reconstruction (Figure S1b [↗](#)). We then examined the oligomerization properties of various MORC2 constructs. MORC2¹⁻⁹⁰⁰ (FLACC3-IBD domain) was found to exist as a monomer, whereas MORC2⁹⁰¹⁻¹⁰³² and MORC2⁹⁰¹⁻¹⁰⁰³ exhibited dimeric properties, confirming the critical role of the CC3 domain (residues 901-1003) in mediating MORC2 dimerization (Figure 1c [↗](#), and Figure S1c [↗](#)).

To gain further structural insights, we determined the 3.1 Å crystal structure of the CC3 dimer using X-ray diffraction (Table S1). The structure revealed that the dimer interface is stabilized predominantly by hydrophobic interactions. Three key hydrophobic residues—L911, L915, and F922—form distinct layers within the dimer interface (layers 1-3, Figure 1e [↗](#), and Figure S1e, f [↗](#)). The side chains of these residues are buried within the dimerization interface, creating a hydrophobic core critical for maintaining the stability of the dimer.

We systematically introduced single-point mutations in these hydrophobic residues to assess their contributions to dimerization. SLS analysis demonstrated that the wild-type (WT) MORC2⁹⁰¹⁻¹⁰³² (with Trx-tag) fragment predominantly exists as a dimer (~61 kDa). In contrast, all three key mutants (L911Q, L915Q, and F922Q) displayed altered molecular weight distributions indicative of disrupted dimer formation and enhanced higher-order oligomerization (more than 61 kDa) (Figure S1g [↗](#)). However, none of these mutations completely disrupted dimerization or converted the dimer to a monomeric state (Figure S1g [↗](#)). Further analyses of additional single-site mutants (e.g., I908Q, Y921A, F951Q, Y954A, L958Q) showed minimal effects on dimer stability, underscoring the central importance of L915 and its neighboring hydrophobic layers in stabilizing the CC3 dimer (Figure S1g [↗](#)). Moreover, we examined the dimeric status of truncated CC3 constructs (residues 901-975), which confirmed that the CC3 N-terminal region is sufficient to drive dimer formation (Figure 1f [↗](#)). To delineate the structural basis of these interactions, we mapped the positions of hydrophobic residues within the CC3 backbone (Figure S1e [↗](#)). The distances between interacting residues, such as I908-I908' (3.4 Å), L911-L915' (3.7 Å), F922-F922' (3.9 Å), and L958-L958' (4.0 Å), revealed a tightly packed hydrophobic network. The robustness of this network ensures the integrity of the CC3 dimer, even under varying salt concentration and protein concentration tested *in vitro* conditions. These findings collectively suggest that the CC3 domain plays a critical role in MORC2 dimerization. The dimeric assembly of CC3 is essential for maintaining the structural integrity of the protein, which likely underpins its functional activities, including associations with chromatin¹⁴.

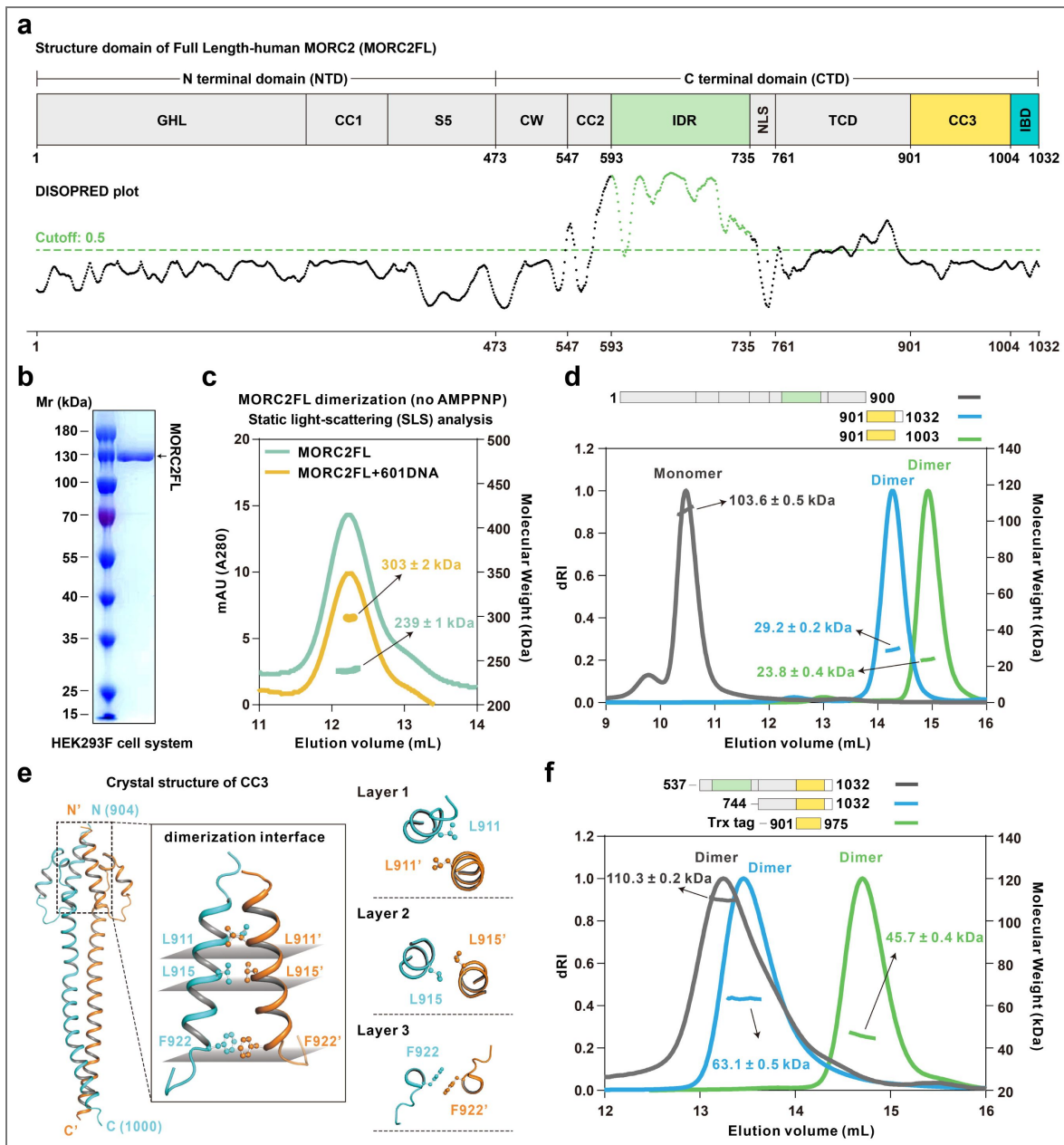


Figure 1. The CC3 domain serves as the structural basis for full-length MORC2 dimerization.

(a) The domain organization of human MORC2FL, highlighting the ATPase module (GHL, CC1 and S5 domains, collectively termed NTD), the C-terminal domains (CTD) with coiled-coil regions (CC2 and CC3), CW-type zinc finger (CW), intrinsically disordered region (IDR, marked by a fresh green line by IUPRED2 prediction), nuclear localization signal (NLS), Tudor-chromodomain (TCD), and a 30-residues tail domain (IBD). (b) Purification and characterization of full-length human MORC2 in HEK293F cell system. SDS-PAGE analysis of the purified protein demonstrates its purity and distribution. (c) Static light-scattering (SLS) analysis shows that MORC2 forms dimers in the absence of ATP, with a molecular weight of 239 ± 1 kDa in standard buffer (mint green line) and 303 ± 2 kDa in the presence of 601 DNA (sunflower yellow line). (d) SLS analysis reveals the oligomerization states of MORC2 truncations. Full-length MORC2 lacking CC3-IBD (1-900) is monomeric, while CC3-IBD (901-1032) and CC3 alone (901-1003) form stable dimers. (e) Crystal structural of the CC3 dimer. Hydrophobic residues contributing to dimer stability (L911, L915, and F922) are highlighted using ball-and-stick representations. Top view of the hydrophobic core of the CC3 dimer. Layers of hydrophobic contacts stabilize the dimer interface, illustrated with paired residues (e.g., Layer 1: L911-L911', Layer 2: L915-L915', and Layer 3: F922-F922'). (f) Additional SLS analysis confirms that fragments spanning residues 537-1032, 744-1032, and Trx-901-975 (CC3 truncation) adopt dimeric conformations.

MORC2 forms condensates both *in vivo* and *in vitro*

In the predicted structure of MORC2FL, we observed an extended unstructured region at the C-terminus of the protein (Figure S1b). This region is classified as an intrinsically disordered region (IDR), a protein architecture frequently reported to facilitate multivalent interactions and can act as a scaffold for phase separation^{25, 28}. Specifically, IUPred2A analysis predicts that the residues 593-735 predominantly exhibit high disorder scores, prompting us to define this segment as the IDR domain (Figure 1a).

In vitro, sedimentation assays showed that when the salt concentration was reduced to 150 mM, more than half of the EGFP fused MORC2FL (EGFP-MORC2FL) precipitated into the pellet fraction (Figure 2a, b). Fluorescence imaging revealed that at this near-physiological ionic strength, EGFP-MORC2FL spontaneously demixed to form well-defined spherical droplets (Figure 2c, d). These droplets exhibited a characteristic polydispersity in size and showed significant coarsening as protein concentration increased (Figure S2a). Sedimentation assays further showed that the pellet fraction of EGFP-MORC2FL after centrifugation was concentration-dependent (Figure S2b). The EGFP-MORC2FL-enriched condensates exhibited dynamic behavior, including droplet fusion (Figure S2c) and fluorescence recovery after photobleaching (FRAP) (Figure 2e, f), indicating a dynamic state.

Using antibody labeling, we observed that endogenous MORC2 forms discrete, dense punctate structures within the nuclei of HeLa cells (Figure 2g). The specificity of these puncta was confirmed by their absence in MORC2-knockout (KO) HeLa cells (Figure S2d, e), suggesting that MORC2 may natively organize into higher-order nuclear condensates under physiological conditions. To further characterize these assemblies, we transfected HeLa cells with EGFP-MORC2FL and observed the formation of prominent, droplet-like nuclear condensates (Figure 2h). Notably, consistent with the endogenous staining patterns, EGFP-MORC2FL condensates preferentially localized to regions of reduced chromatin density, as visualized by the H2A-mCherry signal. This distinct spatial distribution implies a potential role for MORC2 condensates in organizing or regulating the heterochromatin landscape. FRAP analysis of these EGFP-MORC2FL puncta revealed fluorescence recovery on a timescale of seconds (Figure 2i, j). Furthermore, these puncta underwent active fusion events (Figure S2f), a hallmark of dynamic liquid-like condensates with a highly mobile internal environment.

To determine whether MORC2 forms biomolecular condensates under physiological conditions, we generated mice expressing EGFP-MORC2 protein from the endogenous locus (Figure S3a). We focus on neurons in the brain and spinal cord as missense mutations in *MORC2* cause neuropathies including Charcot-Marie-Tooth and spinal muscular atrophy disease^{11,13}. Imaging of NeuN-positive neurons revealed that EGFP-MORC2FL condensates occur in neurons under endogenous expression level (Figure 2k). FRAP analysis in freshly prepared brain slices further demonstrated rapid and partial fluorescence recovery, supporting the dynamic nature of endogenous MORC2 assemblies (Figure 2l). In addition, unlike MORC3, which shows cell cycle-dependent nuclear puncta formation²⁹, MORC2 condensates were absent in mitotic cells (Figure S3b). Together, these results indicate that MORC2 forms phase-separated biomolecular condensates under physiological conditions *in vivo* and *in vitro*.

The MORC2 C-terminal domain (CTD) drives condensate formation

To identify the key determinants governing the assembly of MORC2 nuclear condensates, we partitioned the protein into two functional modules based on its domain architecture: the N-terminal domain (NTD, residues 1–472), comprising the ATPase and CC1 motifs, and the C-terminal domain (CTD, residues 473–1032), which encompasses the IDR and multiple coiled-coil regions. A series of EGFP-tagged truncation constructs were then expressed in HeLa cells to systematically map the regions essential for condensate formation (Figure 3a). Deletion of the CTD completely abolished puncta formation, resulting in a diffuse localization across the nucleus and cytoplasm (Figure S4a). By contrast, NTD deletion had negligible effects on nuclear localization or the capacity for condensation. To exclude the possibility that the NTD's diffuse pattern was merely due

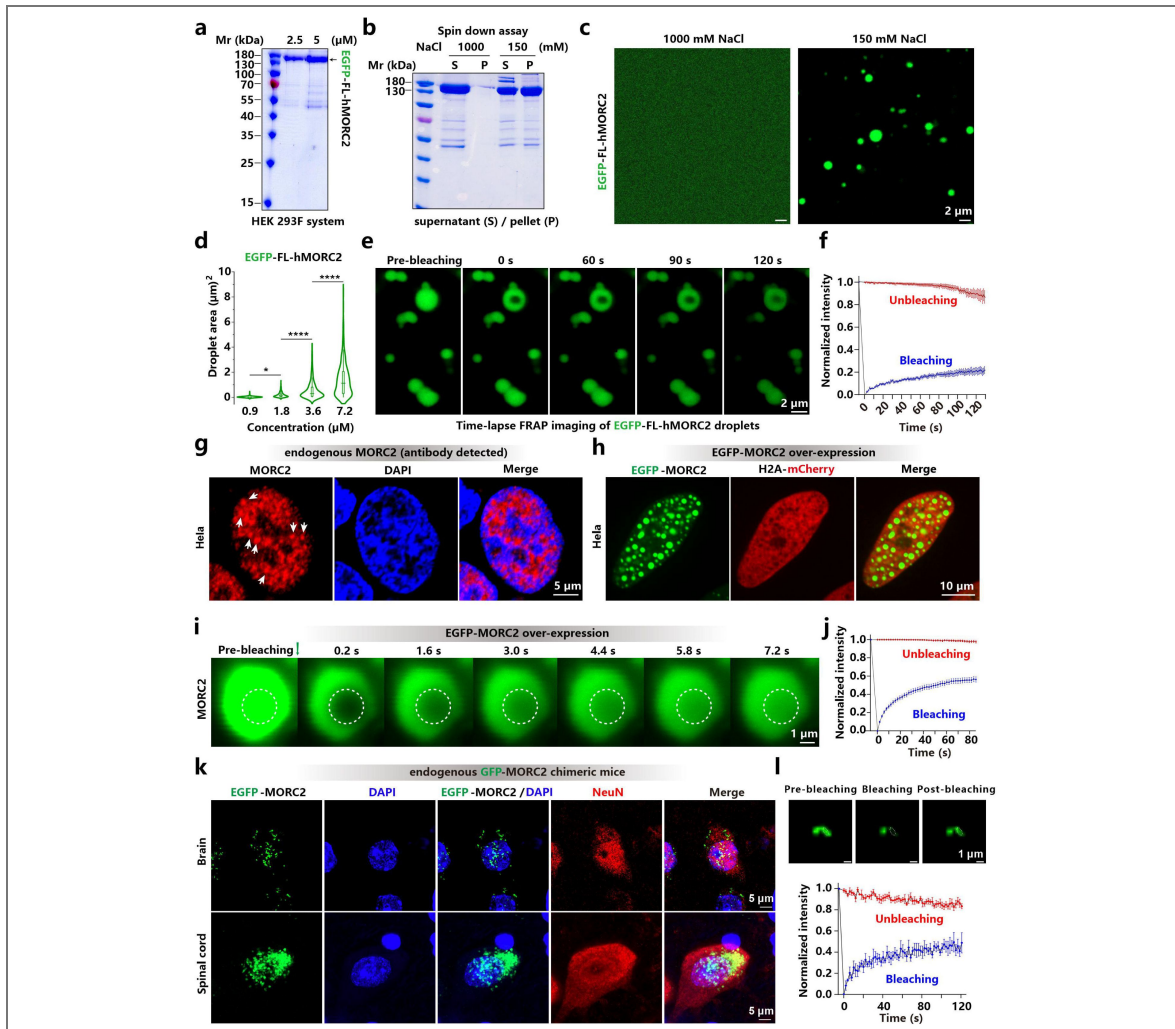


Figure 2. Endogenous MORC2 organizes into dynamic condensates within the nucleus

(a) SDS-PAGE analysis of EGFP-MORC2FL protein purified from HEK293F cells, showing protein concentration at 2.5 μ M and 5 μ M. (b) Sedimentation assay shows the distribution of EGFP-MORC2FL (7.2 μ M) between supernatant (S) and pellet (P) fractions under 1000 mM (control) and 150 mM NaCl, indicating its propensity for phase separation. (c) EGFP-MORC2 (7.2 μ M) undergoes phase separation into droplets in a buffer containing 150 mM NaCl, but not in 1000 mM NaCl buffer. Scaler bar: 2 μ m. (d) The proteins were examined in a buffer containing 150 mM NaCl, and phase separation was assessed by fluorescence microscopy with 488 nm excitation for EGFP. Quantification of droplet areas from (Figure S2a), presented as violin plots with box plots indicating medians. Data are derived from ≥ 3 independent images and reflect the size distribution of phase-separated droplets. Data are presented as mean $\pm$ SEM; one-way ANOVA with Tukey's post hoc test. **** $p < 0.0001$; * $p < 0.05$. (e) Time-lapse FRAP imaging of EGFP-MORC2FL droplets formed in vitro, demonstrating fluorescence recovery over time. Scaler bar: 2 μ m. (f) Quantitative analysis of fluorescence recovery from (e), showing the dynamic properties of in vitro MORC2 droplets. Data represent mean $\pm$ SEM, with $n \geq 3$ replicates. (g) Immunostaining of endogenous MORC2 in HeLa cells revealed punctate localized within the nucleus, likely representing condensates involved in transcriptional regulation (white arrows). Nuclei are counterstained with DAPI to visualize chromatin-rich regions. Scaler bar: 5 μ m. (h) Live-cell imaging of transiently transfected EGFP-MORC2FL showed its assembly into dispersed, nearly spherical condensates in the nucleus, while stable expression of H2A-mCherry served to mark chromatin-enriched domains. Scaler bar: 10 μ m. (i) Time-lapse FRAP analysis of EGFP-MORC2FL condensates in the nuclei of transiently transfected HeLa cells revealed rapid fluorescence recovery within seconds, indicative of their dynamic, nature. Scaler bar: 1 μ m. (j) Quantitative analysis of fluorescence recovery from (i), showing the dynamic properties of in vitro MORC2 droplets. Data represent mean $\pm$ SEM, with $n \geq 10$ replicates. (k) Sections of brain and spinal cord from endogenous EGFP-MORC2 chimeric mice show EGFP-MORC2 condensation distribution in NeuN-positive neurons. Scaler bar: 10 μ m. (l) Representative FRAP images from of FRAP experiments performed on fresh 250- μ m brain slices of endogenous EGFP-MORC2 chimeric mice demonstrate the dynamic nature of endogenous MORC2 condensates. Data are presented as mean $\pm$ SEM, with $n = 3$ droplets analyzed. Scaler bar: 5 μ m.

to its loss of nuclear entry, we appended an exogenous nuclear localization signal (NLS, PKKKRKV³⁰) to the NTD; however, while this redirected the protein to the nucleus, it failed to restore puncta (Figure 3a), confirming that the CTD is the primary driver of MORC2 condensation. To assess whether the CTD is intrinsically prone to undergo condensation *in vitro*, we purified a CTDΔCW (residues 537–1032) variant from *E. coli*, as the full CTD exhibited suboptimal biochemical stability (Figure S5a, b). At 150 mM NaCl, Cy3-labeled CTDΔCW readily formed discrete condensates in a concentration-dependent manner (Figure 3b, c), and displayed a sedimentation profile characteristic of phase-separated assemblies (Figure 3d, e). Fluorescence microscopy and static light scattering (SLS) confirmed that Cy3 labeling did not perturb its condensation behavior, morphology, or molecular weight distribution (Figure S5c–f). Further dissection of the CTD revealed that deletion of either the CC2 or TCD domains did not impair the formation of nuclear puncta (Figure S4b). In contrast, removal of the IDR-NLS region (residues 593–761) abolished both puncta formation and nuclear localization (Figure S4b). Notably, appending an exogenous NLS (PKKKRKV) to CTDΔIDR-NLS restored its nuclear entry but failed to rescue condensate assembly (Figure S4c), identifying the IDR as a prerequisite for condensation. Similarly, deletion of the CC3-IBD segment maintained nuclear localization but eliminated puncta formation, indicating that this region is also indispensable (Figure S4b). Intriguingly, neither the IDR-NLS fragment nor the CC3-IBD fragment alone (even when NLS-fused) was sufficient to trigger condensation (Figure S4d), implying a cooperative mechanism between these modules. Consistent with these cellular findings, *in vitro* assays showed that neither the MORC2ΔCC3-IBD (residues 1–900) nor the isolated CC3-IBD fragment could undergo condensation (Figure S6a–b). However, mixing the IDR with the IBD fragment reconstituted robust condensate formation at higher concentration (Figure S6c), confirming their synergistic role. Finally, the CW domain appears to modulate condensate morphology, as its deletion led to enlarged and frequently fused puncta. We conclude that the IDR, CC3, and IBD constitute the minimal structural core required for MORC2 condensation, while the CW domain serves to restrict condensate overgrowth and maintain proper morphology (Figure 3a, and Figure S4e).

Multivalent interactions between the IDR and IBD promote MORC2 condensation

Next, we elucidated the mechanistic basis for MORC2 assembly mediated by the IDR and CC3-IBD. We hypothesized that multivalent interactions between these modules drive the formation of a condensed phase that coexists with a dilute aqueous phase. Using nuclear magnetic resonance (NMR) spectroscopy, we characterized the interaction between the IDR and the C-terminal segment (residues 901–1032). ¹⁵N-HSQC titration assays using the ¹⁵N-labeled CC3 (residues 901–1003) or the 1004–1032 fragment revealed that residues 1004–1032 constitute the primary binding site for the IDR, which we termed the IDR-Binding Domain (IBD), whereas the CC3 region exhibited negligible interaction (Figure 4a–c).

Further subdivision of the IDR into three segments—IDRa (residues 593–643), IDRb (residues 644–694), and IDRc (residues 695–735)—pinpointed IDRa as the key region responsible for interaction with IBD (Figure 4a, d–f). As is typical for IDR-mediated interactions involved in condensation, the observed chemical shift perturbations (CSPs) were relatively small but highly region-specific. Notably, the CSP profile of IDRa closely mirrors that of the full-length IDR, whereas IDRb and IDRc showed minimal perturbations (Figure S7). Detailed sequence analysis revealed that the CSPs are clustered around charged residues (D, E, K, R) and their neighboring residues (Y, T, N, I, L, P). This suggests that the association is driven by a concerted action of multiple non-covalent forces: beyond the foundational electrostatic attraction, the involvement of aromatic (Y) and aliphatic (I, L, P) residues points to the contribution of π - π stacking, cation- π interactions, and hydrophobic effects, while polar residues (T, N) likely facilitate intermolecular hydrogen bonding. This hierarchical assembly, characterized by both charge complementarity and hydrophobic-like contributions, aligns with the ‘sticker-and-spacer’ model. In this framework, these multivalent residues act as ‘stickers’ that provide the requisite valency to scaffold a robust yet dynamic network, while the intervening sequences serve as ‘spacers’ to maintain the fluidity necessary for

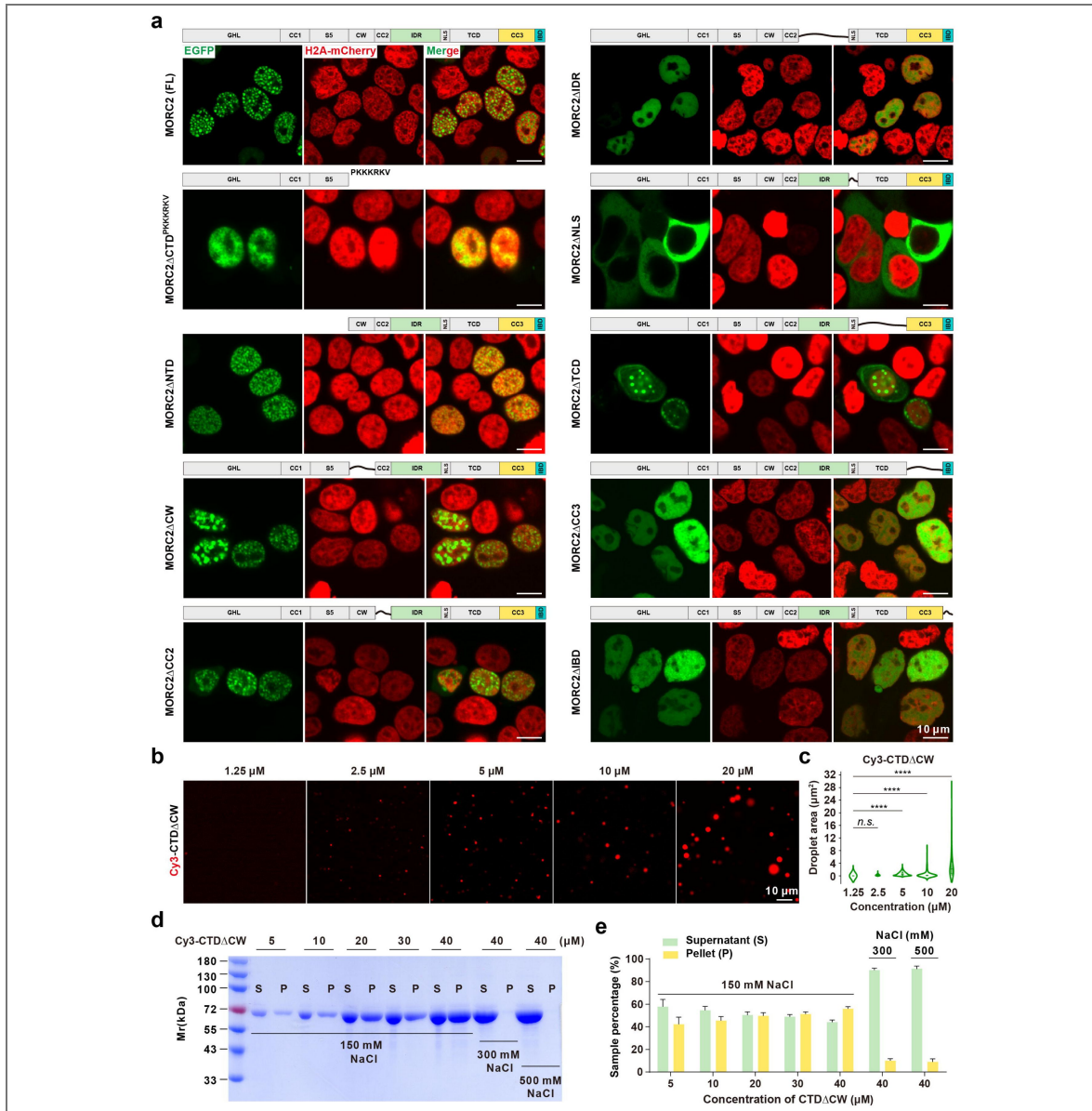


Figure 3. multiple domains regulate phase separation of MORC2.

(a) Sequential deletion analysis revealed that the NTD is dispensable for condensate formation, as EGFP-tagged NTD alone failed to form condensates and was excluded from the nucleus. Fusion with a canonical nuclear localization signal ("PKKKRKV") restored nuclear localization but did not rescue condensate formation. In contrast, the CTD alone was sufficient to form nuclear condensates. Deletion of the CW domain resulted in enlarged condensates in a subset of cells, while removal of CC2 or the TCD had no apparent effect on condensate formation. Strikingly, deletion of either the IDR or CC3 completely abolished nuclear condensate assembly. Deletion of the IBD markedly reduced the frequency of condensate formation. Loss of the intrinsic NLS led to cytoplasmic localization of MORC2; however, no condensates were detected in the cytoplasm under transient transfection conditions in HeLa cells. Together, these results define the IDR and CC3 as the minimal and essential elements for MORC2-mediated phase separation, and implicate the CW domain, IBD, and nuclear microenvironment in the fine-tuning of condensate assembly. Scaler bar: 10 μm. (b) In vitro phase separation of Cy3-labeled CTDΔCW at concentrations from 1.25 μM to 20 μM in 150 mM NaCl buffer, visualized as spherical condensates. Scaler bar: 10 μm. (c) Quantification of droplet areas from (b), displayed as violin plots with box plots indicating median values. Data are presented as mean ± SEM; one-way ANOVA with Tukey's post hoc test. ****p < 0.0001, n.s. not significant. (d) SDS-PAGE analysis of Cy3-CTDΔCW protein distribution between supernatant (S) and pellet (P) fractions after centrifugation at increasing concentrations (5 μM to 40 μM) in 150 mM NaCl buffer. At 40 μM, increasing the NaCl concentration to 300 mM or 500 mM reduced droplet formation, indicating salt sensitivity. (e) Quantification of Cy3-CTDΔCW showing the percentage of soluble protein from densitometric analysis of S/P fractions in panel (d), based on three independent biological replicates.

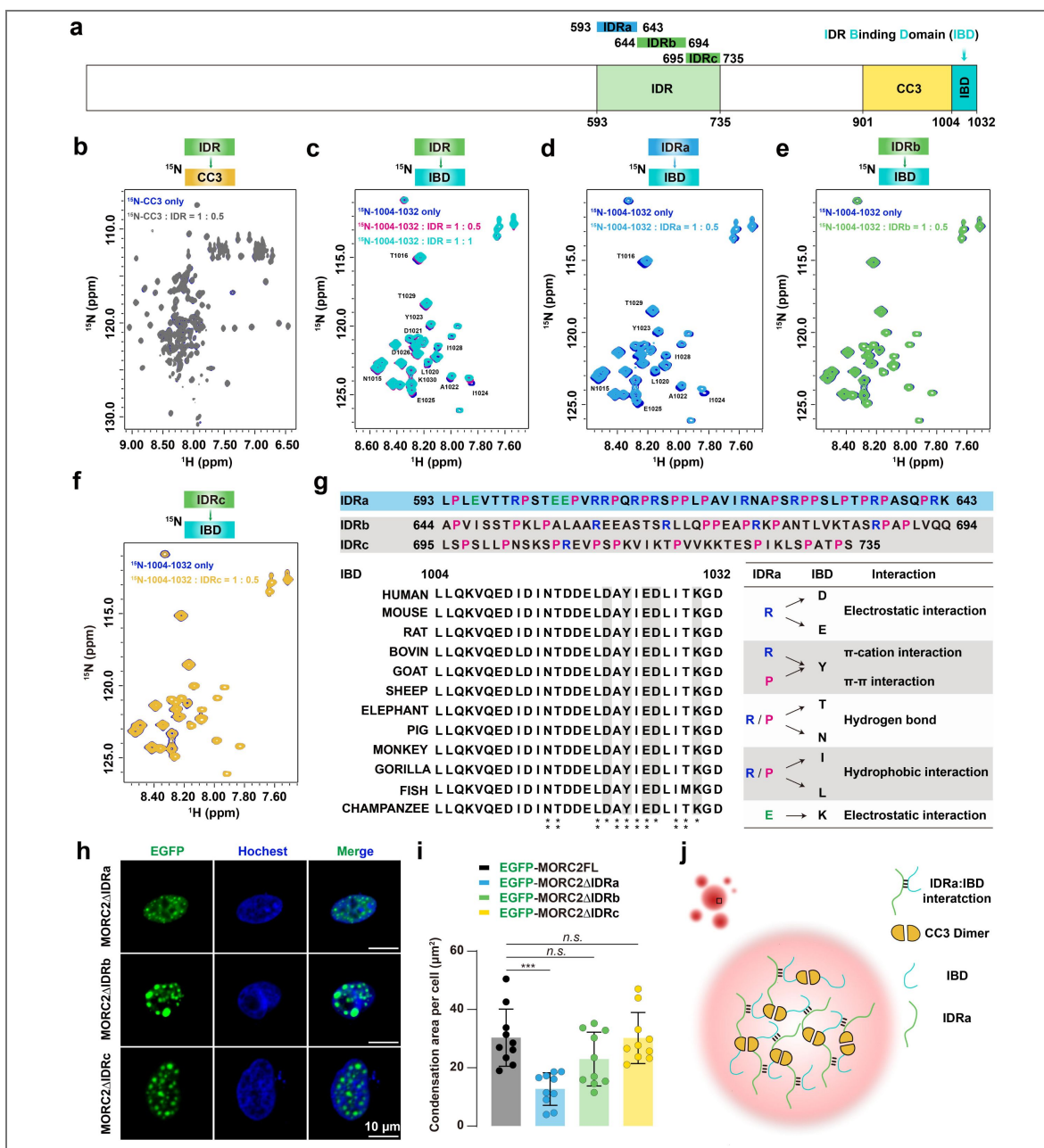


Figure 4. Multivalent interactions between IDR and IBD drive MORC2 condensation.

(a) Domain organization of MORC2 C-terminal regions analyzed by NMR titration, including the IDR, CC3, and IBD. (b) ^{15}N -labeled CC3 (residues 901–1003) shows no chemical shift upon titration with IDR (residues 593–735), indicating no detectable interaction. (c) ^{15}N -labeled IBD (residues 1004–1032) exhibits clear chemical shift perturbations (CSPs) upon IDR titration, confirming a direct, specific interaction. (d–f) NMR titrations of ^{15}N -labeled IBD with IDR subregions: IDRa (593–643), IDRb (644–694), and IDRc (695–735). IDRa induces clear CSPs (d), while IDRb (e) and IDRc (f) show minimal or no shifts, identifying IDRa as the primary IBD-binding segment. (g) Sequence analysis of IDR subregions reveals that IDRa is enriched in proline and arginine residues, enabling electrostatic interactions. Sequence alignment of IBD across species highlights conserved residues involved in IDRa binding. A summary table lists representative residues from IDRa and IBD exhibiting CSPs in titration assays (marked with asterisks). (h) Representative confocal images of HeLa cells expressing EGFP-tagged MORC2ΔIDRa, ΔIDRb, or ΔIDRc constructs. Scale bar: 10 μm. (i) Quantification of droplet area per cell in (h). Deletion of IDRa significantly impairs condensate formation. n = 10 cells per condition. Data are presented as mean ± SEM; statistical analysis: one-way ANOVA with Tukey's post hoc test. ***p < 0.001; n.s. not significant. (j) Working model of MORC2 condensates. CC3 dimerization serves as a structural scaffold, while weak, transient, but specific multivalent interactions between IDRa and IBD cooperatively promote condensate formation.

the condensed state (Figure 4g). Collectively, these quantitative CSP data support a model of weak, dynamic, and region-specific interactions, a hallmark of IDR-driven condensation mechanisms^{31–33}.

To assess the role of IDRa-IBD multivalent interactions in cells, we transfected HeLa and HEK293T cells with MORC2FL variants harboring individual deletions of the IBD, IDRa, IDRb, or IDRc regions. Deletion of the IBD abolished condensate formation in both HeLa cells (Figure 3a, and Figure S4e) and HEK293T cells (Figure S8). Similarly, deletion of IDRa significantly impaired condensation, resulting in a marked decrease in both the number and area of nuclear puncta, whereas IDRb or IDRc deletions had minimal impact (Figure 4h, i). Based on these findings, we propose a model in which multivalent ‘sticker’ interactions between IDRa and the IBD drive MORC2 intermolecular assembly, a process further potentiated by CC3-mediated dimerization, which collectively provides the valency threshold required for condensation (Figure 4j).

DNA binding promotes MORC2 condensation and stimulates ATPase activity

Notably, MORC2 condensates form exclusively within the nuclei of transfected HeLa cells, implying that chromatin-associated factors—most likely DNA—serve as essential cues for their nucleation or stabilization. While the N-terminal CC1 domain was previously identified as a DNA-binding module¹³, we hypothesized that the full-length MORC2 architecture utilizes additional interfaces to facilitate complex chromatin interactions.

To test this, we performed a comprehensive domain screen using electrophoretic mobility shift assays (EMSA). Beyond confirming the robust DNA-binding activity of the CC1 domain (Figure 5a, b), our screen revealed that the CC2, IDR, and TCD (Tudor Chromo-like domain) also possess significant DNA-binding capabilities (Figure 5c–e). In contrast, the CC3–IBD fragment showed no affinity for DNA (Figure 5f). These results indicate that MORC2 employs a multivalent DNA-binding architecture distributed across both its N-terminal and C-terminal regions.

Building on this, we investigated whether these multivalent DNA-protein interactions could scaffold MORC2 condensation *in vitro*. At 0.9 μ M, MORC2FL alone failed to form discrete assemblies; however, the addition of 25 nM 601 DNA induced robust condensate formation, demonstrating that DNA acts as a molecular seed to promote MORC2 phase separation (Figure 5g, h). Interestingly, while the NTD (residues 1–536) binds DNA, it remains diffuse (Figure S6d,e), whereas the CTD Δ CW (residues 537–1032) undergoes DNA-driven condensation (Figure 5i). Systematic evaluation across a matrix of protein and DNA concentrations yielded a phase diagram mapping the transitions between clear solutions, liquid-like condensates, and amorphous precipitates, consistently showing that DNA significantly lowers the concentration threshold for MORC2 condensation (Figure S9). This functional divergence suggests a division of labor within MORC2: while multiple domains provide DNA affinity, the CTD provides the essential protein-protein interaction network required for higher-order assembly.

Finally, we explored how this assembly impacts the MORC2 enzymatic core. The addition of 601 DNA markedly stimulated the ATPase activity of wild-type MORC2, but not the ATP-binding-deficient N39A mutant (Figure 5j). This reveals a synergistic interplay: DNA binding does not merely recruit MORC2 to chromatin; it acts as a molecular scaffold that facilitates condensation, which in turn may optimize the productive conformation or local density required for allosteric activation of the ATPase domain. In conclusion, these findings highlight a coordinated mechanism that integrates DNA binding, condensation, and enzymatic activity, providing a biophysical basis for the multivalent assembly and allosteric regulation of MORC2 on chromatin.

Condensate formation and dynamics are required for MORC2-mediated transcriptional regulation

Establishing a causal link between biomolecular condensation and biological function—particularly in transcriptional regulation—remains a central challenge. A key question is whether MORC2 condensates represent discrete functional entities or merely inactive byproducts. To

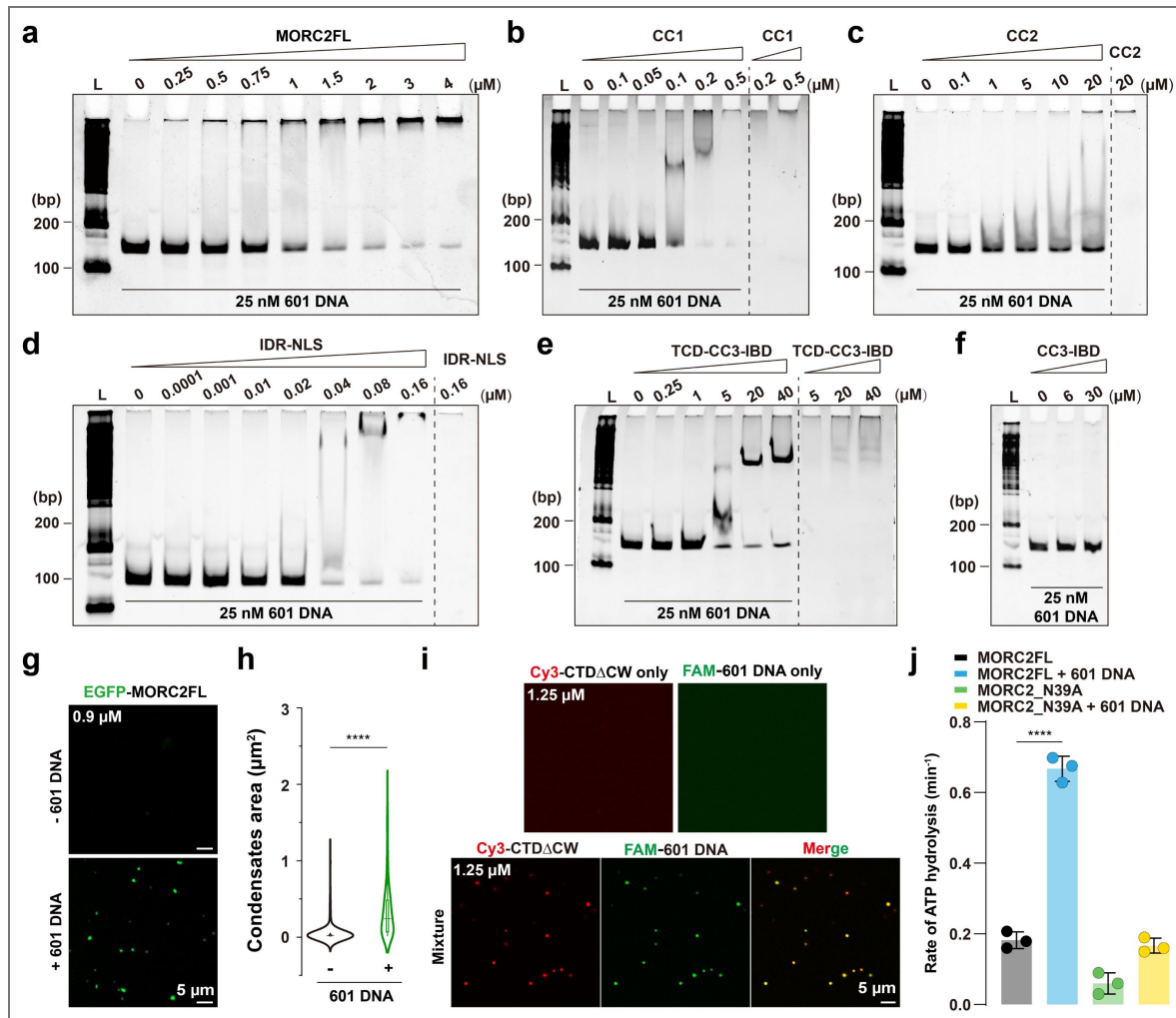


Figure 5. DNA binding enhances MORC2 condensation and modulates its ATPase activity.

(a) Electrophoretic mobility shift assay (EMSA) showing that MORC2FL binds 25 nM 601 DNA, as evidenced by shifts in DNA mobility corresponding to protein-DNA complexes. (b) EMSA highlighting the strong DNA-binding affinity of the CC1 domain, corroborating previous reports that identify this region as a key mediator of DNA interaction. (c-f) EMSA results demonstrating the binding capacities of individual MORC2 domains to 25 nM 601 DNA: CC2 (c), IDR (d), TCD-CC3-IBD (e), and CC3-IBD (f). The IDR exhibits robust DNA-binding activity, CC2 and TCD-CC3-IBD show weaker interactions, and CC3-IBD alone fails to bind DNA. These findings identify CC1, CC2, IDR, and the TCD as DNA-binding domains of MORC2. (g) Confocal imaging showing that 601 DNA promotes MORC2 phase separation. At 0.9 μ M, EGFP-MORC2FL alone does not form droplets, but the addition of 25 nM 601 DNA induces distinct condensate formation. Scaler bar: 5 μ m. (h) Quantification of MORC2 condensate sizes with or without 601 DNA. Violin plots illustrate the distribution of droplet sizes, with box plots indicating median values. Data are presented as mean $\pm$ SEM; unpaired student t-test. **** $p < 0.0001$. (i) Fluorescence microscopy showing Cy3-labeled CTD Δ CW (1.25 μ M) coalescing with 25 nM FAM-labeled 601 DNA. DNA is recruited into MORC2 droplets, forming distinct condensates. Scaler bar: 5 μ m. (j) In the presence of 10 nM 601 DNA, the ATP hydrolysis activity of MORC2FL is significantly enhanced. The N39A mutant, which is deficient in ATP binding, serves as a negative control; nevertheless, a slight but detectable increase in ATPase activity is observed upon DNA binding. ATP hydrolysis activity of MORC2FL is significantly increased in the presence of 10 nM 601 DNA. The N39A mutant, which is deficient in ATP binding, serves as a negative control. Data are presented as mean $\pm$ SEM; statistical analysis: one-way ANOVA with Tukey's post hoc test. **** $p < 0.0001$.

address this, we sought to decouple condensate formation from internal dynamics, allowing for a rigorous interrogation of their functional relevance. While deletion of the CC3 domain abolishes MORC2 condensation, CC3 is also indispensable for dimerization, confounding functional interpretations. Furthermore, although deleting IDR_a markedly weakens MORC2 condensation, a tool to specifically manipulate the solubility and internal mobility of these assemblies was previously lacking.

To overcome these limitations, we employed a micropeptide-based ‘killswitch’ (KS) strategy fused to the C-terminus of MORC2 (Figure 6a). This approach selectively reduces condensate dynamics without disrupting protein expression, folding, or domain architecture³⁴. Unlike the ΔCC3 or ΔIDR_a mutants, the MORC2+KS variant robustly form nuclear condensates but exhibits markedly reduced internal dynamics, as demonstrated by FRAP analyses showing minimal fluorescence recovery (Figure 6b, c). This strategy allowed us to perturb the material properties of the condensates independently of structural integrity.

To evaluate how the modulation of MORC2 condensates influences its gene regulatory capacity, we compared the transcriptional consequences of rescuing MORC2-knockout HeLa cells with MORC2^{FL}, condensation-deficient mutants (ΔCC3 and ΔIDR_a), and the dynamics-defective MORC2+KS (Figure 6d). Bulk RNA-seq was performed 48 hours post-transfection. We hypothesized that impairing either the assembly or the fluidity of the condensates would compromise MORC2-dependent gene regulation. Notably, despite being expressed at substantially higher levels than MORC2^{FL} (Figure 6e), all three mutants showed a striking and consistent failure to restore the MORC2-dependent transcriptional profile (Figure 6f–h). This defect was particularly pronounced for transcriptionally repressed genes, including two sets of high-confidence MORC2 targets identified in prior studies^{12, 22} (Figure 6i, and Figure S10). Gene Ontology analysis revealed that the MORC2-regulated genes are involved in DNA damage responses, viral infection, and cell cycle regulation (Figure 6g). These findings demonstrate that neither increased protein abundance nor the mere presence of static, condensate-like structures is sufficient to restore MORC2 function.

Collectively, these data support a model in which the dynamic nature of MORC2 condensates is essential for full transcriptional activity. While soluble MORC2 complexes are likely involved in initial target recognition, our results indicate that proper condensate formation—and critically, internal fluidity—are required for effective transcriptional control. The inability of the MORC2+KS mutant to rescue transcriptional defects, despite forming visible nuclear assemblies, strongly argues against a model where MORC2 condensates are merely inert byproducts of its activity.

Pathogenic MORC2 variants differentially modulate condensation, DNA binding, and ATPase activity

MORC2 is essential for gene silencing at H3K9me3-marked loci and retrotransposon repression^{6,35–37}. Pathogenic mutations in its GHL-type ATPase domain have been linked to severe neuropathies and cancer^{38–42}. To explore whether these mutations influence the condensation behavior and biochemical activities of the full-length protein, we examined a panel of disease-associated variants, including those linked to Charcot–Marie–Tooth disease type 2Z (CMT2Z: E236G, R252W, Q400R, D466N) and spinal muscular atrophy (SMA: S87L, S218L, F256L, R266A, T424R) (Figure 7a).

In HeLa cells, all variants retained the ability to form nuclear puncta. However, the CMT2Z-linked E236G and, notably, the SMA-linked T424R mutants significantly increased the proportion of cells exhibiting nuclear condensates (Figure 7b). These observations suggest that the N-terminal region plays a nuanced role in modulating the phase boundary of MORC2. Specifically, previous structural studies indicated that the T424R mutation has minimal effects on the overall N-terminal architecture but primarily alters the assembly and dissociation kinetics of the N-terminal dimer¹³. This suggests that the conformational cycling or dimerization state of the N-terminus directly contributes to the macroscopic regulation of MORC2 condensates. Interestingly, while deletion of the N-terminal region did not abolish puncta formation, FRAP analyses revealed a marked loss of

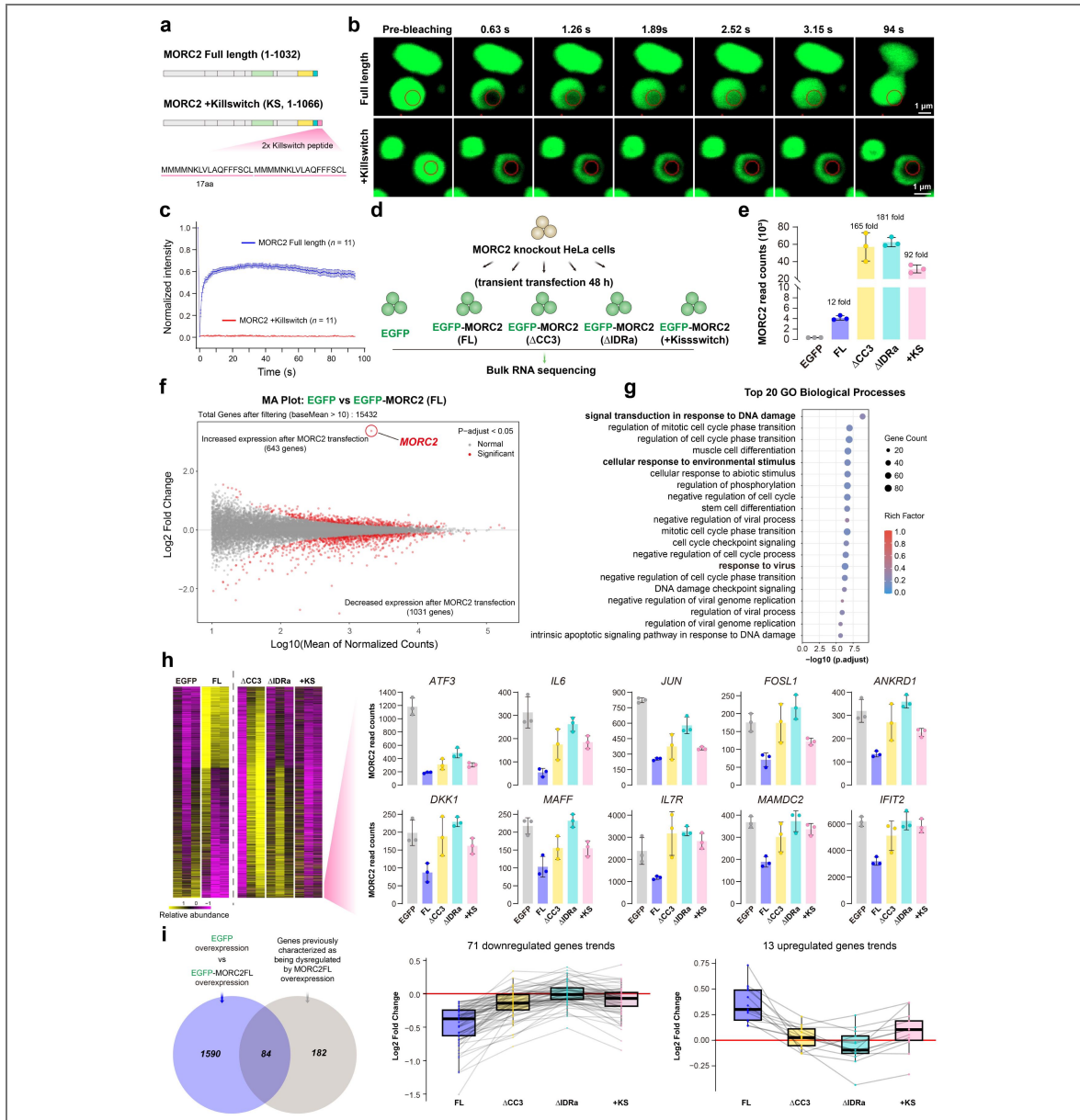


Figure 6. Dynamic MORC2 condensation is required for its transcriptional regulatory activity.

(a) Schematic diagram showing the fusion of two 17-amino acid killswitch sequences to the C terminus of MORC2. **(b)** Live-cell FRAP images of HeLa cells overexpressing EGFP-MORC2 full length (FL) and EGFP-MORC2-Killswitch (+KS). Scale bar: 1 μ m. **(c)** FRAP analysis of EGFP-MORC2 condensates in the nucleus. Data represent mean $\pm$ SEM, with $n = 11$ replicates. **(d)** Schematic of the rescue experiment strategy. HeLa cells with CRISPR-Cas9-mediated MORC2 knockout were reconstituted with EGFP, EGFP-MORC2FL, or phase separation-deficient mutants EGFP-MORC2 Δ CC3 and EGFP-MORC2 Δ IDR, and dynamics-defective +KS mutant. **(e)** RNA-seq-derived read counts for the MORC2 gene in EGFP, EGFP-MORC2FL, EGFP-MORC2 Δ CC3, EGFP-MORC2 Δ IDR, and EGFP-MORC2FL+KS expressing cells. **(f)** The effect of MORC2 rescue on the transcriptome. MA plot showing all significantly differentially expressed genes between EGFP-MORC2FL and EGFP transfection in MORC2-knockout HeLa cells ($n = 1,074$; DESeq2 adjusted $p < 0.05$), highlighted in red. **(g)** GO-based biological process terms enriched among all differentially expressed genes. Data represent mean $\pm$ SD, with $n = 3$ replicates. **(h)** Abundance trajectories of all regulated genes, normalized to their respective mean values. Shown is a selected panel of ten downregulated genes that are influenced by MORC2 rescue but show reduced responsiveness to all three mutants. **(i)** Venn diagram illustrating the overlap between differentially expressed genes identified following EGFP-MORC2FL overexpression in this study and previously characterized MORC2-regulated targets. Fold-change ($\log_2$ fold change) analysis of the 84 overlapping genes showing significant expression changes upon MORC2FL overexpression, compared with their corresponding fold changes following overexpression of the three MORC2 mutants. Dashed lines connect the same gene across the four transfection conditions.

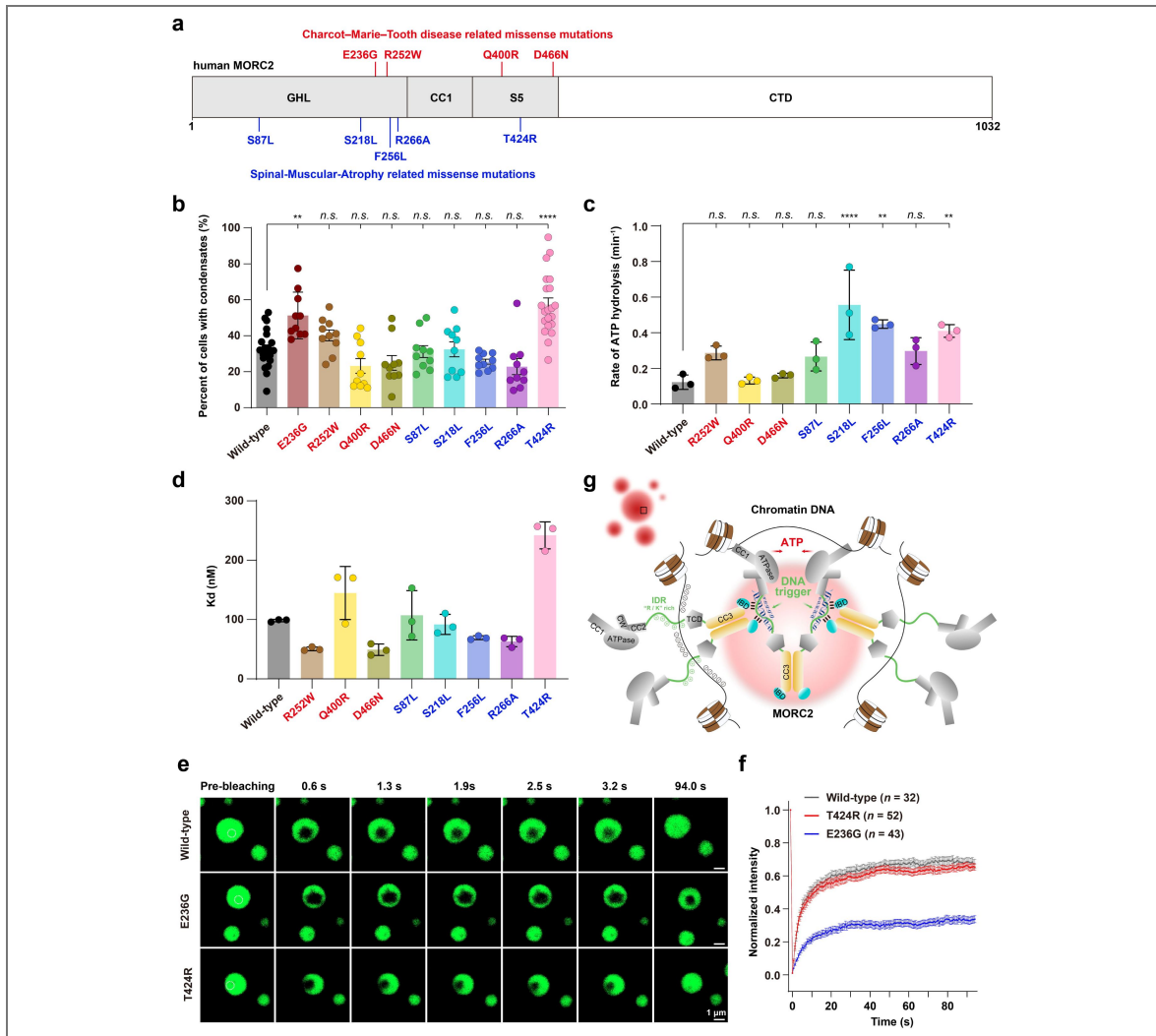


Figure 7. Pathogenic variants of MORC2 alter condensates behavior, DNA binding, and ATPase activity.

(a) Schematic illustration of pathogenic MORC2 mutations associated with Charcot-Marie-Tooth disease type 2Z (CMT2Z, red) and spinal muscular atrophy (SMA, blue), mapped onto the MORC2 domain structure. (b) Quantification of HeLa cells transfected with WT MORC2 or nine pathogenic mutations exhibiting nuclear condensates was performed for each field of view. E236G and T424R shows the most pronounced enhancement of phase separation. $n \geq 10$ fields. Data are presented as mean $\pm$ SEM; one-way ANOVA with Tukey's post hoc test. **** $p < 0.0001$, ** $p < 0.01$, n.s. not significant. (c) ATPase activity of WT MORC2 and eight pathogenic variants, measured under revised conditions. S218L, F256L, and T424R exhibit significantly elevated activity. We were unable to purify MORC2 constructs bearing the E236G mutation from either HEK293F cells, suggesting that it may cause misfolding of the ATPase module. Data are presented as mean $\pm$ SEM; one-way ANOVA with Tukey's post hoc test. **** $p < 0.0001$, ** $p < 0.01$, n.s. not significant. (d) Fluorescence polarization (FP) analysis of DNA binding affinities to the 601 DNA sequence. All protein variants were evaluated under uniform assay conditions, including those pertinent to phase separation. Dissociation constants (K_d) were determined by fitting the FP data to a log(agonist) vs. response model and are reported as mean $\pm$ SEM: WT (98 ± 1 nM), R252W (51 ± 2 nM), Q400R (145 ± 26 nM), D466N (49 ± 6 nM), S87L (107 ± 24 nM), S218L (92 ± 10 nM), F256L (69 ± 2 nM), R266A (63 ± 5 nM), and T424R (242 ± 13 nM). Several variants, including R252W, D466N, F256L, and R266A, demonstrated increased binding affinity relative to WT, while others, such as T424R and Q400R, exhibited markedly reduced binding. (e). Live-cell FRAP images of HeLa cells overexpressing EGFP-MORC2 full length (FL), E236G, and T424R mutants. Scale bar: 1 μ m. (f). FRAP analysis of EGFP-MORC2 condensates in the nucleus. Data represent mean $\pm$ SEM, with $n \geq 32$ replicates. (g) A Mechanistic Model of MORC2-Mediated Chromatin Condensation. MORC2 functions as a multivalent DNA-binding machine. The C-terminal CC3-IBD and IDR serve as the structural engine for phase separation, while DNA acts as a scaffold that lowers the phase boundary. The N-terminal ATPase domain undergoes conformational cycling (Open/Closed), providing the dynamic fluidity necessary for the assembly to remain within a functional 'viscoelastic window'. In this window, MORC2 achieves the local density required for DNA compaction and gene silencing.

condensate fluidity compared with the full-length protein (Figure S11 [↗](#)). This highlights the N-terminus as a critical regulator of the material properties of MORC2 assemblies. Consequently, we evaluated the dynamics of E236G and T424R condensates; FRAP measurements showed substantially reduced recovery for E236G, indicating decreased internal fluidity, whereas T424R exhibited minimal effects (Figure 7e, f [↗](#)).

In vitro ATPase assays revealed that T424R significantly increases enzymatic activity (Figure 7c [↗](#)), consistent with its reported role in promoting ATPase dimer cycling¹³. Additionally, S218L and F256L variants enhanced ATP hydrolysis, whereas others, such as Q400R and D466N, showed activity comparable to WT (Figure 7c [↗](#)). Due to severe solubility issues, E236G could not be biochemically characterized *in vitro* (Figure S12e [↗](#)).

We next assessed DNA binding affinity of MORC2 variants using fluorescence polarization (FP) assays. Despite subtle differences, none of the variants showed statistically significant changes in DNA-binding capacity (Figure 7d [↗](#)), indicating that these mutations primarily perturb enzymatic turnover and condensation dynamics rather than DNA recruitment.

In summary, our results support a model in which MORC2 mediates transcriptional regulation through its intrinsic ability to undergo biomolecular condensation. The N-terminal ATPase domain undergoes conformational cycling between open and closed states, while the C-terminal CC3-mediated dimer acts as a stable scaffold for condensate formation (Figure 7g [↗](#)). MORC2 condensation is further fine-tuned by multivalent interactions between the IBD and IDR_a regions, as well as by DNA binding. Together, these features enable MORC2 to bridge distal chromatin regions and establish repressive condensates, underscoring its central role in chromatin organization and genome stability.

Discussion

Biophysical Principles of MORC2 Organization

While the MORC family of ATPases is recognized as a central player in epigenetic regulation⁴³, genome stability^{44,45}, and meiotic progression^{46,47}, the biophysical principles governing their spatiotemporal organization have remained elusive. In this study, we establish a causal link between MORC2 condensation and its transcriptional regulatory function. By integrating high-resolution structural insights with functional assays, we reveal that MORC2 does not act merely as a solitary chromatin remodeler^{3,5,48}; instead, it functions as phase separation-driven DNA compaction machinery. This discovery demonstrates how mesoscale material properties—specifically the fluidity and density of protein-DNA assemblies—are directly translated into precise gene regulation.

A Modular Engine for Condensate Assembly

The architectural basis of MORC2 assembly relies on a finely tuned multi-domain interplay. The C-terminal region, comprising the structured CC3-IBD and the intrinsically disordered IDR, provides the mechanistic engine for this process. CC3 serves as a critical structural scaffold (Figure 1e [↗](#), and 3a), mediating homodimerization that effectively doubles the molecular valency, thereby facilitating multivalent “sticker” interactions between the IDR and the defined IBD (Figure 4 [↗](#)). We found that mutations selectively abrogating condensate formation, or those that arrest condensate dynamics without blocking assembly, impair MORC2’s ability to repress core target genes (Figure 6 [↗](#)). This underscores that high internal fluidity is functionally paramount, ensuring the rapid exchange kinetics required for continuous ATP hydrolysis and dynamic chromatin engagement.

The “Viscoelastic Window” and Molecular Stoichiometry

A defining insight from our study is that the mere presence of nuclear puncta is insufficient for MORC2’s repressor activity. Instead, its function is governed by the precise material state of the condensates. We demonstrate that DNA acts as an essential functional scaffold (Figure 5 [↗](#)), driving condensation potentially mediated by nonspecific electrostatic interactions with the

arginine/lysine-rich IDR. Crucially, these assemblies are exquisitely sensitive to molecular stoichiometry. This establishes a narrow, functionally permissive “viscoelastic window”: Too Diffuse: MORC2 fails to achieve the local density necessary for mechanical DNA compaction. Too Rigid: If the system crosses the threshold into a solid-like, arrested state (as seen in the MORC2 +KS mutant or E236G variant), it loses the fluidity required for enzymatic turnover and co-repressor engagement. Within the nucleus, the CW domain likely serves as a regulatory node, orchestrating the compositional regulation of these condensates by reading specific epigenetic marks to fine-tune local valency and morphology (Figure 3 [↗](#) and Figure S3 [↗](#)).

Mechanistic Insights into Pathogenic Variants

Our analysis of disease-associated variants—particularly E236G (CMT2Z) and T424R (SMA)—highlights how mutations in the ATPase domain can pleiotropically affect both enzymatic turnover and phase behavior (Figure 7 [↗](#)). While T424R primarily accelerates the ATPase dimer cycle¹³, E236G triggers a transition toward a less dynamic, “hardened” condensate state. These results emphasize that disease mutations do not simply “turn off” the protein; they perturb the biophysical context of the assembly. The observation that some variants exhibit elevated ATPase activity *in vitro* suggests that in the native cellular environment, condensation-mediated local enrichment may be required to properly couple ATP hydrolysis to productive chromatin remodeling.

Our findings clarify previous discrepancies regarding MORC2’s DNA binding. By utilizing the 601 nucleosome-positioning sequence, we demonstrate a much more robust interaction than previously captured with short, random dsDNA. Furthermore, the use of full-length MORC2 reveals that C-terminal regulatory domains are essential for maintaining enzymatic homeostasis—a feature masked in truncated constructs. Whether MORC2-mediated silencing is achieved through direct modulation of ATPase cycles or an independent mechanical compaction mechanism remains an open question.

While our study establishes the essential role of MORC2 condensation in transcriptional control, this exploration represents an initial step toward deciphering the complex coordination between enzymatic turnover and phase behavior. Future research is required to resolve how the ATP-dependent “DNA clamp” cycle is spatially coupled to the physical compaction of chromatin, and whether additional co-repressors are selectively partitioned into these dynamic assemblies to maintain epigenetic stability.

Materials and Methods

Plasmid construction, cell culture and transfection

WT MORC2 (UniProt: Q9Y6X9) cDNA was amplified from a cDNA library derived from HeLa cells via reverse transcription. The MORC2 cDNA was then subcloned into the p3xFLAG-Myc-CMV-24 vector between BglII and KpnI restriction sites using seamless cloning. The seamless cloning process was facilitated by the ClonExpress II One-Step Cloning Kit (Vazyme, #C112-02) according to the manufacturer’s instructions. An EGFP sequence was subsequently inserted into the BglII site at the N-terminal of MORC2 to generate the EGFP-MORC2 construct. To generate deletion mutant constructs of MORC2 for intracellular studies, specific regions of MORC2 were targeted and deleted using a polymerase chain reaction (PCR)-based approach. An 18- or 21 bp homologous sequence originating from one of the forward or reverse primers was introduced into the other primer, and the resulting PCR products were subjected to homologous recombination cloning, facilitated by the 18-bp homologous sequence. For protein expression and purification, MORC2FL and truncated MORC2 were cloned into prokaryotic expression vectors with MBP or Trx fusion proteins for bacterial expression.

In addition, MORC2FL and a series of truncation mutants were purified using a HEK293F eukaryotic expression system for *in vitro* phase separation and static light scattering (SLS) assays. A260/280 measurements, agarose gel analysis, or EMSA information on protein purifications and their experimental applications in this study is provided in Supplementary Table 2 [↗](#). HeLa,

HEK293T cells were cultured in DMEM medium (Biological Industries, DMEM, high re maintained in a humidified incubator at 37°C with 5% CO₂. Cell transfection was performed using Lipofectamine 3000 (Invitrogen) according to the manufacturer's instructions. Briefly, cells were seeded in appropriate culture plates the day before transfection and allowed to reach 40-60% confluency. The separately diluted Lipofectamine 3000 reagent and plasmids in Opti-MEM reduced serum medium (Gibco) were mixed, and the transfection complexes were added to cells after 20 minutes. After a 6-hour incubation, the transfection medium was replaced with fresh complete DMEM medium.

Recombinant protein expression, purification

Plasmids containing His-tagged genes were transformed into *E. coli* BL21 (DE3) cells and cultured in Luria-Bertani medium with 50 mg/L ampicillin. When the culture reached an optical density (OD) of 0.6-0.8 at 600 nm, the temperature was lowered to 16°C. Isopropyl β-D-1-thiogalactopyranoside (IPTG) (0.25 mM) was added to induce protein expression for 18-20 hours. Cells were harvested by centrifugation at 13,000 × g for 15 minutes, then resuspended in Buffer A (50 mM Tris-HCl, pH 8.0, 1000 mM NaCl, 10 mM imidazole, 1 mM dithiothreitol (DTT), and 1 mM phenylmethylsulfonyl fluoride (PMSF)). The cell suspension was snap-frozen and stored at -80°C for subsequent protein purification. HEK293F suspension-adapted cells were cultured in conical flasks at 37°C, 120 rpm, and 5% CO₂, until a density of 1.0 × 10⁶ cells/mL was reached. The cells were transiently transfected with 1 mg of MORC2-6 × His plasmid using 2 mL of branched polyethyleneimine (1 mg/mL). After 48 hours of incubation, cells were harvested by centrifugation at 3,000 × g for 5 minutes, resuspended in Buffer A, snap-frozen, and stored at -80 °C for further purification.

All purification steps were performed at 4°C to preserve protein activity. Frozen cell suspensions were rapidly thawed in water, lysed by sonication (200 W for ~0.3 hours), and centrifuged at 20,000 × g for 30 minutes at 4°C. The supernatant was filtered through a 0.22 μm syringe filter to obtain soluble material. A His-Trap Ni-NTA 6FF chromatography column was equilibrated with five column volumes of Buffer A (Buffer A: 50 mM Tris-HCl, pH 8.0, 1000 mM NaCl). The clarified extract was loaded onto the column, washed with Buffer A to remove non-specific proteins, and eluted with a linear gradient of nickel elution buffer (Buffer B: 50 mM Tris-HCl, pH 8.0, 1 M NaCl, 300 mM imidazole, 1 mM DTT). Eluted fractions were analyzed by SDS-PAGE and Coomassie staining. The protein of interest was concentrated, exchanged into Buffer C (50 mM Tris-HCl, pH 8.0, 1000 mM NaCl, 1 mM EDTA, 1 mM DTT) using an Amicon Ultra centrifugal filter unit, following overnight treatment with Benzonase nuclease to eliminate nucleic acid contamination or HRV 3C Protease to cut His-tag. The purification process was subsequently refined through gel filtration using a Superdex 6 Increase column (GE Healthcare) for the MORC2FL. Following this, fragments were conjugated with a GPGS linker, and the resulting mixture was subjected to a size-exclusion chromatography process (Superdex 200/75 column, GE Healthcare) to yield a final solution comprising 50 mM Tris-HCl, pH 8.0, 500 mM NaCl, 1 mM EDTA, and 1 mM DTT. Peak fractions were analyzed by SDS-PAGE and Coomassie staining, confirming the successful purification of highly homogeneous MORC2FL (1-1032) and various fragments (Figure S7 [\[4\]](#)), then flash-frozen in liquid nitrogen and stored at -80°C.

Crystallization, data collection and processing

The CC3 domain (901-1003) and various mutations of MORC2 constructs were cloned into a modified pET32M vector (with Trx-tag) using standard PCR-based methods and confirmed by DNA sequencing. Due to the lack of homologous structures, protein crystals with heavy atom derivatives are required to determine their phases. Therefore, we used the methionine-deficient strain B834 to express the target protein with selenomethionine (SeMet) in an inorganic selenosubstituted medium. Recombinant proteins were expressed in *E. coli* B834 cells in M9 medium at 16°C. MORC2 CC3 fused with a GPGS linker were finally changed by step of size-exclusion chromatography (Superdex 75 increase column, GE Healthcare) into 50 mM Tris-HCl, pH 8.0, 100 mM NaCl, 1 mM EDTA and 1 mM DTT. The freshly purified protein was concentrated to

10–20 mg/mL. Crystals were obtained by the static sitting drop method in 0.1M Citric acid (pH 3.5) and 25% w/v PEG 3350 at 25°C. Glycerol (20%) was added as the cryo-protectant. A 3.1-Å resolution X-ray dataset was collected at the beamline BL18U1 of the Shanghai Synchrotron Radiation Facility (SSRF). The diffraction data were processed and scaled by HKL2000 (<http://www.hkl-xray.com/>). Using the structure of auto-build model by SAD data and Alphafold2 predicted model as the search model, the initial structural model was solved by molecular replacement in PHASER (<https://www.ccp4.ac.uk/>). Further manual model adjustment and refinement were completed iteratively using COOT and PHENIX. The final structure was validated by PISA and MolProbity (<https://www.phenix-online.org/>). The final refinement statistics of the complex structure are listed in Supplementary Table 1. All structure figures were prepared using the programme PyMOL (<http://pymol.sourceforge.net/>).

Protein labeling with fluorophore

The Cy3 NHS ester (AAT Bioquest) fluorophores were dissolved in DMSO at 10 mg/mL. Purified proteins were prepared in reaction buffer at a concentration of 10 mg/mL. The fluorophores were added to the protein solution at a 1:1 molar ratio. The reaction was conducted with shaking at room temperature for 1 hour in the dark. After the reaction, labeling was quenched with 200 mM Tris-HCl, pH 8.3, and the labeled proteins were desalted using a HiTrap desalting column (GE Healthcare). Fluorescence labeling efficiency was assessed using a Nanodrop 2000 spectrophotometer (ThermoFisher #ND-2000). The final labeling efficiency of each protein was adjusted to 1% by mixing the labeled protein with an excess of unlabeled protein (1:99 ratio).

Lentiviral mediated H2A-mCherry inducible expression in HeLa cells

Stable expression of H2A-mCherry in HeLa cells was achieved through lentiviral-mediated transduction. First, H2A-mCherry gene sequence was inserted into the inducible lentiviral vector pTRIPZ to generate the expression plasmid pTRIPZ-H2A-mCherry. Lentiviral particles were generated by triple transfection of HEK293T cells with pTRIPZ-H2A-mCherry, PMD2.G, and PSPAX2 plasmids using Lipofectamine 3000. The viral supernatant was collected 48 hours after transfection, filtered with 0.45-µm filter to remove cell debris, and used to transduce HeLa cells. Following transduction, HeLa cells were selected with 50 µg/mL hygromycin to establish stable cell lines. H2A-mCherry expression was induced with 1 µg/mL doxycycline.

Live cell imaging and immunofluorescence staining

For live-cell imaging, HeLa cells were seeded onto 35 mm culture dishes with glass-bottom plates (Biosharp) and transfected with EGFP-MORC2 or its mutant constructs. Images were captured 12 hours post-transfection using a Nikon confocal microscope equipped with an incubator system (37°C, 5% CO₂). Nuclei were stained with Hoechst 33342 or visualized using mCherry-labeled H2A fusion protein. For immunofluorescence, cells were fixed with 4% paraformaldehyde, permeabilized with 0.1% Triton X-100, and blocked with 10% goat serum. Primary antibodies were incubated overnight at 4°C, followed by incubation with fluorescence-conjugated secondary antibodies for 2 hours at room temperature in the dark. Fluorescence images were acquired using a Nikon confocal microscope.

CRISPR-Cas9 mediated MORC2 knockout cells

MORC2 knockout (KO) HeLa cell lines were generated using the CRISPR-Cas9 system. A CRISPR single-guide RNA (sgRNA) targeting exon 5 of MORC2 (5'-ACACCTGAGTCTACTCAGAT-3') was designed using the online tool (<http://crispor.tefor.net/>). Two DNA oligos with complementary sgRNA targeting sequence and additional bases were designed as previously described^{49,50}. These oligos were annealed and cloned into pLentiCRISPR v2 (Addgene plasmid #52961) to generate the MORC2 knockout vector. After transfection and puromycin selection, MORC2-deleted cells were obtained. Individual cells were sorted into separate wells of a culture plate to establish monoclonal cell lines using flow cytometry. To validate the knockout cell lines, genomic DNA was

extracted from both WT and KO cells. The specific regions targeted by the sgRNA were amplified using PCR with primers: 5'-CAACATTCTCGAGCTGGACCTACAG-3' (forward) and 5'-CCACGACAAGACTGGAAACGTGACTC-3' (reverse). The PCR products were purified and subjected to Sanger sequencing, revealing the deletion and insertion mutations in the target sites of the MORC2 allelic genes (Figure S2d [↗](#)).

Generation of endogenous EGFP-MORC2 chimeric mice

CRISPR-Cas9 was used to generate EGFP-MORC2 chimeric mice. The EGFP and linker sequence were inserted at the N-terminus of the mouse *MORC2a* gene. Briefly, a single-guide RNA (sgRNA) was designed to target exon 1 of *MORC2a*, with the sequence GACTGAAGACTCATTGCTGTCA. A donor DNA fragments containing the EGFP-linker region flanked by homologous arms was constructed. Fertilized mouse zygotes were collected, and a mixture of Cas9 mRNA (50 ng/μL), sgRNA (25 ng/μL), and donor DNA (50 ng/μL) was microinjected into pronucleus of the zygotes. The injected embryos were transferred into the oviducts of pseudopregnant recipient female mice. Resulting pups were genotyped by PCR and Sanger sequencing to confirm transgene integration (Figure S2f [↗](#)). The genotyping primers for the EGFP-MORC2 allele were as follows: Left-F1: GCGGTGGTTGAGTTCCAATTCC; Left-R1: CTCAGGGTCAGCTGCCGTAG; Right-F1: GAACGGCATCAAGGTGAACTTC; Right-R1: TGTATCCAAGGTAATCTCTGTGTA.

Mice were bred at the Animal Center of the University of Science and Technology of China (USTC). Animals were housed under controlled conditions (22[°C, 12-hour light/dark cycle) with free access to food and water ad libitum. All experimental procedures were approved by the Institutional Animal Care and Use Committee (IACUC) of the University of Science and Technology of China (Hefei, China).

Fluorescence recovery after photobleaching (FRAP)

The FRAP assay was performed to investigate the dynamics of protein mobility within condensates, either in *in vivo* or *in vitro*. Specific regions of interest (ROIs) approximately 2 μm² in size were selected and subjected to photobleaching using 100% laser power (488 nm) for approximately 20 ns. One droplet was left unbleached as a control. Fluorescence recovery within the ROI was monitored for 100-300 seconds. FRAP was performed using a 63x oil immersion objective mounted on a Nikon confocal microscope (in cells) or a Zeiss LSM 880 (*in vitro*). Fluorescence intensity was normalized to the pre-bleach signal to analyze the recovery kinetics. For brain slice FRAP, organotypic brain slices were prepared from adult chimeric mice expressing endogenous EGFP-MORC2, following a slightly modified protocol previously described⁴⁹. After decapitation, brains were rapidly extracted and placed in ice-cold dissection medium consisting of Hibernate A, 2% B27 supplement, 2[mM L-glutamine, and 1% penicillin–streptomycin. The cerebellum and midbrain were removed, and the remaining cerebral hemispheres were coronally sectioned at 250[μm using a McIlwain tissue chopper. Slices were gently separated in chilled dissection medium and transferred onto glass-bottom dishes containing culture medium composed of Neurobasal A, 2% B27 supplement, 2[mM L-glutamine, and 1% penicillin–streptomycin. Imaging was initiated immediately after brain slice preparation. FRAP experiments were performed using the Nikon confocal microscope. Photobleaching was carried out with 5-7 laser pulses (50[μs dwell time), and images were acquired at 2-s intervals. Post-bleach images were acquired 20[s after photobleaching.

Sedimentation assay

To assess concentration-dependent partitioning behavior, purified proteins at the indicated concentrations were incubated under the same buffer conditions used for *in vitro* phase separation assays and equilibrated for 10 min at room temperature. Samples were then centrifuged at 16,000 × g for 10 min to separate the supernatant and pellet fractions. Following centrifugation, the supernatant was carefully collected, and the pellet was resuspended in an equal volume of buffer. Equal volumes of supernatant and pellet fractions were analyzed by SDS–PAGE and quantified by densitometry.

Electrophoretic mobility shift assays (EMSA)

Widom 601 DNA (147 bp, 25 nM) was amplified by PCR and incubated with increasing concentrations of MORC2 or its domain constructs in binding buffer (20 mM Tris-HCl, pH 7.5, 45 mM KCl, 4 mM MgCl₂, 5% glycerol, 5 mM DTT). The reaction mixture was loaded onto a 6% polyacrylamide gel and electrophoresed at 120 V for 70 minutes. DNA was visualized using SYBR Gold stain, and images were captured using the ChemiDoc MP Imaging System.

Size-exclusion chromatography coupled with static light scattering

The analysis was performed using an AKTA FPLC system (GE Healthcare) coupled to a static light scattering detector (miniDawn, Wyatt) and a differential refractive index detector (Optilab, Wyatt). Protein samples (70 μM) were filtered and applied to a Superdex 200 increase or Superose 6 increase column, pre-equilibrated with a column buffer containing 50 mM Tris-HCl, pH 8.0, 150 mM NaCl or 1000 mM NaCl (MORC2FL), 1 mM EDTA, and 1 mM DTT. Data were analyzed using ASTRA6 software (Wyatt).

In vitro phase separation and fluorescence microscopy imaging

Droplet formation was monitored using differential interference contrast (DIC) and fluorescence microscopy. To induce phase separation, protein was added to phase separation buffer (50 mM Tris-HCl, pH 8.0, 150 mM NaCl, 1 mM DTT). For consistency, all MORC2 samples were diluted to a final concentration of 4 μM with phase separation buffer containing 200 mM NaCl, unless otherwise specified. All preparation steps were performed at 4°C to preserve protein activity. The phase separation mixtures were then equilibrated at room temperature for 20 min. A 5–10 μL aliquot of each sample was placed onto a custom-made glass slide, covered with an 18 mm diameter coverslip, and incubated at room temperature for 20 min before imaging.

Protein isotope labeling and NMR Spectroscopy

Isotopically labeled protein samples were expressed in minimal M9/H₂O or M9/D₂O medium. Uniformly ¹⁵N-labeled (or ¹³C- or ¹⁵N-labeled) protein samples were prepared in M9/H₂O medium supplemented with 1 g/L of ¹⁵NH₄Cl and 2 g/L of glucose (or ¹³C-glucose). The cells were harvested at an OD₆₀₀ of approximately 1.0. Isotope-labeled precursors and amino acids were purchased from Cambridge Isotope Laboratories (CIL). Isotopically labeled samples were prepared in 50 mM Tris-HCl, pH 8.0, 100 mM NaCl, and 7% D₂O. Protein concentrations ranged from 0.5 to 1.0 mM. NMR experiments were performed using Bruker Avance III 600 MHz spectrometers equipped with a cryo-probe. NMR spectra of MORC2 and its complex were recorded at 298 K. Backbone assignments of MORC2^{1004–1032} were obtained using SOFAST versions of standard triple-resonance experiments, including HNCACB and CBCACONH, recorded at 298 K. Residual assignment ambiguities were resolved using ¹⁵N-edited HMQC-NOESY-HMQC spectra. All NMR spectra were processed using NMRPipe and analyzed with NMRViewJ software (<http://www.onemoonscientific.com>). CSPs were calculated in Hz at 600 MHz using the following equation:

$$\Delta\nu = \left((\Delta^1\text{H} \times 600)^2 + (\Delta^{13}\text{C} \times 150)^2 \right)^{1/2}$$

Fluorescence polarization Assay (FP)

Fluorescence polarization (FP) assays were performed at room temperature (25°C) using a BioTek H1 fluorescence spectrophotometer to evaluate the binding affinity between MORC2 protein and 5-carboxyfluorescein (5-FAM)-labeled 601 DNA (FAM-601) across a concentration range of 0–2000 nM. The 147-bp Widom 601 DNA fragment was generated via PCR amplification using a 5-FAM-labeled primer. The labeled primer was dissolved in 0.1 M NaHCO₃ buffer, pH 8.3, and the amplified product was purified by agarose gel extraction to eliminate residual primers and nonspecific amplification products.

Fluorescence measurements were carried out in assay buffer (50 mM Tris-HCl, pH 8.0, 150 mM NaCl, 1 mM EDTA, 1 mM DTT, and 10% glycerol) containing purified FAM-601 DNA at a final concentration of 0.4[ng/μL. Excitation and emission wavelengths were optimized according to the instrument manual, with excitation set at 485[nm and emission at 525[nm. Fluorescence polarization values were recorded in triplicate, and control measurements were included to correct for background signals.

Data were analyzed to determine the equilibrium dissociation constant (K_d) for the MORC2-FAM-601 interaction by fitting data to a log(agonist) vs. response model. In parallel, the effect of small molecules on this binding interaction was assessed by introducing compounds into the assay mixture and monitoring changes in fluorescence polarization.

ATPase activity measurements

ATPase activity was measured⁵² using 1[μM MORC2 or its mutant variants at 37°C in 50[mM Tris-HCl, pH 8.0, 150[mM NaCl, 8[mM MgCl₂, 10% glycerol and 5[mM ATP. All experiments were repeated independently at least three times.

RNA-seq and bioinformatic analysis

MORC2-knockout HeLa cells were seeded at equal density and transfected with 3[μg of plasmid DNA and 12[μg of PEI (polyethylenimine) per dish. After 6[hours, the transfection medium was replaced with fresh complete medium, and cells were incubated for an additional 48[hours. Cells were then washed twice with ice-cold PBS and lysed directly in RNA Isolator (R401, Vazyme) on ice. Total RNA was extracted using the RNeasy Mini Kit (Qiagen, 74004) according to the manufacturer's protocol. RNA quality was assessed using NanoDrop 2000 (Thermo Fisher Scientific), Agilent 2100 Bioanalyzer, and LabChip GX (PerkinElmer). For library preparation, 1[μg of total RNA was used per sample, employing the VAHTS Universal V6 RNA-seq Library Prep Kit for MGI®. The resulting double-stranded DNA libraries were denatured, circularized, and digested to obtain single-stranded circular DNA. DNA nanoballs (DNBs) were generated from these templates through rolling circle amplification (RCA). Sequencing was performed on the MGI DNBSEQ-T7 platform (MGI Tech Co., Ltd) using combinatorial Probe-Anchor Synthesis (cPAS) chemistry.

Before performing differential expression analysis, lowly expressed genes were filtered out by retaining only those with a raw count of ≥10 in at least three samples (baseMean). Differential gene expression analysis between sample groups was carried out using the DESeq2 package (v1.26.0). Genes with *p-adjust* < 0.05 were defined as differentially expressed genes (DEGs). These DEGs were subsequently subjected to Gene Ontology (GO) enrichment analysis in DAVID (Knowledgebase v2025_1).

Statistical Analysis

All statistical analyses were performed in GraphPad Prism. All data underwent normality testing (D'Agostino & Pearson test and Anderson-Darling test) in advance. Two-tailed unpaired Student's *t*-test post-hoc Welch's correction and one-way ANOVA with Tukey's post hoc test was used when data were distributed normally. The significance level was set as *p* < 0.05. All statistical results are provided as Source Data.

Supplemental Tables and Figures

Data collection	
Data sets	MORC2-CC3
Space group	<i>I</i> 41
Wavelength (Å)	0.97915
Unit Cell Parameters (Å)	a=130.253, b=130.253, c= 40.895
	$\alpha=\beta=\gamma=90^\circ$
Resolution range (Å)	50-3.05 (3.10-3.05)
No. of unique reflections	6774 (333)
Redundancy	12.6 (10.5)
I/ σ	25.897 (2.600)
Completeness (%)	100.0 (100.0)
R _{merge} ^a (%)	12.0 (121.4)
CC _{1/2} (last resolution shell) ^b	0.703
Structure refinement	
Resolution (Å)	32.56-3.10
R _{cryst} ^c /R _{free} ^d (%)	19.81 / 24.80
Rmsd bonds (Å) / angles (°)	0.004 / 0.627
Average B factor (Å ²) ^e	57.55
No. of atoms	
Protein atoms	1543
Water	6
Ligands	0
Ramachandran plot regions ^e	
Favored (%)	98.99
Allowed (%)	1.01
Outliers (%)	0

Numbers in parentheses represent the value for the highest resolution shell.

(a) $R_{\text{merge}} = \sum |I_i - \bar{I}| / \sum I_i$, where I_i is the intensity of measured reflection and $\bar{I}$ is the mean intensity of all symmetry-related reflections.

(b) $CC_{1/2}$ were defined by Karplus and Diederichs.

(c) $R_{\text{cryst}} = \sum ||F_{\text{calc}}| - |F_{\text{obs}}|| / \sum F_{\text{obs}}$, where F_{obs} and F_{calc} are observed and calculated structure factors.

(d) $R_{\text{free}} = \sum_T ||F_{\text{calc}}| - |F_{\text{obs}}|| / \sum F_{\text{obs}}$, where T is a test data set of about 5% of the total unique reflections randomly chosen and set aside prior to refinement.

(e) B factors and Ramachandran plot statistics are calculated using MOLPROBITY.

Table S1. Statistics of X-ray Crystallographic Data Collection and Model refinement

<i>MORC2 fragment in vitro</i>	<i>Expression system</i>	<i>Ext. coefficient</i>	<i>A260/280 ratios</i>	<i>Experimental applications</i>
<i>MORC2FL (1-1032)</i>	<i>E. coli</i>	88240	0.59	EMSA, Agarose gel analysis (AGA)
<i>CTD (473-1032)</i>	<i>E. coli</i>	46410	0.71	EMSA; Phase separation
<i>CTDΔCW (537-1032)</i>	<i>E. coli</i>	26930	0.58	EMSA; Phase separation; FPLC-MALS
<i>TCD-CC3-IBD (744-1032)</i>	<i>E. coli</i>	26930	0.67	EMSA; FPLC-MALS
<i>CC3-IBD (901-1032)</i>	<i>E. coli</i>	7450	0.51	EMSA; FPLC-MALS, AGA
<i>CC1 (279-362)</i>	<i>E. coli</i>	1490	0.53	EMSA
<i>CC2 (551-603)</i>	<i>E. coli</i>	0	ND	EMSA
<i>CC3 (901-1003)</i>	<i>E. coli</i>	5960	0.49	FPLC-MALS NMR; Structure determination, AGA
<i>IDR (593-735)</i>	<i>E. coli</i>	0	ND	EMSA
<i>IBD (1004-1032)</i>	<i>E. coli</i>	1490	0.58	NMR
<i>IDRa (593-643)</i>	<i>E. coli</i>	0	ND	NMR
<i>IDRb (644-694)</i>	<i>E. coli</i>	0	ND	NMR
<i>IDRc (695-735)</i>	<i>E. coli</i>	0	ND	NMR
<i>Trx-CC3</i>	<i>E. coli</i>	21555	0.6	FPLC-MALS
<i>Trx-CC3^{trun}_901-975</i>	<i>E. coli</i>	14105	0.58	FPLC-MALS
<i>Trx-CC3-IBD_L915Q</i>	<i>E. coli</i>	21555	0.91	FPLC-MALS
<i>Trx-CC3-IBD_F922Q</i>	<i>E. coli</i>	21555	0.59	FPLC-MALS
<i>Trx-CC3-IBD_L911Q</i>	<i>E. coli</i>	21555	0.6	FPLC-MALS
<i>Trx-CC3-IBD_I908Q</i>	<i>E. coli</i>	21555	0.6	FPLC-MALS
<i>Trx-CC3-IBD_F951Q</i>	<i>E. coli</i>	21555	0.54	FPLC-MALS
<i>Trx-CC3-IBD_Y921A</i>	<i>E. coli</i>	20065	0.56	FPLC-MALS
<i>Trx-CC3-IBD_F954Q</i>	<i>E. coli</i>	21555	0.53	FPLC-MALS
<i>Trx-CC3-IBD_L958Q</i>	<i>E. coli</i>	21555	0.58	FPLC-MALS
<i>MORC2FL (1-1032)</i>	<i>HEK293F</i>	88240	0.6	FPLC-MALS; Phase separation; FP; ATPase activity, AGA
<i>MORC2ΔCC3-IBD (1-900)</i>	<i>HEK293F</i>	80790	0.56	FPLC-MALS
<i>MORC2_N39A</i>	<i>HEK293F</i>	88240	0.79	FP; ATPase activity
<i>MORC2_R252W</i>	<i>HEK293F</i>	93740	0.66	FP; ATPase activity
<i>MORC2_Q400R</i>	<i>HEK293F</i>	88240	0.65	FP; ATPase activity
<i>MORC2_D466N</i>	<i>HEK293F</i>	88240	0.58	FP; ATPase activity
<i>MORC2_S87L</i>	<i>HEK293F</i>	88240	0.75	FP; ATPase activity

<i>MORC2_S218L</i>	<i>HEK293F</i>	<i>88240</i>	<i>0.72</i>	<i>FP; ATPase activity</i>
<i>MORC2_F256L</i>	<i>HEK293F</i>	<i>88240</i>	<i>0.6</i>	<i>FP; ATPase activity</i>
<i>MORC2_R266A</i>	<i>HEK293F</i>	<i>88240</i>	<i>0.75</i>	<i>FP; ATPase activity</i>
<i>MORC2_T424R</i>	<i>HEK293F</i>	<i>88240</i>	<i>0.55</i>	<i>FP; ATPase activity</i>
<i>EGFP-MORC2FL</i>	<i>HEK293F</i>	<i>110130</i>	<i>0.59</i>	<i>Phase separation; FRAP, AGA</i>

Table S2. Summary of MORC2 protein constructs used in this study, including expression systems, extinction coefficients, A260/280 ratios, and corresponding experimental applications.

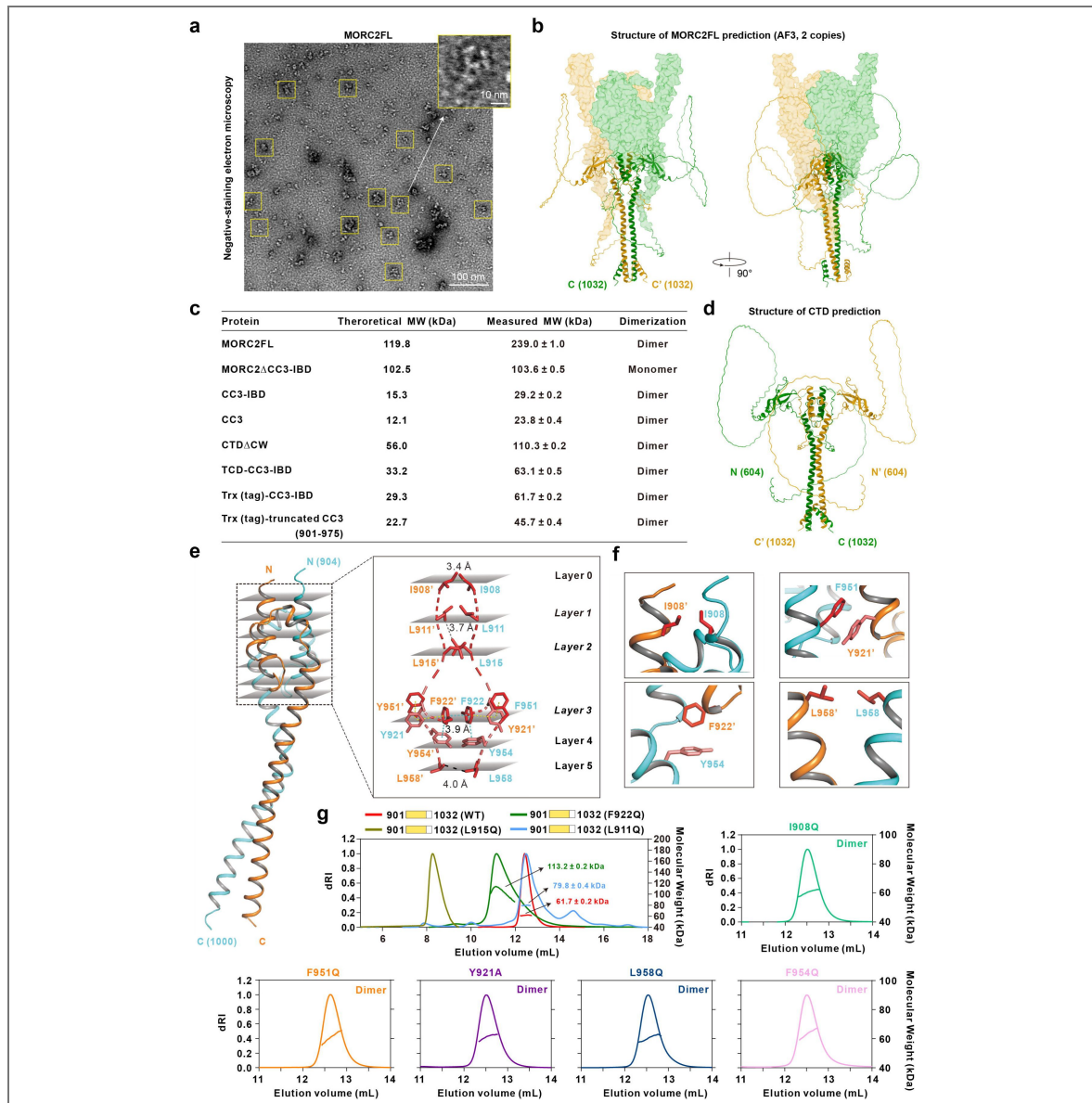


Figure S1. Full-length MORC2 is a dimeric protein, with dimerization mediated by its C-terminal CC3 domain.

(a) Negative-stain electron microscopy revealed that MORC2FL particles purified from HEK293F cells appeared poorly resolved and exhibited considerable morphological heterogeneity. And a zoomed-in particle was to comparing morphology from negative stain EM with AlphaFold predicted structure. Scale bar: 100 nm and 10 nm. (b) Structural prediction of MORC2FL using AlphaFold3 under a two-copy modeling condition revealed the presence of an extended disordered region at the C-terminus and suggested a potential dimerization interface within this region. (c) Summary of theoretical and measured molecular weight of MORC2 constructs, including MORC2FL, MORC2ΔCC3-IBD, CC3-IBD, CC3, and CC3 mutants. Data are derived from recombinant protein purification and static light-scattering (SLS) analysis. (d) AlphaFold3-based modeling of MORC2 with the N-terminal domain removed revealed structural features of the C-terminus, including extensive disorder and a putative dimerization interface. (e) Molecular model of the CC3 dimer highlights hydrophobic core residues (I908, L911, L915, F921, F922, F951, Y954). Key intermolecular distances include I908-I908' (3.4 Å), L911-L915' (3.7 Å), F922-F922' (3.9 Å) and L958-L958' (4.0 Å), supporting dimer stability. (f) Molecular model of the CC3 dimer formation interface show the additional hydrophobic interactions. (g) Point mutations in key residues (L911, L915, F922) affect dimerization. Additional mutants (I908Q, F951Q, Y921A, Y954A, L958Q) show no significant change in molecular weight compared to WT, confirming their minimal impact on dimer formation. SLS analysis shows molecular weight distributions, with WT CC3-IBD forming dimers (~60 kDa) as indicated by red curves, while mutants exhibit higher molecular weights indicative of aggregation. Constructs were tagged with a Trx tag (14 kDa) for solubility.

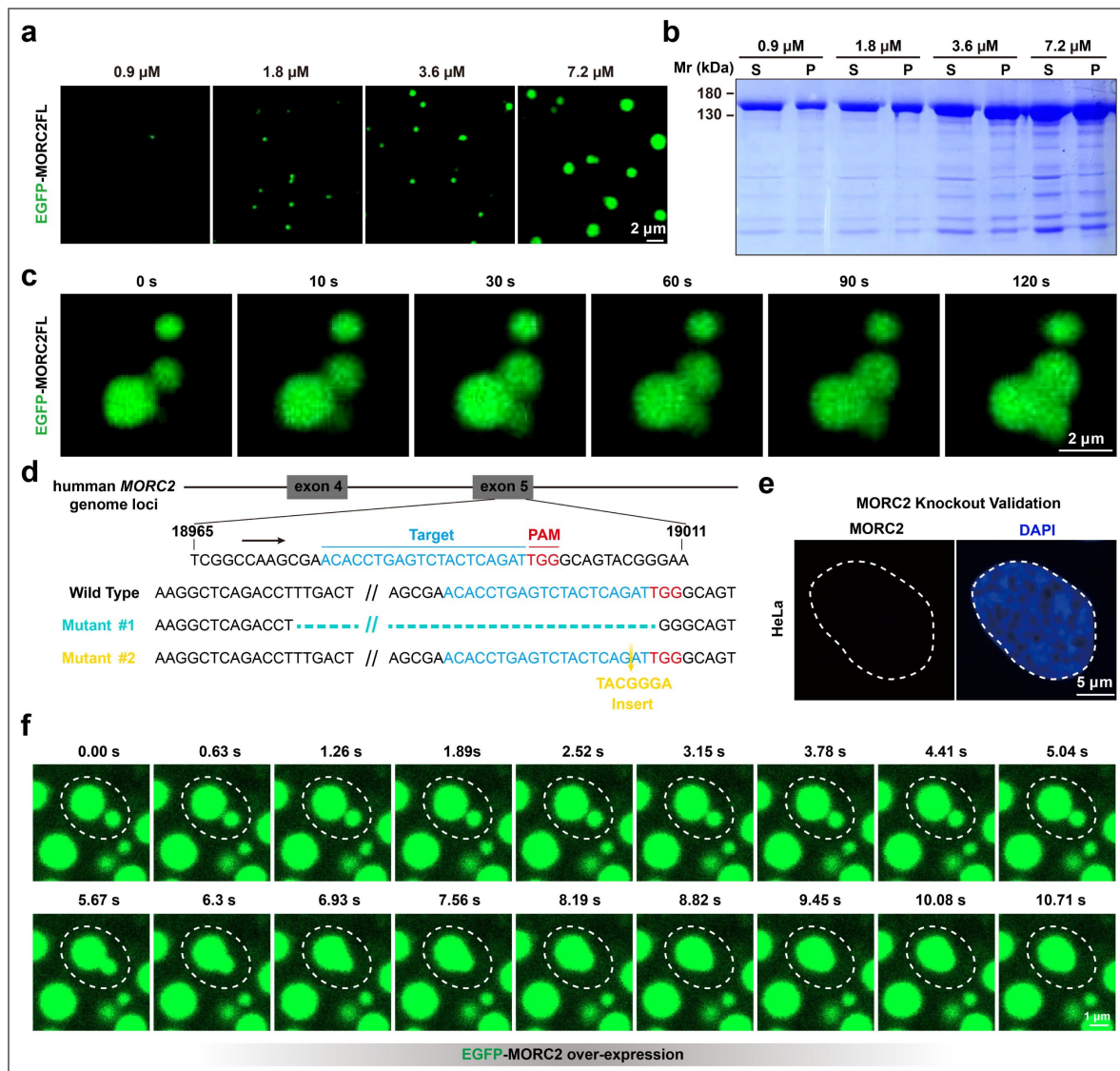


Figure S2. In vitro and in vivo evidence that MORC2 undergoes phase separation.

(a) Concentration gradients of EGFP-MORC2FL proteins (0.9, 1.8, 3.6, and 7.2 μ M) were analyzed for in vitro phase separation. The proteins were examined in a buffer containing 150 mM NaCl, and phase separation was characterized by fluorescence microscopy with 488 nm excitation for EGFP. Scaler bar: 2 μ m. (b) SDS-PAGE analysis of EGFP-MORC2FL in sedimentation-based assays. Protein distribution between supernatant (S) and pellet (P) fractions is shown for increasing concentrations (0.9, 1.8, 3.6, and 7.2 μ M). (c) Time-lapse imaging of EGFP-MORC2FL phase separation over 120 s highlights dynamic droplet fusion. Scaler bar: 2 μ m. (d) Schematic of CRISPR-Cas9 strategy for generating MORC2 knockout (KO) HeLa cells. sgRNA targeting sequences and protospacer adjacent motifs (PAM) are highlighted. Genomic sequencing confirms insertion-deletion mutations introduced at target sites in MORC2 allelic genes. (e) Immunostaining images of WT and MORC2 KO HeLa cells validate the specificity of the custom-made MORC2 antibody. Scaler bar: 5 μ m. (f) The fusion of condensates of EGFP-MORC2 over expression in HeLa cell.

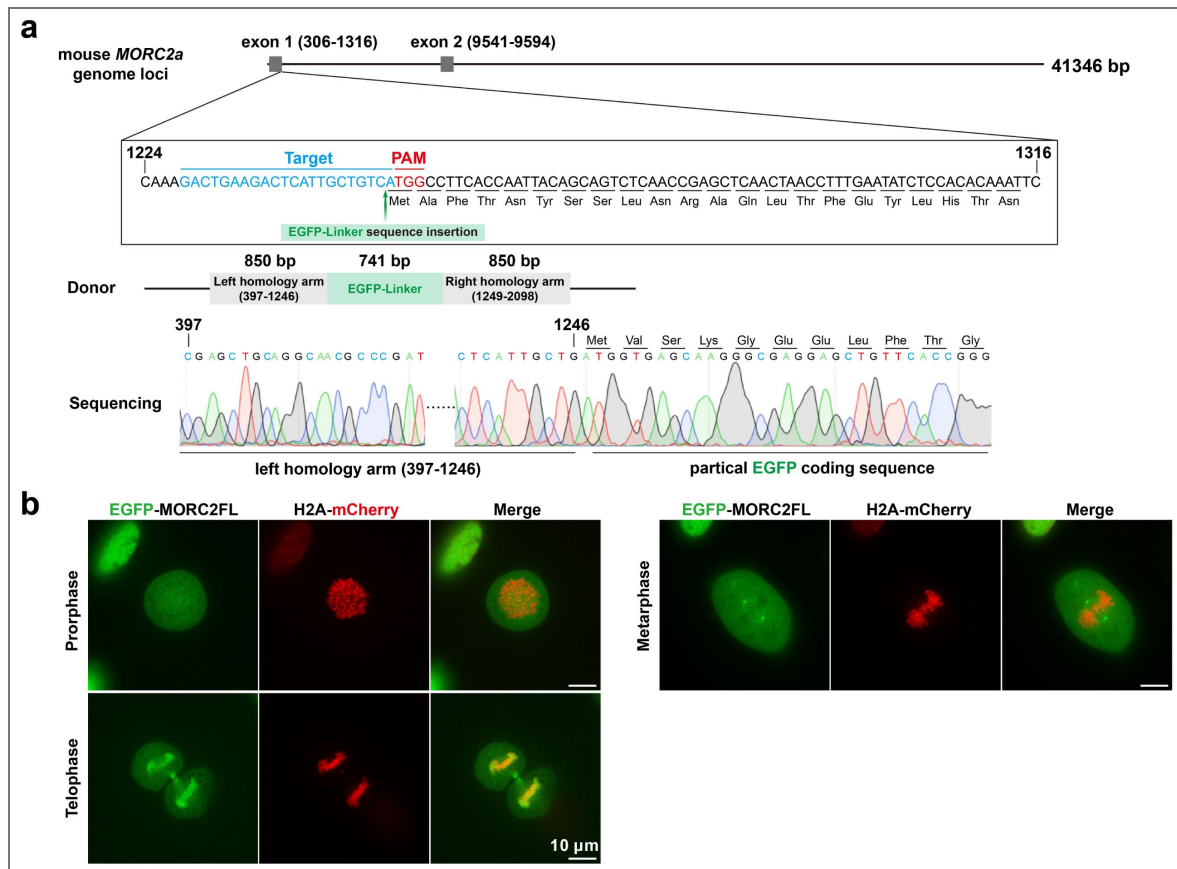


Figure S3. Genomic loci of MORC2a in chimeric mice and its subcellular localization during mitosis.

(a) Generation of endogenous EGFP-MORC2 chimeric mice. Genomic sequencing confirms insertion-EGFP mutations introduced at target sites in MORC2 allelic genes. (b) Live-cell imaging revealed that EGFP-MORC2FL exhibits dynamic subcellular localization throughout mitosis, without forming detectable condensates. Scale bar: 10 μ m.

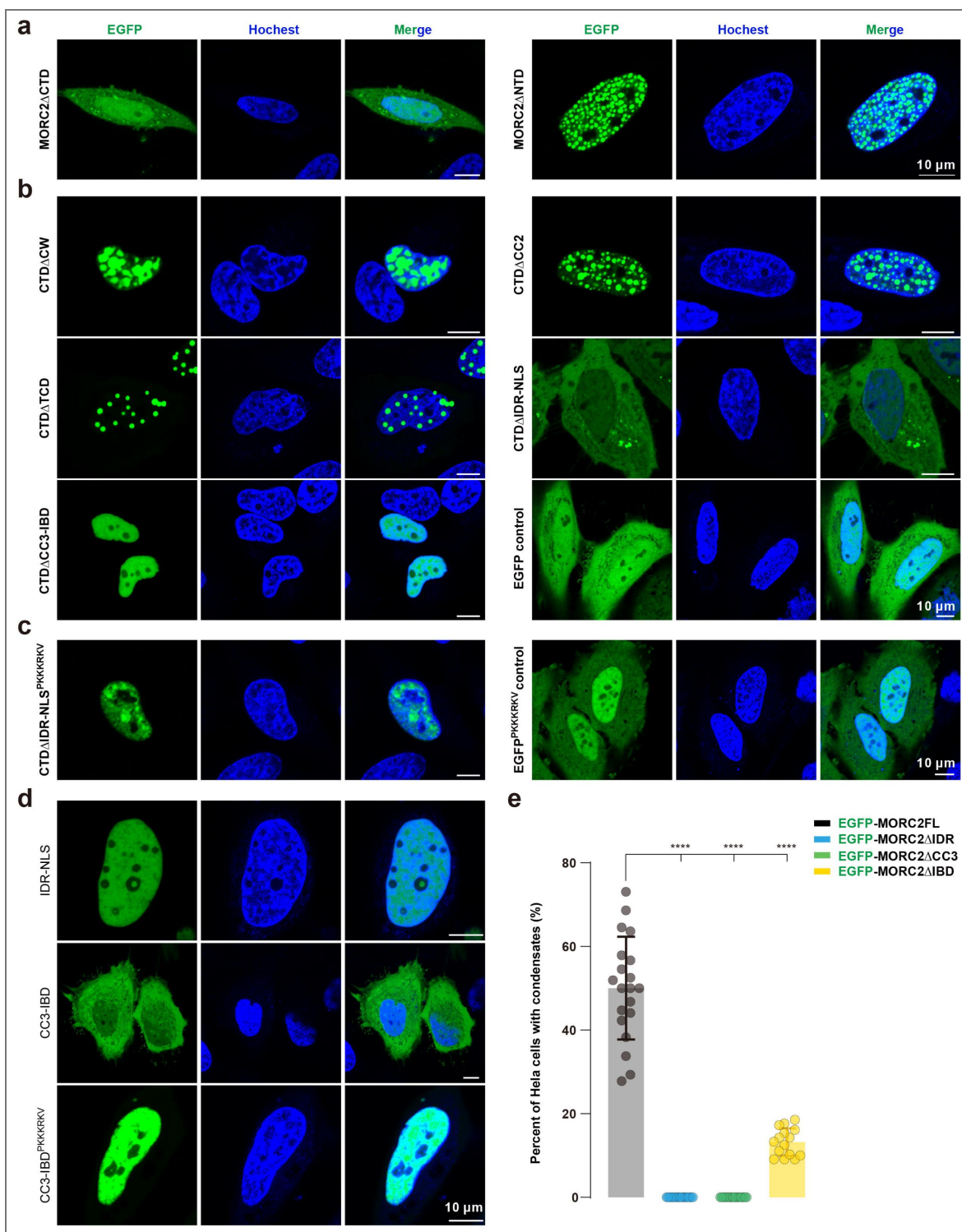


Figure S4. The C-terminal domain (CTD) of MORC2 mediates nuclear condensate formation.

(a) EGFP-tagged MORC2 CTD forms nuclear condensates in HeLa cells, whereas the NTD does not. Scaler bar: 10 μ m. (b) Truncation analysis of EGFP-tagged CTD reveals that both the CC3-IBD and IDR-NLS regions are important for condensate formation. Scaler bar: 10 μ m. (c) EGFP-tagged CTD constructs lacking the IDR-NLS region but fused to an exogenous N-terminal nuclear localization signal ("PKKKRKV") successfully localize to the nucleus but fail to form condensates, indicating that the IDR is essential for phase separation. Scaler bar: 10 μ m. (d) Targeted expression of IDR-NLS and CC3-IBD fragments in the nucleus failed to induce condensate formation, indicating that these domains alone are not sufficient to drive phase separation. Scaler bar: 10 μ m. (e) The statistical data for droplet area in (Figure 3a) demonstrate that the deletion of IDR or CC3 is imperative for condensate formation. The data are expressed as the mean $\pm$ standard error of the mean (SEM). $n \geq 15$ fields; one-way ANOVA with Tukey's post hoc test. **** $P < 0.0001$.

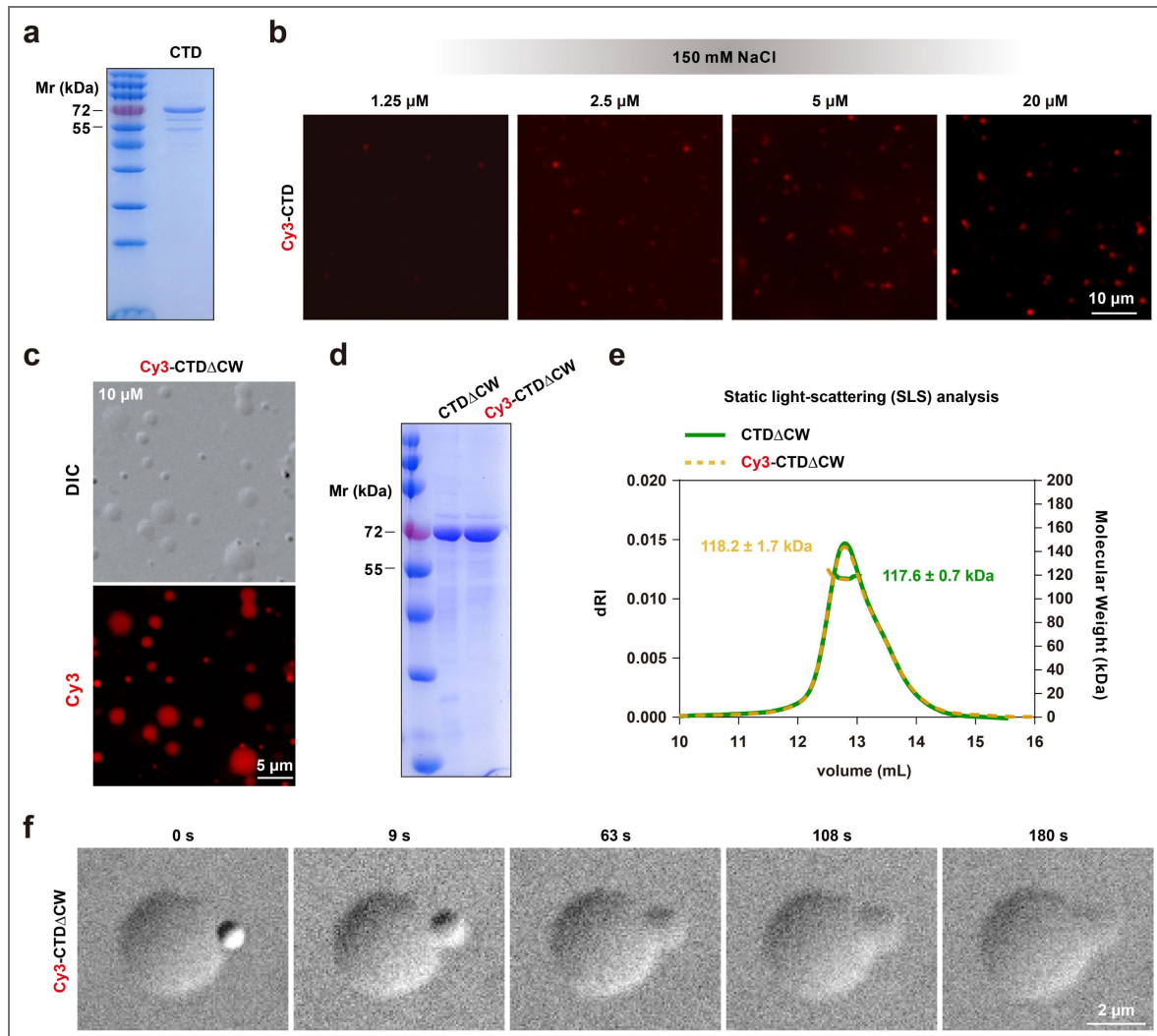


Figure S5. The biophysical behavior of CTD and CTD Δ CW in vitro.

(a) SDS-PAGE analysis of Cy3-labeled CTD showing expected size and degradative behavior. (b) Fluorescence microscopy images of Cy3-labeled CTD at increasing protein concentrations (1.25 μ M, 2.5 μ M, 5 μ M, and 10 μ M) in standard phase separation buffer (50 mM Tris-Cl, pH 8.0, 150 mM NaCl). Despite increasing concentration, CW-CTD formed only small, dense puncta that did not exhibit robust size expansion or morphological changes. Scaler bar: 10 μ m. (c) Representative bright-field and Cy3 fluorescence microscopy images of labeled CTD Δ CW protein droplets, showing intact condensate morphology under standard phase separation conditions. Scaler bar: 5 μ m. (d) SDS-PAGE analysis comparing Cy3-labeled and unlabeled CTD Δ CW, indicating no degradation or mobility shift. (e) Static light scattering (SLS) profiles of Cy3-labeled and unlabeled CTD Δ CW proteins reveal similar molecular weight distributions and hydrodynamic behavior. (f) Time-lapse imaging of Cy3-labeled CTD Δ CW over 180 s reveals dynamic droplet behavior in vitro. Scaler bar: 2 μ m.

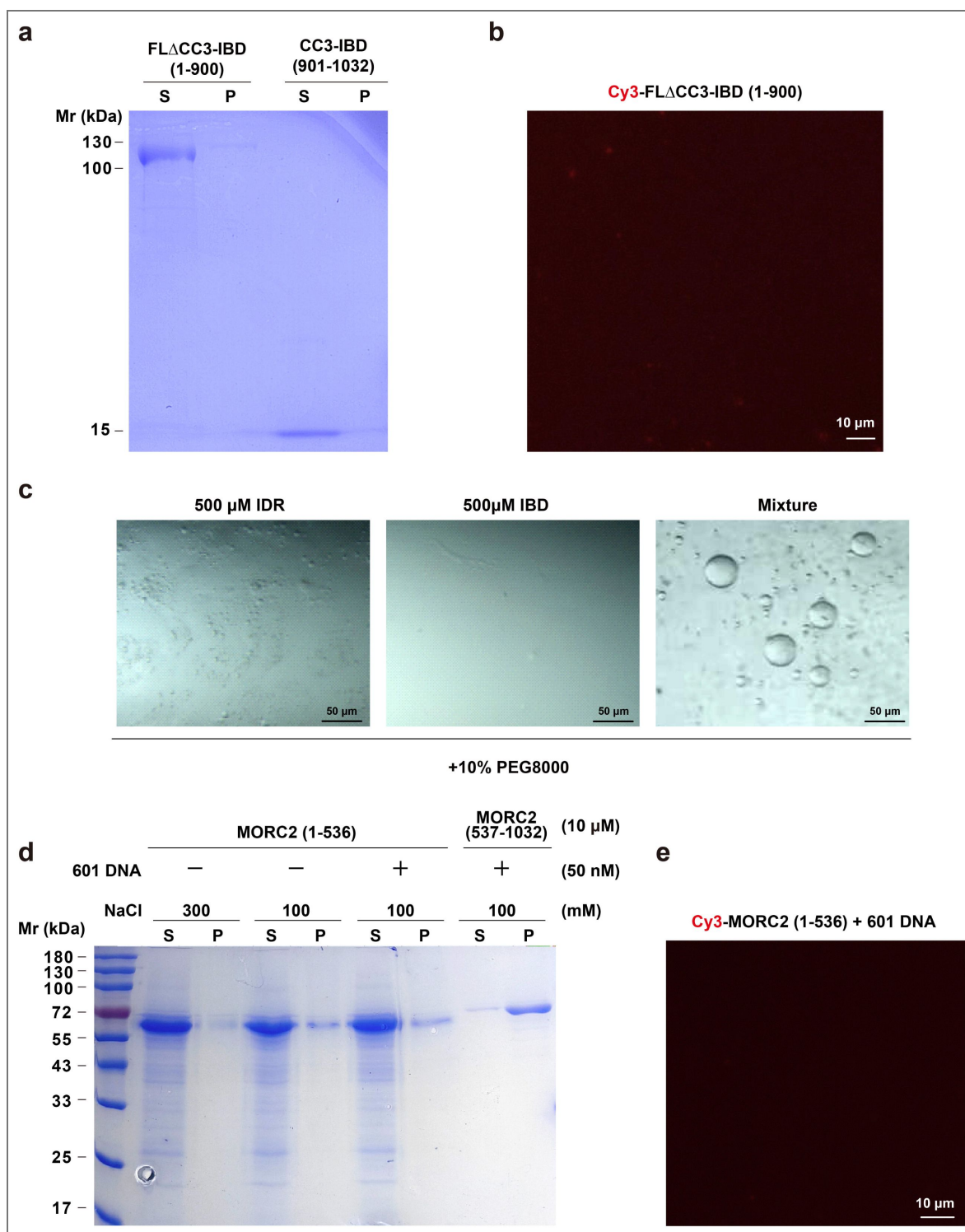


Figure S6. In vitro assays to investigate isolated fragment fails to induce phase separation.

(a) Sedimentation assay showing domain containing 1-900 (5 μM) and 1-900 (5 μM) is not induced phase separation in 100 mM NaCl buffer. (b) Representative images showing that Cy3-labeled MORC2 1-900 (5 μM) could not undergo phase separation. (c) In vitro assays to investigate the contribution of IDR-IBD interactions to phase separation. (d) Sedimentation assay showing domain containing 1-536 (10 μM) and 537-1032 (10 μM) is not induced phase separation in 100 mM NaCl buffer. (e) Representative images showing that Cy3-labeled MORC2 1-536 (10 μM) could not undergo phase separation in including 50 nM 601 DNA.

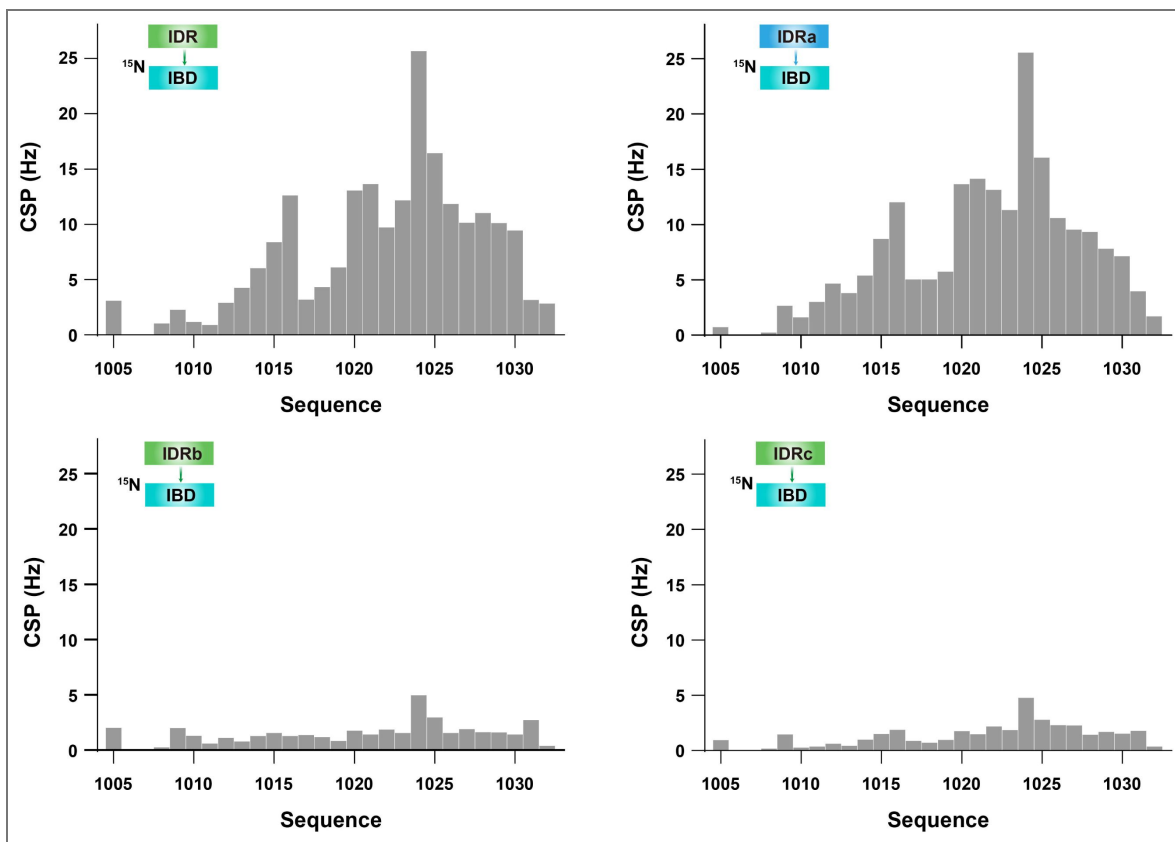


Figure S7. NMR titration about IDR-IBD.

The corresponding CSP plots in the process of NMR titration related to [Figure 4](#)

Figure S8. Phase diagram of MORC2-DNA.

DNA-dependent phase behavior of the protein in vitro. Phase diagram showing the formation of liquid droplets and amorphous structures as a function of DNA and protein concentrations. Each circle represents an individual experimental condition. Black regions indicate regimes favoring liquid droplet formation, whereas the darker shaded region at high DNA and protein concentrations denotes the formation of amorphous assemblies. At low protein concentration (0.5 μM) and low DNA levels, no phase separation was observed in shaded regions.

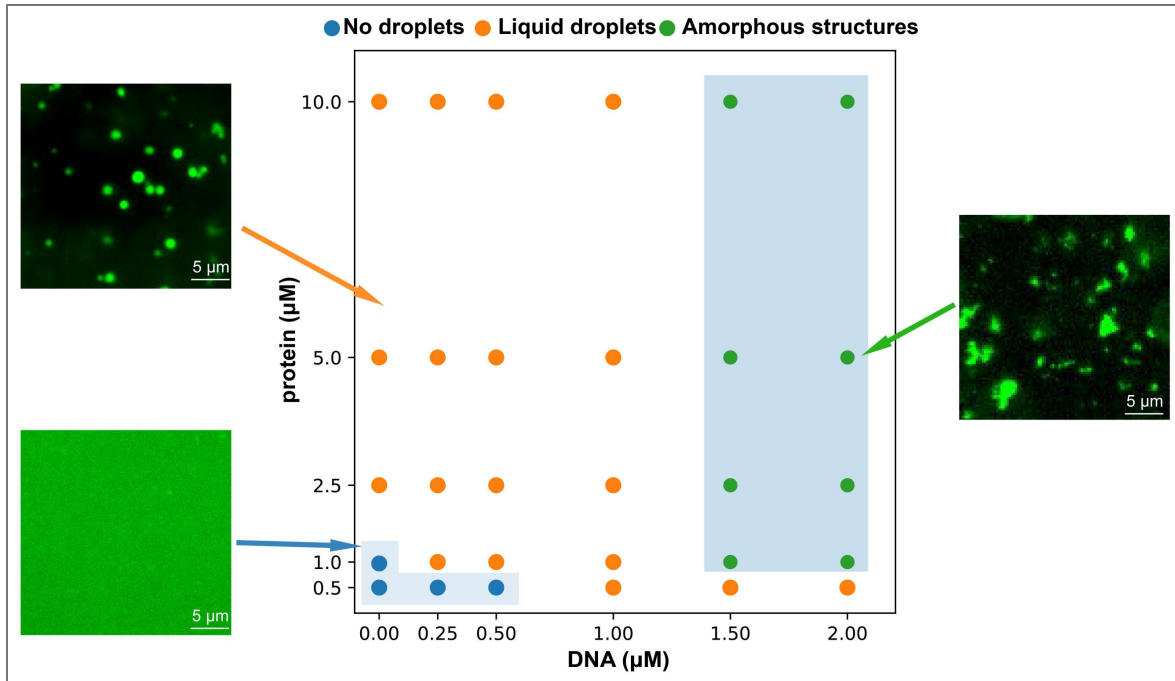
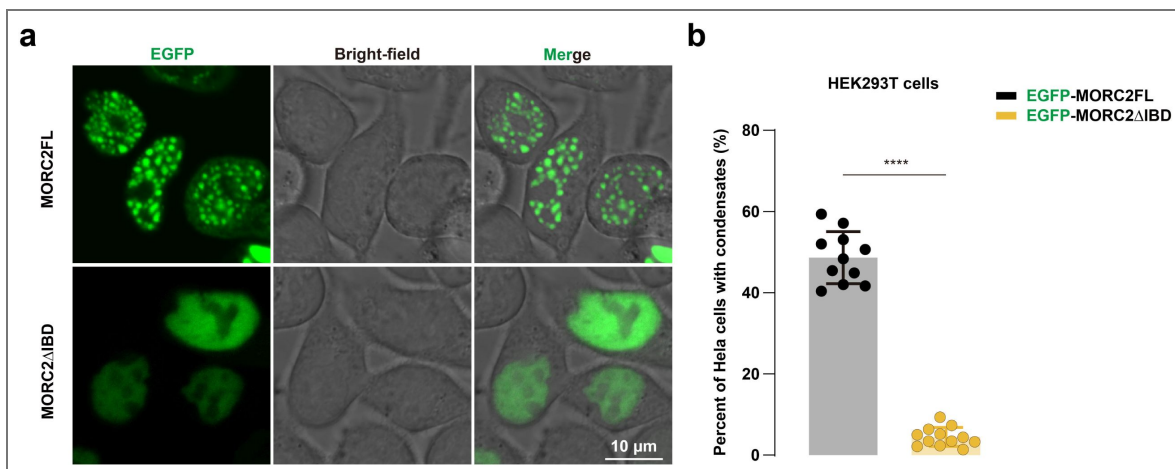


Figure S9. IBD-deficient mutant (ΔIBD) in HEK293T cells

(a) Representative single-cell images of EGFP-MORC2-expressing HEK293T cells. Scaler bar: 10 μm . (b) Quantification of the percentage of cells showing visible nuclear condensates in each construct. Data are presented as mean $\pm$ SEM; $n \geq 11$ replicates. Statistical significance was assessed using unpaired Student's t-test. **** $p < 0.0001$.



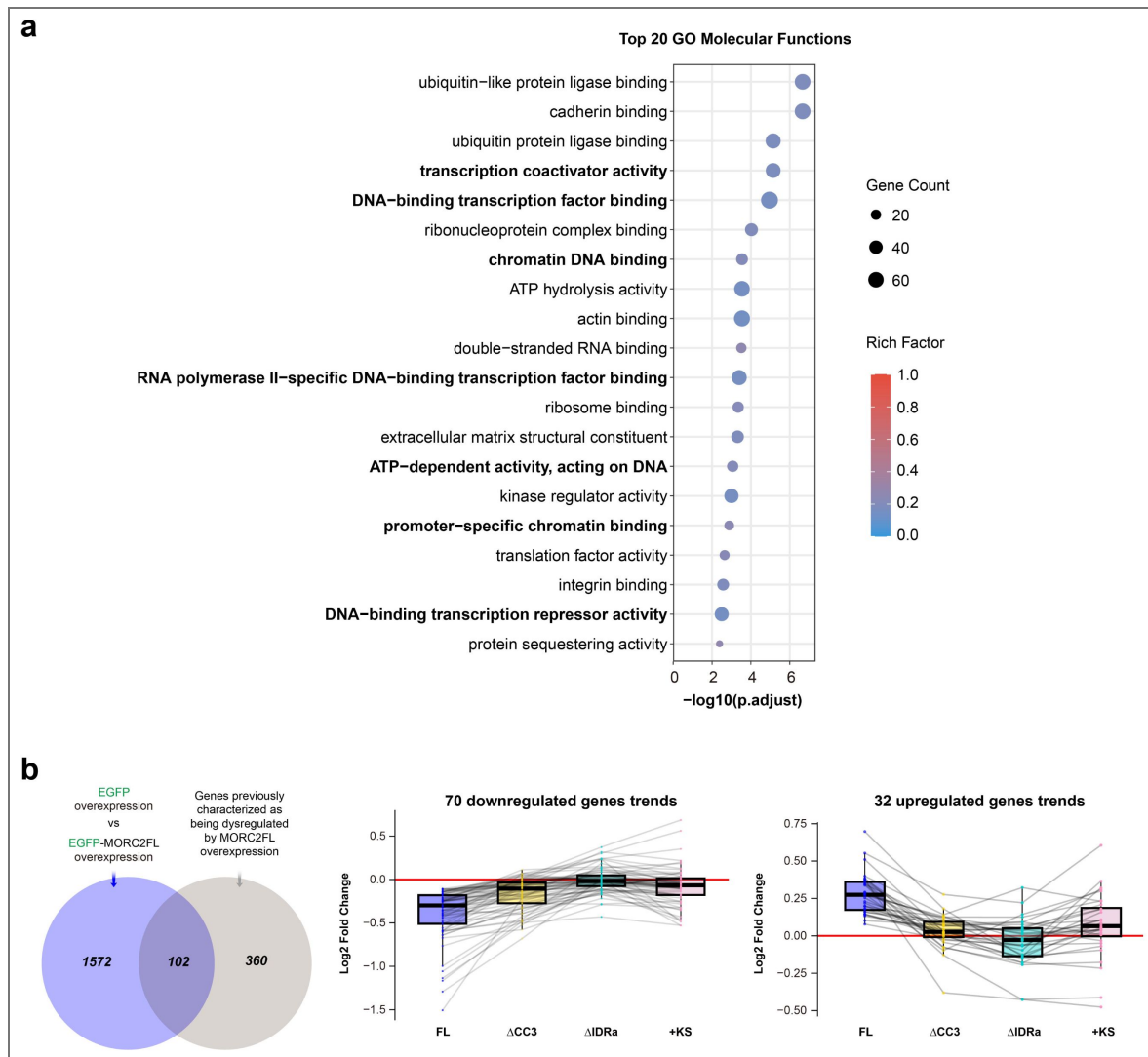


Figure S10. Condensation and dynamics-defective MORC2 mutants impair transcriptional regulation

(a) GO-based molecular function terms enriched among all differentially expressed genes upon EGFP-MORC2FL overexpression. **(b)** Venn diagram illustrating the overlap between differentially expressed genes identified in this study and previously characterized targets regulated in MORC2 knockout HeLa cells¹². Fold-change ($\log_2$ fold change) analysis of the 102 overlapping genes showing significant expression changes upon MORC2FL overexpression, compared with their corresponding fold changes following overexpression of the three MORC2 mutants. Dashed lines connect the same gene across the four transfection conditions.

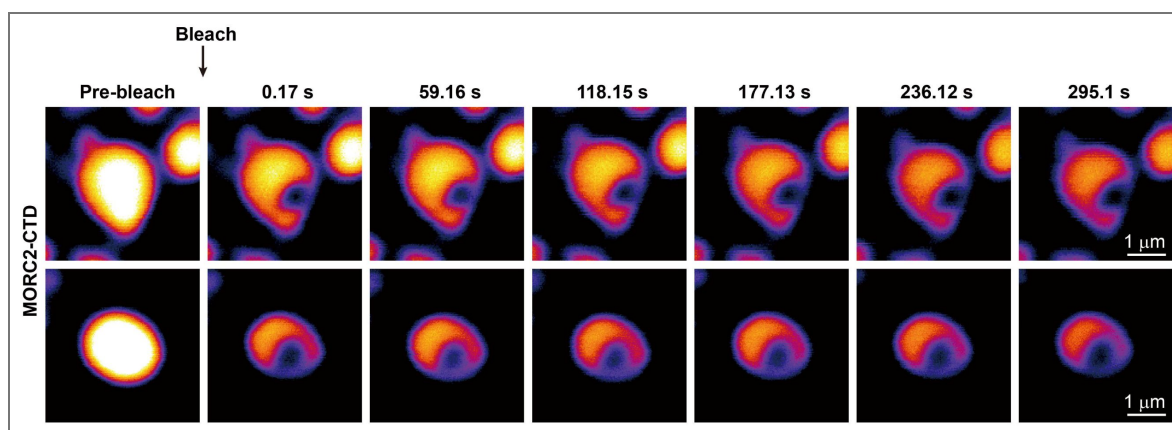


Figure S11. Compared to the wild-type protein, N-terminally truncated MORC2-CTD forms nuclear puncta in HeLa cells with significantly reduced internal dynamics, as revealed by FRAP assays.

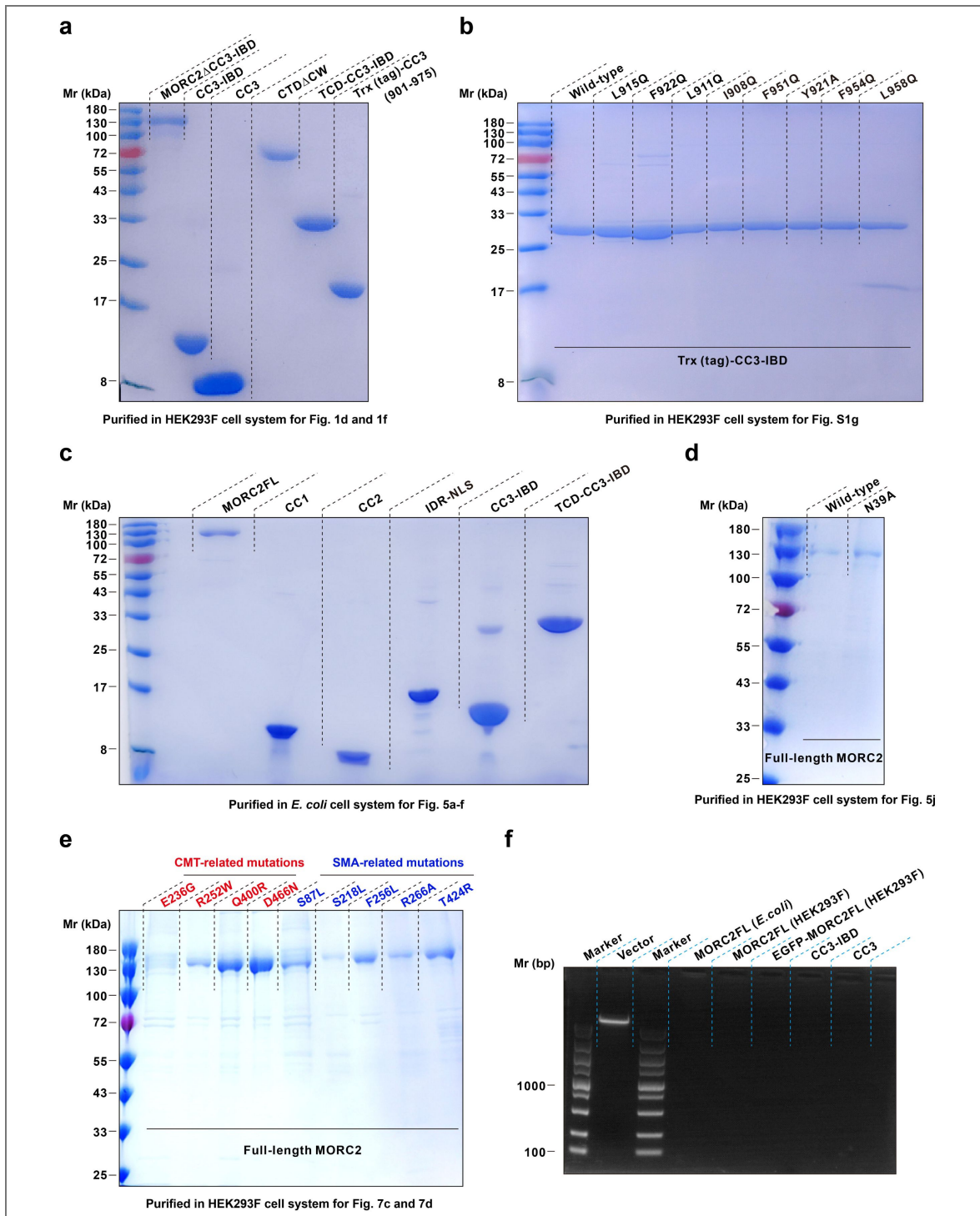


Figure S12. Representative SDS-PAGE gels for all purified protein constructs.

(a) MORC2 fragments used for conformational analysis by FPLC-MALS (related to Figure 1 [□](#)). (b) MORC2 CC3 domain fragments used for dimerization analysis by FPLC-MALS (related to Figure S1 [□](#)). (c) MORC2 fragments expressed in *E. coli* for EMSA-based DNA-binding assays (related to Figure 5 [□](#)). (d) MORC2 fragments expressed in HEK293F cells for ATPase activity measurements with or without 601 DNA (related to Figure 5 [□](#)). (e) Disease-associated MORC2 mutants expressed in HEK293F cells for ATPase and DNA-binding assays, while E236G mutant exhibited poor biochemical properties, precluding further analysis in vitro (related to Figure 7 [□](#)). (f) Agarose gel analysis for MORC2 constructs to confirm the absence of nucleic acids. The pET32 vector as the positive control, the protein preparation for analysis is 0.05 mg.

Data availability

All data needed to evaluate the conclusions in the paper are present in the paper and/or the Supplementary Materials. Raw gene expression data are available in Source Data. The atomic coordinates of the MORC2 CC3 domain have been deposited to the Protein Data Bank under the accession code: 9KQL.

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Additional information

Author contributions

Y.Z., W.X., W.D., C.H., C.W., and Y.B. designed the experiments. Y.B., Y.Z., W.X., and W.D. conducted the experiments. Y.Z. determined the crystal structure. Y.W., W.J. and F.Z. assisted with the experiments and reagents. All authors analyzed the data. Y.Z., Y.B., C.H., and C.W. wrote the manuscript with input from all authors. All authors approved the final version of the manuscript. C.H., and C.W supervised the project.

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Additional files

[PDB X-ray Structure Validation Report](#) 

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Peer reviews

Reviewer #1 (Public review):

This work demonstrates that MORC2 undergoes phase separation (PS) in cells to form nuclear condensates, and the authors demonstrate convincingly the interactions responsible for this phase separation. Specifically, the authors make good use of crystallography and NMR to identify multiple protein:protein interactions and use EMSA to confirm protein:DNA interactions. These interactions work together to promote in vitro and in cell phase separation and boosted ATPase activity by the catalytic domain of MORC2.

Moreover, the authors show solid evidence supporting their important claim that MORC2 PS is important for MORC2-mediated gene regulation. Exploring causal links between PS and function is an important need in the phase separation field, particularly as regards the role of condensates in gene regulation, and is a non-trivial matter. It is crucial and challenging to properly explore the alternative possibility that soluble complexes, existing in the same conditions as phase-separated condensates, are the functional species. The authors have attempted to address this concern by manipulating the physical nature of the MORC2 condensates using a killswitch (KS) peptide (MORC2 +KS), finding that reducing condensates dynamics results in a cellular phenotype very similar to that of the phase separation-deficient MORC2 condensates. While not fully ruling out the alternative, soluble-complex hypothesis, this experiment suggests that function is indeed localized inside the MORC2 condensates, and that perturbing the condensate can be functionally equivalent to removing condensate formation.

The authors also look at several disease related mutants of MORC2. While most of these do not seem to have an obvious connection to the phase separation data, it is quite interesting that one mutant, E236G, displays similar intra-condensate dynamics compared to MORC2 +KS, strengthening the claim that MORC2 phase separation is important for function and suggesting that the observations in this paper may indeed have some disease relevance.

Strengths

Static light scattering and crystallography are nicely used to demonstrate the dimerization of MORC2FL and to discover the structure of the CC3 domain dimer, presumably responsible for the dimerization of MORC2FL (Figure 1).

Extensive use of deletion mutants in multiple cell lines is used to identify regions of MORC2 that are important for forming condensates in the nucleus: the IBD, IDR, and CC3 domains are found to both be essential for condensate formation, while the CW domain plays an unknown role in condensate morphology (Figure 3). The authors use NMR to further identify that the IBD domain seems to interact with the first third of the centrally located IDR, termed IDR_a, but not with the latter two thirds of the IDR domain (Figure 4). This leads them to propose that phase separation is the product of IBD:IDR_a interaction, CC3 dimerization, and an unknown but important role for the CW domain.

Based on the observation that removal of the NLS resulted in diffuse cytoplasmic localization, they hypothesized that DNA may play an important role in MORC2 PS. EMSA was used to demonstrate interaction between DNA and several MORC2 domains: CC1, CC2, IDR, and TCD-CC3-IBD. Further in vitro microscopy with purified MORC2 showed that DNA addition significantly reduces MORC2 saturation concentration (Figure 5).

These assays convincingly demonstrate that MORC2 phase separates in cells and identifies the protein domains and interactions responsible for this phenomenon.

Weaknesses

The connection between condensates and function, while improved from the original manuscript, still has some weak points.

The central experiment demonstrating that MORC2 condensates mediate function takes the form of RNA-Seq in MORC2 KO HeLa cells (Figure 6), rescued with WT, condensate-deficient mutants, and a KS peptide mutant that reduces dynamics by increasing homotypic protein interactions. The observation that rescuing with MORC2 +KS is ineffective, in a manner similar to rescue with condensate-deficient MORC2 mutants, suggests that unperturbed condensates are important for function. An alternative possibility, however, is that condensates are non-functional bystanders, and that the increased homotypic interactions present in MORC2 +KS result in stronger MORC2 +KS recruitment to condensates, reducing

the pool of functional, dilute phase MORC2 +KS and squashing function via sequestration. Similar ideas have been explored by others for transcription factors (e.g. Chong et al, Mol Cell, 2022). This possibility is neither discussed nor ruled out. The absence of microscopy data showing similar localization of MORC2 and MORC2 +KS (particularly the amount of diffuse MORC2 outside condensates) amplifies this concern.

The RNA-Seq data presented in Figure 6h also has some concerning qualities. Inter-replicate variability is higher than ideal, particularly for MORC2 deltaCC3. This may be a product of the transient transfection system used for these experiments, which inherently results in stochasticity. Specific sets of genes regulated by MORC2 are consistent with the main conclusion (Figure 6i, individual genes in 6h, showing that all mutants are more similar to one another than to WT MORC2), but global transcription shifts seem quite different between MORC2 condensate-deficient mutants and MORC2 +KS (Figure 6h heatmap), suggesting much more than simple condensate disruption is taking place. Together, this weakens the conclusion that MORC2 condensates are the functional form of MORC2.

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Reviewer #2 (Public review):

Summary:

The study by Zhang et al. focuses on how condensation of a chromatin-associated protein MORC2 regulates gene expression. Their study shows that MORC2 forms dynamic nuclear condensates in cells. In vitro, MORC2 phase separation is driven by dimerization and multivalent interactions involving the C-terminal domain but interplay with other parts of MORC2 too. A key finding is that the intrinsically disordered region (IDR) of MORC2 exhibits strong DNA binding. They report that DNA binding enhances MORC2's phase separation and its ATPase activity, offering new insights into how MORC2 contributes to chromatin organization and gene regulation. Authors correlate MORC2's condensate forming ability and material properties with its gene silencing function using a few variants. Moreover, they investigate the effect of disease-linked mutations in the N-terminal domain of MORC2 on its ability to form cellular condensates, ATPase activity and DNA-binding. Their work implies that proper material properties of MORC2 condensates may be important to their biological function.

Strengths:

The authors determined a 3.1 Å resolution crystal structure of the dimeric coiled-coil 3 (CC3) domain of MORC2, revealing a hydrophobic interface that stabilizes dimer formation. They present extensive evidence that MORC2 phase separates across multiple contexts, including in vitro, in cellulo, and in vivo. Through systematic cellular screening, they identified the C-terminal domain of MORC2 as a key driver of condensate formation. Biophysical and biochemical analyses further show that the IDR within the C-terminal domain interacts with the C-terminal end region (IBD) and also exhibit strong DNA-binding capacity (using 601 DNA), both of which promote MORC2 phase separation. Together, this study emphasizes that interactions mediated by multiple domains-CC3, IDR, and IBD- drives MORC2 phase separation. Additionally, the work uses a unique kill-switch peptide fused to the MORC2 sequence to disrupt its material properties – this permits the authors to examine the link between material properties and transcription function. The study is overall strengthened by (1) the combination of variants tested both in vitro and in cellulo, and (2) the systematic examination of domain contributions that highlight the multivalent interactions at play mediating MORC2 condensation.

Weaknesses:

The employed MORC2 variants have enabled the beginning of an investigation linking condensation and biological function, but more work will be needed to really dissect the contribution of condensation to DNA-binding, ATPase activity, and gene silencing. A systematic investigation of differential material properties on MORC2 condensates will be needed to assess the link to biological function, especially as the authors' work is reminiscent of how the liquidity of *Caulobacter crescentus* PopZ condensates tunes bacterial fitness.

<https://doi.org/10.7554/eLife.108479.2.sa2>

Reviewer #3 (Public review):

Summary:

The manuscript by Zhang et al. demonstrates that MORC2 undergoes liquid-liquid phase separation (LLPS) to form nuclear condensates critical for transcriptional repression. Using a combination of in vitro LLPS assays, cellular studies, NMR spectroscopy, and crystallography, the authors show that a dimeric scaffold formed by CC3 drives phase separation, while multivalent interactions between an intrinsically disordered region (IDR) and a newly defined IDR-binding domain (IBD) further promote condensate formation. Notably, LLPS enhances MORC2 ATPase activity in a DNA-dependent manner and contributes to transcriptional regulation, establishing a functional link between phase separation, DNA binding, and transcriptional control.

Strengths:

The manuscript is well organized and logically structured. It provides valuable mechanistic insights into MORC2 function, and the majority of the conclusions are well supported by the data presented.

<https://doi.org/10.7554/eLife.108479.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1 (Public review):

Summary:

This work demonstrates that MORC2 undergoes phase separation (PS) in cells to form nuclear condensates, and the authors demonstrate convincingly the interactions responsible for this phase separation. Specifically, the authors make good use of crystallography and NMR to identify multiple protein: protein interactions and use EMSA to confirm protein: DNA interactions. These interactions work together to promote in vitro and in cell phase separation and boost ATPase activity by the catalytic domain of MORC2.

However, the authors have very weak evidence supporting their potentially valuable claim that MORC2 PS is important for the appropriate gene regulatory role of MORC2 in cells. Exploring causal links between PS and function is an important need in the phase separation field, particularly as regards the role of condensates in gene regulation, and is a non-trivial matter. Any study with convincing data on this matter will be very important. For this reason, it is crucial to properly explore the alternative possibility that soluble complexes, existing in the same conditions as phase-separated condensates, are the functional species. It is also critical to keep in mind that, while a specific protein

domain may be essential for PS, this does not mean its only important function pertains to PS.

In this study, the authors do not sufficiently explore the role that soluble MORC2 complexes may play alongside MORC2 condensates. Neither do they include enough data to solidly show that domain deletion leads to phenotypes via a loss of phase separation per se, rather than the loss of phase separation being a microscopically visible result, not cause, of an underlying shift in protein function. For these reasons, the authors' conclusions regarding the functional role of MORC2 condensates are based on incomplete data. This also dampens the utility of this work as a whole, since the very nice work detailing the mechanism of MORC2 PS is not paired with strong data showing the importance of this observation.

We thank the reviewer for this thoughtful and constructive critique. We agree that establishing a causal link between phase separation (PS) and biological function—particularly in transcriptional regulation—is a central and non-trivial challenge in the condensate field. We also appreciate the reviewer's emphasis on two critical alternative interpretations: (i) that soluble MORC2 complexes, rather than condensates, may represent the primary functional species, and (ii) that loss of phase separation upon domain deletion could reflect a downstream consequence of altered protein function rather than its cause.

To address these concerns, we have performed a series of new experiments specifically designed to decouple condensate formation, and condensate dynamics, thereby allowing us to more rigorously interrogate the functional relevance of MORC2 condensates.

First, to overcome the limitation of domain deletions which may affect MORC2 function beyond phase separation we introduced a micropeptide-based kill switch (KS) to the C terminus of MORC2. This strategy has recently emerged as a powerful approach to selectively reduce condensate dynamics without disrupting protein expression, folding, or domain architecture [1]. Importantly, unlike CC3 or IDR Δ deletions, MORC2+KS robustly form nuclear condensates but exhibits markedly reduced internal dynamics, as demonstrated by FRAP analyses showing minimal fluorescence recovery after photo bleaching (Figure 6a-c). This strategy therefore allows us to perturb condensate material properties independently of MORC2 domain integrity.

Second, we systematically compared the transcriptional consequences of rescuing MORC2-knockout HeLa cells with MORC2^{FL}, condensation-deficient mutants (Δ CC3 and Δ IDR Δ), and the dynamics-defective MORC2+KS (Figure 6d). Despite being expressed at substantially higher levels than MORC2^{FL} (Figure 6e), all three mutants showed a striking and consistent failure to restore MORC2-dependent transcriptional regulation (Figure 6f-h). This effect was particularly pronounced for transcriptionally repressed genes, including two sets of high-confidence MORC2 targets reported in prior studies (Figure 6i and FiguresS10). These findings demonstrate that neither increased protein abundance nor the mere presence of condensate-like structures alone is sufficient to restore MORC2 function.

Third, our data instead support a model in which both soluble MORC2 complexes and dynamic MORC2 condensates are required for full transcriptional regulation activity. While soluble MORC2 is likely involved in target recognition and complex assembly, our results indicate that proper condensate formation—and critically, condensate dynamics—are essential for effective transcriptional repression and activation. The inability of the MORC2+KS mutant to rescue transcriptional defects, despite intact condensate formation, points away from a model in which MORC2 condensates represent only microscopically visible byproducts of MORC2 activity.

We believe these new data strengthen the manuscript by pairing the detailed mechanistic dissection of MORC2 phase separation with direct functional evidence, enhancing the

conceptual impact and biological significance of the study.

Strengths:

Static light scattering and crystallography are nicely used to demonstrate the dimerization of MORC2FL and to discover the structure of the CC3 domain dimer, presumably responsible for the dimerization of MORC2FL (Figure 1).

Extensive use of deletion mutants in multiple cell lines is used to identify regions of MORC2 that are important for forming condensates in the nucleus: the IBD, IDR, and CC3 domains are found to be essential for condensate formation, while the CW domain plays an unknown role in condensate morphology (Figure 3). The authors use NMR to further identify that the IBD domain seems to interact with the first third of the centrally located IDR, termed IDRa, but not with the latter two-thirds of the IDR domain (Figure 4). This leads them to propose that phase separation is the product of IDB:IDRa interaction, CC3 dimerization, and an unknown but important role for the CW domain.

Based on the observation that removal of the NLS resulted in diffuse cytoplasmic localization, they hypothesized that DNA may play an important role in MORC2 PS. EMSA was used to demonstrate interaction between DNA and several MORC2 domains: CC1, CC2, IDR, and TCD-CC3-IBD. Further in vitro microscopy with purified MORC2 showed that DNA addition significantly reduces MORC2 saturation concentration (Figure 5).

These assays convincingly demonstrate that MORC2 phase separates in cells, and identify the protein domains and interactions responsible for this phenomenon, with the notable caveat that the role of the CW domain here is left unexplored.

We appreciate the reviewer for their positive and detailed assessment of the strengths of our study. Our understanding of the CW domain's function remains preliminary. Although we observed that the CW domain can influence condensate size, the IDR, IBD, and CC3 domains constitute the core structural elements driving phase separation. Consequently, the CW domain was not a primary focus of the current study. Nonetheless, investigating its functional contributions represents an interesting avenue for future work.

Weaknesses:

Although the authors demonstrated phase separation of MORC2FL, their evidence that this plays a functional role in the cell is incomplete.

Firstly, looking at differentially upregulated genes under MORC2FL overexpression, the authors acknowledge that only 10% are shared with differentially regulated genes identified in other MORC2FL overexpression studies (Figure 6c, d). No explanation is given for why this overlap is so low, making it difficult to trust conclusions from this data set.

We thank the reviewer for raising this important concern. In response, we have improved the quality and robustness of our RNA-seq analysis by repeating the experiments with optimized sample handling and increased sequencing depth. Using this updated dataset, we identified a considerably higher overlap between MORC2-regulated genes in our study and those reported previously.

Specifically, we observed 84 overlapping genes with the study by Nikole L. Fendler et al. [2], corresponding to approximately 32% of the MORC2-regulated genes reported in that work (Figure 6i). In addition, we identified 102 overlapping genes with the dataset reported by Iva A. Tchasovnikarova et al. [3], representing approximately 22% of the genes identified in that study (Figure S10b).

We note that complete concordance with previous reports is not expected, given substantial differences in experimental design. For example, Fendler et al. employed a doxycycline-inducible MORC2 expression system [2], whereas our study relies on transient overexpression in MORC2-knockout HeLa cells. In contrast, Tchasovnikarova et al. compared transcriptomes between MORC2 knockout and wild-type cells [3], rather than MORC2 rescue conditions. Moreover, RNA-seq results are inherently influenced by cell line batch variability, sequencing depth, and analysis pipelines, all of which differ across studies.

Taken together, we consider an overlap in the range of ~20–30% to be reasonable and biologically meaningful in the context of these experimental differences, and we believe that the revised RNA-seq data provide a more reliable foundation for our conclusions regarding MORC2-dependent transcriptional regulation.

Secondly, of the 21 genes shared in this study and in earlier studies, the authors note that the differential regulation is less pronounced when a phase-separation-deficient MORC2 mutant is overexpressed, rather than MORC2FL (Figure 6e). This is taken as evidence that phase separation is important for the proper function of MORC2. However, no consideration is made for the alternative possibility that the mutant, lacking the CC3 dimerization domain, may result in non-functional complexes involving MORC2, eliminating the need for a PS-centric conclusion. To take the overexpression data as solid evidence for a functional role of MORC2 PS, the authors would need to test the alternative, soluble complex hypothesis. Furthermore, there seems to be low replicate consistency for the MORC2 mutant condition (Figure S6a), with replicate 3 being markedly upregulated when compared to replicates 1 and 2.

We thank the reviewer for raising these important concerns. In the revised manuscript, we have substantially strengthened both the experimental evidence and the data presentation to directly address the alternative “soluble complex” interpretation as well as the issue of replicate consistency. Specifically, we now provide data that clarify the functional impact of phase-separation-deficient MORC2 mutants and explicitly show replicate-level RNA-seq analyses. The Figure 6 and Figure S10 support these improvements and enhance both the robustness and transparency of our transcriptional analyses. Collectively, these revisions directly address the reviewer’s concerns regarding the functional interpretation of MORC2 phase separation.

Thirdly, the authors close by examining the in-cell PS capabilities and ATPase activity of several disease-associated mutants of MORC2 (Figure 7). However, the relevance of these mutants to the past 6 figures is unclear. None of these mutations is in regions identified as important for PS. Two of the mutations result in a higher percentage of the cell population being condensate-positive, but this is not seemingly connected to ATPase activity, as only one of these two mutants has increased ATPase activity. Figure 7 does not add any support to the main hypotheses in the paper, and nowhere in the paper do the authors investigate the protein regions where the mutations in Figure 7 are found.

We thank the reviewer for raising this point regarding Figure 7. At the current stage, the results for disease-associated mutations are primarily descriptive. While we observed that certain mutations clustered at the N-terminus can affect MORC2 condensate formation, ATPase activity, and DNA binding, we did not identify a mechanistic explanation for these correlations. Notably, the T424R mutation, previously reported to significantly enhance ATPase activity [4], also increased both intracellular condensate formation and *in vitro* DNA binding in our experiments. In contrast, other mutations did not show such consistent effects. Previous studies have established that MORC2’s ATP-binding and DNA-binding activities are independent [4]. Our results further suggest that MORC2’s phase separation behavior is independent of both ATP and DNA binding affinity, although existing evidence hints at potential cross-regulatory interactions among these three functions.

We would also like to emphasize an additional observation that may help contextualize the relevance of N-terminal mutations. Although deletion of the MORC2 N-terminus does not prevent the remaining C-terminal region from forming nuclear condensates, these C-terminal condensates exhibit a marked loss of fluorescence recovery in FRAP assays (Figure S11). This finding suggests that while the N-terminus is not strictly required for condensate assembly, it plays an important role in regulating condensate fluidity. Accordingly, disease-associated mutations distributed across the N-terminal region may influence MORC2 function by modulating condensate material properties rather than condensate formation per se. Based on this hypothesis, we evaluated the fluidity of condensates formed by the E236G and T424R mutants. FRAP measurements indicated substantially reduced fluorescence recovery in E236G, whereas T424R exerted minimal effects (Figure 7e, f).

Overall, our interpretation of the results in Figure 7 is still at a preliminary stage. Nevertheless, the role of the MORC2 N-terminus in modulating condensate fluidity, together with the observed impairment caused by the E236G mutation, appears to be robust, although the underlying mechanism remains to be elucidated. We have incorporated additional discussion on this point and consider it an important direction for future study.

Reviewer #1 (Recommendations for the authors):

(1) Why does MORC2 overexpression lead to changes in gene regulation that are so different from past MORC2 overexpression studies? This is unsettling to me.

(2) Likewise, why is replicate 3 for the MORC2 Δ CC3 variant so different from replicates 1 and 2? Perhaps repeating this experiment would be helpful, both for showing better repeatability and perhaps as regards pulling out a stronger phenotype.

We have repeated the experiments and obtained improved data quality.

(3) A better explanation of the relevance of Figure 7 to the story of the rest of the paper, especially the phase-separation of MORC2, would be important to improving this paper.

We thank the reviewer for this suggestion. We have performed additional experiments and expanded the discussion.

(4) Are expression levels of mutant proteins in Figure 7 uniform between mutants? If not, is it possible that expression levels might account for the difference in condensate-positive cells between mutants?

We cannot fully exclude the possibility that differences in expression levels may contribute to the observed differences among mutants. In our experiments, equal amounts of plasmid DNA were used for transfection across all conditions. Although we did not directly quantify post-transfection protein expression levels by immunoblotting or similar approaches, even if certain mutations were to affect protein expression, it would be technically challenging to further optimize the strategy to fully normalize expression levels across mutants.

Importantly, we note that MORC2 does not form condensates in all transfected cells, even when EGFP fluorescence indicates robust expression levels that are comparable to, or even exceed, those observed in condensate-positive cells. This observation suggests that high expression alone is not sufficient to drive MORC2 phase separation in cells. Therefore, we do not favor the interpretation that the E236K and T424R mutations enhance MORC2 condensation simply by increasing MORC2 protein expression levels.

Minor:

(1) I would suggest considering using the term "dynamic" rather than "liquid-like", as FRAP is technically a measurement of the dynamicity of a protein within a volume, rather

| *than a measurement of the actual fluidity of that volume.*

We thank the reviewer for this helpful suggestion. We agree that FRAP measurements primarily report protein mobility and condensate dynamics rather than the physical fluidity of the condensates. We have therefore revised the manuscript to replace “liquid-like” with “dynamic” where conclusions are based on FRAP analyses.

| *(2) A further investigation of the role of the CW domain would be very interesting, since it clearly has a major role in condensate morphology. Perhaps CW confers important heterotypic interactions which contribute to compositional control of the MORC2 condensates, and thus function and morphology? However, due to the complexity of this specific question and the potentially marginal improvement offered by this paper, I do not think this is a critical addition.*

We thank the reviewer for this insightful suggestion. We have noted this possibility in the Discussion as an important avenue for future investigation.

(3) Why is TCD not tested alone by EMSA for affinity to DNA in Figure 5?

Our inference regarding the DNA-binding capacity of the TCD domain was based on comparative EMSA analyses. Specifically, we found that the TCD–CC3–IBD fragment was able to bind DNA, whereas the CC3–IBD fragment alone showed no detectable DNA binding. From this comparison, we inferred that the TCD domain is responsible for the observed DNA-binding activity.

Because the TCD domain does not affect MORC2 condensate formation, it was not a central focus of the present study, which primarily aims to elucidate the mechanisms underlying MORC2 phase separation and its functional relevance. For this reason, we did not further test TCD alone by EMSA in Figure 5.

Reviewer #2 (Public review):

Summary:

The study by Zhang et al. focuses on how phase separation of a chromatin-associated protein MORC2, could regulate gene expression. Their study shows that MORC2 forms dynamic nuclear condensates in cells. In vitro, MORC2 phase separation is driven by dimerization and multivalent interactions involving the C-terminal domain. A key finding is that the intrinsically disordered region (IDR) of MORC2 exhibits strong DNA binding. They report that DNA binding enhances MORC2's phase separation and its ATPase activity, offering new insights into how MORC2 contributes to chromatin organization and gene regulation. The authors try to correlate MORC2's condensate-forming ability with its gene silencing function, but this warrants additional controls and validation. Moreover, they investigate the effect of disease-linked mutations in the N-terminal domain of MORC2 on its ability to form cellular condensates, ATPase activity, and DNA-binding, though the findings appear inconclusive in the manuscript's current form.

Thank you for your thorough and constructive review of our manuscript. In response to the concerns raised regarding the functional relevance of MORC2 condensate formation, we have redesigned and expanded the experiments presented in Figure 6 and Figure S6 to directly link MORC2's condensate-forming capacity with its transcriptional regulatory function. These new experiments provide additional controls and validation, strengthening the causal relationship between MORC2 condensate dynamics and gene regulation.

At the current stage, the results for disease-associated mutations are descriptive. While we observed that certain mutations clustered at the N-terminus can affect MORC2 condensate formation, ATPase activity, and DNA binding, we did not identify a mechanistic explanation

for these correlations. Notably, the T424R mutation, previously reported to significantly enhance ATPase activity [4], also increased both intracellular condensate formation and *in vitro* DNA binding in our experiments. In contrast, other mutations did not show such consistent effects. Previous studies have established that MORC2's ATP-binding and DNA-binding activities are independent [4]. Our results further suggest that MORC2's phase separation behavior is also independent of both ATP and DNA binding, although existing evidence hints at potential cross-regulatory interactions among these three functions.

Strengths:

The authors determined a 3.1 Å resolution crystal structure of the dimeric coiled-coil 3 (CC3) domain of MORC2, revealing a hydrophobic interface that stabilizes dimer formation. They present extensive evidence that MORC2 undergoes liquid-liquid phase separation (LLPS) across multiple contexts, including in vitro, in cellulo, and in vivo. Through systematic cellular screening, they identified the C-terminal domain of MORC2 as a key driver of condensate formation. Biophysical and biochemical analyses further show that the IDR within the C-terminal domain interacts with the C-terminal end region (IBD) and also exhibits strong DNA-binding capacity, both of which promote MORC2 phase separation. Together, this study emphasizes that interactions mediated by multiple domains-CC3, IDR, and IBD- drives MORC2 phase separation. Finally, the authors quantified the effect of removing the CC3 on the upregulation and downregulation of target gene expression.

We thank the reviewer for their appreciation of the key findings presented in this manuscript.

Weaknesses:

Though the findings appear compelling in isolation, the study lacks discussion on how its findings compare with previous studies. Particularly in the context of MORC2-DNA binding, there are previous studies extensively exploring MORC2-DNA binding (Tan, W., Park, J., Venugopal, H. et al. Nat Commun 2025), and its effect on ATPase activity (ref 22). The contradictory results in ref 22 about the impact of DNA-binding on ATPase activity, and ATPase activity on transcriptional repression, warrant proper discussion. The authors performed extensive in-cellulo screening for the investigation of domain contribution in MORC2 condensate formation, but the study does not consider/discuss the possibility of some indirect contributions from the complex cellular environment. Alternatively, the domain-specific contributions could be quantified in vitro by comparing phase diagrams for their variants. While the basis of this study is to investigate the mechanism of MORC2 condensate-mediated gene silencing, the findings in Figure 6 appear incomplete because the CC3 deletion not only affects phase separation of MORC2 but also dimerization. Furthermore, their investigation on disease-linked MORC2 mutations appears very preliminary and inconclusive because there are no obvious trends from the data. Overall, the discussion appears weak as it is missing references to previous studies and, most importantly, how their findings compare to others'.

We thank the reviewer for their careful assessment of MORC2's DNA-binding properties and its relationship with ATPase and transcriptional activities. We would like to offer the following clarifications to address these concerns, which will also be incorporated into the Discussion section of the revised manuscript.

First, recent work by Tan et al. [5] similarly identified multiple DNA-binding sites in MORC2, consistent with our findings, though there are discrepancies in the precise binding regions. In particular, they reported that isolated CC1 and CC2 domains do not bind 60 bp dsDNA, which contrasts with our observations. We attribute this difference to the types of DNA used in the assays. In our study, we employed 601 DNA, a defined nucleosome-positioning sequence,

which differs substantially from randomly designed short dsDNA. For instance, prior work by Christopher H. Douse et al. [54] also confirmed that MORC2's CC1 domain can bind 601 DNA.

Second, in the study by Fendler et al. [2], DNA binding was reported to reduce MORC2's ATPase activity—an observation that appears inconsistent with the results presented in our Figure 5j. A critical distinction between the two studies lies in the experimental systems used: Fendler et al. [2] employed MORC2 constructs and 35 bp double-stranded DNA (dsDNA), whereas our experiments utilized full-length MORC2 and 601 bp DNA (a sequence with high nucleosome assembly potential). These differences including the absence of potentially regulatory C-terminal regions in the truncated construct and the varying length/structural properties of the DNA substrates introduce variables that substantially complicate direct comparative analysis of ATPase activity outcomes.

Separately, Douse et al. [4] demonstrated that the efficiency of HUSH complex-dependent epigenetic silencing decreases as MORC2's ATP hydrolysis rate increases, implying an inverse relationship between ATPase activity and silencing function. Notably, our current work has not established a direct mechanistic link between MORC2 phase separation and its ATPase activity. Thus, we refrain from inferring that the effect of MORC2 phase separation on transcriptional repression is mediated through modulation of its ATPase function this remains an important question to address in future studies.

Finally, we have redesigned and expanded the experiments presented in Figure 6 and Figure S6 to directly link MORC2's condensate-forming capacity with its transcriptional regulatory function.

Reviewer #2 (Recommendations for the authors):

Major concerns:

(1) Unaddressed discrepancies with the previous study:

(a) Inadequate discussion of Reference 22 and apparent contradictions. Notably, Reference 22 provides evidence for reduced ATPase activity upon DNA binding, in contrast to the current study's observations. Moreover, Reference 22 demonstrates that ATP hydrolysis (ATPase activity) is inversely associated with MORC2-mediated gene silencing, whereas this study concludes that 'the silencing function of MORC2 requires its ATPase activity'. These apparent contradictions warrant a more thorough discussion to reconcile the differences, including potential mechanistic explanations and experimental context that could account for the discrepancies. Additionally, the authors should discuss potential reasons why Ref. 22 may not have observed phase separation during MORC2 biophysical analysis. For instance, in Ref. 22, SEC-MALS was performed at 2 mg/mL (~16 μ M) MORC2 FL in the presence of 150 mM NaCl, conditions that could influence phase behavior based on the current manuscript's results. Addressing whether differences in protein construct, buffer composition, or experimental design might account for this discrepancy would strengthen the discussion.

We thank the reviewer for pointing out the apparent discrepancies between our results and those reported in Ref. 22. We agree that these differences warrant explicit discussion, and we have revised the Discussion accordingly to clarify the experimental and conceptual distinctions between the two studies.

First, regarding the effect of DNA binding on ATPase activity, Ref. 22 examined MORC2 ATPase activity under conditions where MORC2 does not undergo detectable phase separation, whereas our ATPase assays were performed under conditions in which MORC2 readily forms condensates in the presence of DNA. We therefore propose that the observed increase in ATPase activity in our study may reflect a distinct biochemical regime in which phase separation and/or high local protein concentration modulates enzymatic activity.

Importantly, our data do not exclude the possibility that DNA binding per se can inhibit ATPase activity under non-condensing conditions, as reported in Ref. 22.

Second, with respect to transcriptional repression, Ref. 22 reported an inverse correlation between ATP hydrolysis and MORC2-mediated silencing, whereas our study finds that ATPase activity is required for efficient repression. We suggest that these observations are not necessarily contradictory but may reflect different regulatory layers of MORC2 function. Specifically, ATP binding and hydrolysis may be required for MORC2 structural remodeling and chromatin engagement, while excessive or dysregulated ATP hydrolysis could impair stable silencing complexes, as suggested previously [4]. We now explicitly discuss this possibility in the revised manuscript.

Finally, we appreciate the reviewer's suggestion regarding the absence of phase separation in Ref. 22. Indeed, SEC-MALS experiments in Ref. 22 were conducted at $\sim 16 \mu\text{M}$ MORC2 in the presence of 150 mM NaCl (the purification condition is 500 mM NaCl, 10% glycerol), conditions that based on our phase diagrams—are close to or above the saturation concentration but also strongly influenced by ionic strength. This combination of factors explains why the UV peak from SEC-MALS is not indicative of a homogeneous sample [3].

(b) The DNA binding capacity of individual MORC2 domains was tested in Figure 5. IDR appears to be the strongest DNA binder among others. Is this the effect of IDR being isolated from the rest of the protein? A recent paper (Tan, W., Park, J., Venugopal, H. et al. Nat Commun 2025) also investigated DNA binding capacity of different regions of MORC2 using hydrogen-deuterium exchange experiments and EMSA. Interestingly, it can be seen in Figure S9 that the DNA binding capacity of different regions changes when compared together to when in isolation (MORC2 1-603 vs 1-265; 1-495; 496-603). In line with the above, MORC2 IDR's interaction with DNA warrants additional investigation, taking the system as a whole to avoid misinterpretation arising from non-specific interactions.

We appreciate the reviewer's insightful comments regarding domain-specific DNA binding and the potential caveats of studying isolated regions. In Figure 5, our EMSA analyses show that the isolated IDR exhibits the strongest DNA-binding signal among the tested fragments. We agree that this observation may, at least in part, reflect the removal of structural or regulatory constraints imposed by the full-length protein.

Consistent with the reviewer's point, Tan et al. [5] demonstrated that DNA-binding behavior of MORC2 regions differs when analyzed in isolation versus in the context of larger constructs. We have now incorporated this comparison into the Discussion and explicitly note that DNA binding by the IDR should be interpreted as a contextual and potentially cooperative property rather than an autonomous function.

Importantly, our conclusions do not rely on the IDR acting as an independent DNA-binding module *in vivo*. Rather, we propose that the IDR contributes to DNA engagement and phase behavior within the architectural framework of full-length MORC2. We now emphasize this limitation and highlight the need for future studies that probe DNA binding in the context of intact MORC2 or minimally perturbed constructs.

(2) MORC2 DNA binding impacting phase separation and ATPase activity:

While it is clear that MORC2: DNA interaction facilitates MORC2 phase separation, the impact on ATPase activity is not conclusive. First, they observe an opposite trend (compared to ref. 22) for DNA binding on MORC2's ATPase activity. Secondly, it is not clear if the increase in ATPase activity is mediated by DNA binding or phase separation. The ATPase activity was measured at $1 \mu\text{M}$ MORC2 protein concentration in the presence of DNA, where MORC2 appears to phase separate. To draw more definitive conclusions,

additional controls are necessary. Specifically, a phase separation-deficient mutant (from this study) and a DNA-binding-deficient mutant (see ref. 22) should be included to disentangle the contributions of DNA binding and phase separation to ATPase activity. The choice of ATP-binding-deficient mutant N39A as a negative control seems inconclusive in this regard. Additionally, why is there an increase in ATP hydrolysis rate for the ATP-binding-deficient mutant in the presence of DNA, resulting in ATP hydrolysis rates similar to WT MORC2? This raises further questions about the underlying mechanism.

We agree with the reviewer that disentangling the contributions of DNA binding and phase separation to ATPase activity is challenging and that our current data do not fully resolve this issue. As noted, ATPase assays were performed at protein concentrations (1 μ M) where MORC2 undergoes DNA-induced phase separation, making it difficult to distinguish whether enhanced ATP hydrolysis arises directly from DNA binding or indirectly from condensate formation.

We acknowledge that inclusion of additional mutants such as phase separation deficient or DNA-binding deficient variants would provide a more definitive mechanistic separation of these effects. However, generating and validating such mutants in a manner that preserves overall protein integrity is beyond the scope of the current study. Accordingly, we have revised the text to present our findings more cautiously and to frame the observed ATPase enhancement as a correlation rather than a causal mechanism.

Regarding the ATP-binding-deficient N39A mutant, we agree that its behavior in the presence of DNA raises interesting mechanistic questions. We now explicitly note this unexpected observation and discuss possible explanations, including partial ATP binding, altered oligomeric states, or indirect effects mediated by condensate formation.

(3) Dissecting the domain-specific contribution in MORC2 phase separation:

(a) While in cellulo data indicate that the presence of IDR, NLS, CC3, and IBD is all essential for MORC2 condensate formation, it is not clear if this is the effect of the complex cellular environment or whether it is intrinsic for MORC2 phase separation ability. In lines 256-259, the authors suggest IDR α interaction with IBD may serve as a nucleation mechanism for LLPS. In other places, it has been mentioned that CC3 dimerization acts as a scaffold for condensate formation. It is not clear if all of these are essential for MORC2 phase separation, or one of them is essential while the other domain(s) facilitates the phase separation. Though Figure 3 provides a qualitative overview of the contribution of different regions in MORC2 phase separation in cellulo-influenced by the complex cellular environment and substrate interactions, the absolute domain contribution in phase separation would be better studied in vitro by quantitatively comparing phase diagrams (for example, c-sat vs temperature) of different domain deletion constructs.

We thank the reviewer for highlighting the distinction between intrinsic phase separation propensity and cellular context dependent effects. Our in cellular screening was designed to identify regions required for condensate formation under physiological conditions, where chromatin, binding partners, and macromolecular crowding are present. We agree that this approach does not directly quantify the intrinsic phase separation contribution of individual domains.

While CC3 dimerization, IDR-IBD interactions, and nuclear localization all contribute to condensate formation, our data do not imply that these elements are mechanistically equivalent. Rather, we propose that CC3 provides a structural scaffold, while IDR-mediated interactions lower the energetic barrier for condensation. We have revised the manuscript to

clarify this hierarchical model and to avoid implying that all domains contribute equally or independently.

We agree that quantitative *in vitro* phase diagrams would provide valuable insight into intrinsic domain contributions. Whereas the MORC2 Δ CC3-IBD (1–900) and CC3-IBD (900–1032) fragment fails to induce phase separation, the IDR mix CC3–IBD fragment drives robust phase separation; additionally, phase separation is entirely abrogated in the absence of domain–domain interactions. These observations collectively verify that phase separation is contingent on specific domain combinations and their interactions.

(b) Similarly, for line 228-231: 'Notably, condensates formed exclusively in the nucleus and not in the cytoplasm of transfected HeLa cells, suggesting that chromatin-associated nuclear factors, such as DNA, may contribute to the nucleation or stabilization of MORC2 condensates.' This is an important observation made by the authors. Since MORC2 readily phase separates in vitro under physiological conditions, it is important to discuss why MORC2 does not make condensates in the cytoplasm (in the case of MORC2 Δ NLS). In this regard, how does the concentration of overexpressed EGFP-MORC2 constructs compare with in vitro tested droplets of MORC2?

We thank the reviewer for highlighting this important conceptual point. Although MORC2 readily undergoes phase separation *in vitro* under physiological buffer conditions, the absence of condensate formation in the cytoplasm of cells expressing MORC2 Δ NLS underscores the importance of the nuclear environment in promoting MORC2 assembly.

The cytoplasm differs fundamentally from the nucleus not only in overall molecular composition but also in the availability of high-valency scaffolds such as chromatin. We propose that chromatin-associated components, particularly DNA, provide a platform that locally concentrates MORC2 and increases its effective valency, thereby facilitating nucleation or stabilization of condensates in the nucleus. In contrast, the cytoplasm lacks such scaffolds, even when MORC2 is expressed at appreciable levels. In cultured cells, MORC2 is seldom observed in the cytoplasm. While specific experimental contexts may facilitate its cytoplasmic localization, such observations are rarely reported [6]. In transfection-based systems, MORC2 predominantly displays droplet-like behavior in the nucleus. Notably, in endogenous EGFP–MORC2 chimeric mice, we detected punctate MORC2 structures in the neuronal cytoplasm of the brain and spinal cord. The functional significance and biophysical state of cytoplasmic MORC2 remain largely unexplored.

With respect to protein concentration, while EGFP-MORC2 is robustly expressed in cells, direct comparison between cellular expression levels and the protein concentrations used *in vitro* is inherently challenging. Importantly, *in vitro* phase separation is driven by bulk protein concentration under defined conditions, whereas in cells, effective local concentration and interaction valency are strongly shaped by spatial confinement and chromatin association. We have revised the manuscript text to emphasize this distinction and to avoid interpreting nuclear specificity as a purely concentration-dependent phenomenon.

(c) Lines 227-228: '... CW domain restricts condensate overgrowth or fusion', this inference is based on CTD Δ CW puncta being larger in size (Figure 3a). However, in Figure 4h MORC2 Δ IDRb and MORC2 Δ IDRc also result in larger puncta. Making a final conclusion that the CW domain restricts condensate overgrowth or fusion warrants additional investigation.

We thank the reviewer for pointing out the limitation of our original conclusion. We agree that the enlarged puncta in both CTD Δ CW (Figure 3a) indicate that condensate size regulation involves the CW domain was insufficiently rigorous.

Re-analysis of existing data identifies clear phenotypic disparities between the mutants:

MORC2 Δ IDRb/ Δ IDRc mutants show two distinct phenotypes (reduced puncta number with enlarged size, or unchanged puncta number with uniform enlargement), and their total puncta area per cell is comparable to the WT. By contrast, CTD Δ CW mutants display markedly larger puncta relative to the WT. Based on this distinction, we have revised our conclusion to a more cautious formulation: "These observations suggest that the CW domain may participate in regulating initial nucleation size and the exact molecular mechanisms require further investigation."

(4) MORC2 condensate-mediated gene silencing:

This is one of the key investigations of this study where the authors evaluate the ability of MORC2 condensates to regulate gene silencing (transcriptional repression). The major concern here is that the authors are drawing their conclusion based on a CC3 domain deletion mutant of MORC2 and comparing it with wild-type MORC2. Notably, the CC3 domain is responsible for MORC2 dimerization, and as the authors quote, 'The dimeric assembly of CC3 is essential for maintaining the structural integrity of the protein', the absence of CC3 would have a direct impact on its function (such as ATPase activity). With these considerations, it is not clear whether the effect of CC3 domain deletion on gene regulation is an effect of no phase separation or a consequence of loss of function. This necessitates additional validation by including other controls, such as IDRa domain deletion mutant, IDRa domain deletion mutant, where the phase separation is impeded without affecting dimerization.

We appreciate the reviewer's concern regarding the interpretation of CC3 deletion experiments. We agree that CC3 deletion affects both dimerization and phase separation, complicating attribution of gene regulatory effects solely to condensate formation. Our intention was not to claim that loss of repression arises exclusively from impaired phase separation, but rather to demonstrate that disrupting condensate-dynamic capacity correlates with impaired silencing.

To directly address these concerns, we have performed a series of new experiments specifically designed to decouple condensate formation, condensate dynamics, and protein abundance, thereby allowing us to more rigorously interrogate the functional relevance of MORC2 condensates.

First, to overcome the limitation of domain deletions which may affect MORC2 function beyond phase separation we introduced a micropeptide-based kill switch (KS) to the C terminus of MORC2. This strategy has recently emerged as a powerful approach to selectively reduce condensate dynamics without disrupting protein expression, folding, or domain architecture [1]. Importantly, unlike CC3 or IDRa deletions, MORC2+KS robustly form nuclear condensates but exhibits markedly reduced internal dynamics, as demonstrated by FRAP analyses showing minimal fluorescence recovery after photo bleaching (Figure 6a-c). This strategy therefore allows us to perturb condensate material properties independently of MORC2 domain integrity.

Second, we systematically compared the transcriptional consequences of rescuing MORC2-knockout HeLa cells with MORC2^{FL}, condensation-deficient mutants (ACC3 and Δ IDRa), and the dynamics-defective MORC2+KS (Figure 6d). Despite being expressed at substantially higher levels than MORC2^{FL} (Figure 6e), all three mutants showed a striking and consistent failure to restore MORC2-dependent transcriptional regulation (Figure 6f-h). This effect was particularly pronounced for transcriptionally repressed genes, including two sets of high-confidence MORC2 targets reported in prior studies (Figure 6i and Figure S10). These findings demonstrate that neither increased protein abundance nor the mere presence of condensate-like structures alone is sufficient to restore MORC2 function.

Third, our data instead support a model in which both soluble MORC2 complexes and dynamic MORC2 condensates are required for full transcriptional activity. While soluble MORC2 is likely involved in target recognition and complex assembly, our results indicate that proper condensate formation and critically, condensate dynamics are essential for effective transcriptional repression and activation. The inability of the MORC2+KS mutant to rescue transcriptional defects, despite intact condensate formation, points away from a model in which MORC2 condensates represent only microscopically visible byproducts of MORC2 activity.

We believe these new data strengthen the manuscript by pairing the detailed mechanistic dissection of MORC2 phase separation with direct functional evidence, enhancing the conceptual impact and biological significance of the study.

(5) Uncertain impact of pathogenic MORC2 mutations:

Line 356-365: While the statements such as "disease-associated mutations primarily affect enzymatic and phase behaviors rather than DNA affinity" and "these findings provide mechanistic insight into how specific mutations may contribute to distinct pathological outcomes" are conceptually compelling, the data presented in Figure 7b-d do not appear to fully support these conclusions. For many of the mutants, the differences from WT across key parameters-condensation, ATPase activity, and DNA binding-are either modest or statistically insignificant. As such, drawing a unified mechanistic conclusion from these datasets may overstate what the data actually support.

We agree that the effects of disease-associated MORC2 mutations described in Figure 7 are modest and, in some cases, statistically insignificant. Our intention was to document observable trends rather than to propose a unified mechanistic framework. We have revised the manuscript to temper these conclusions and to emphasize the descriptive nature of these data.

(6) Important conceptual clarifications:

(a) Intrinsically disordered regions (IDRs) are not synonymous with phase separation. As the authors show, it is a combination of IDR-mediated interactions and CC3 dimerization that contributes towards the phase separation of MORC2. While IDRs can act as scaffolds for multivalent weak interactions that may promote biomolecular condensate formation, many IDRs serve other roles-such as mediating transient interactions, signaling, or regulatory functions-without undergoing phase separation. Researchers should avoid generalizing the assumption that the mere presence of IDRs in a protein implies its ability for phase separation. In this regard, authors should consider restructuring some of their generalized statements: Line 87-88: 'Recent studies suggest that intrinsically disordered regions (IDRs) can drive liquid-liquid phase separation (LLPS)' and Line 159-161: 'we noticed a long unstructured region at its C-terminus (Figure S1b), a characteristic often associated with proteins capable of phase separation'.

We agree that IDRs are not synonymous with phase separation and have revised the Introduction to avoid generalized statements. The revised text now emphasizes that IDRs can contribute to phase separation in a context-dependent manner and act in concert with structured oligomerization domains such as CC3-IBD.

(b) Liquid-liquid phase separation: I would suggest switching the phrase to just phase separation. The rationale is that the in vitro studies of MORC2 (FRAP, droplet imaging) do not show liquid-like behavior, but perhaps liquid-solid. The FRAP studies suggest liquid-like behavior for some of the constructs. Given the differences in viscoelastic properties across the in vitro and in cellulo studies, it is better to generalize to "phase separation".

Movies for droplet fusion and FRAP, wherever applicable, would be much appreciated. As the nature of in vitro MORC2 droplets appears different than in cells, movie representations of the above would enable readers to better assess the viscoelastic nature of the droplets (whether liquid, gel, etc).

We appreciate the reviewer's insight regarding the viscoelastic properties of MORC2. Our experimental data indeed show a disparity in dynamics between the two environments: while *in vitro* MORC2-FL condensates exhibit relatively low internal mobility, the *in cellulo* MORC2-FL puncta display high dynamics, characterized by rapid internal recovery in FRAP assays and droplet fusion events (Figure S2f).

This contrast suggests that the intracellular microenvironment plays a critical role in regulating the material state of MORC2 condensates. Consequently, we have focused on providing *in vivo* fusion data, as we believe *in vitro* characterizations (such as fusion or FRAP under various artificial conditions) may not faithfully represent the physiological behavior of MORC2. We have revised the manuscript to use the more general term "phase separation" or "condensation" and have added a discussion on these limitations to avoid overinterpreting the material properties observed *in vitro*.

(7) Methods:

(a) Figure 6 S2b: If phase separation occurs at, say, 1.8 μ M protein concentration, this indicates that the protein has reached its saturation concentration (c-sat). Beyond c-sat, any additional protein should partition into the dense phase, while the concentration of the dilute phase remains constant. However, in this figure, the dilute phase concentration appears to increase with increasing total protein concentration, which is inconsistent with expected phase separation behavior. As the methods section does not have any subsection for the sedimentation assay, it becomes difficult to understand how this experiment was performed, whether there is any technical discrepancy in the way soluble and pellet fractions were handled and processed for loading onto the gels. This is also the case with Figure 3d.

We thank the reviewer for carefully examining the sedimentation assay and for raising this important conceptual point. We agree that, for an ideal two-phase system at thermodynamic equilibrium, the concentration of the dilute phase is expected to remain constant once the saturation concentration (c-sat) is reached.

In our study, the sedimentation assay was used as an operational readout to assess concentration-dependent partitioning rather than to quantitatively define equilibrium phase boundaries. The assay involves centrifugation-based separation of supernatant and pellet fractions followed by SDS-PAGE analysis, and therefore does not necessarily report the equilibrium concentrations of coexisting dilute and dense phases. In particular, this approach can be influenced by incomplete physical separation of phases, kinetic trapping, and redistribution of material during handling, especially in systems where condensate maturation or internal reorganization occurs on longer timescales.

Consequently, the apparent increase in the supernatant fraction with increasing total protein concentration likely stems from kinetic limitations and inherent technical constraints of the sedimentation assay, rather than a genuine deviation from classical phase separation behavior. These caveats are now explicitly clarified in the Methods section, with similar limitations of centrifugation-based assays for defining equilibrium phase behavior of biomolecular condensates reported previously.

(b) Figure 4: The NMR comparisons appear to be primarily qualitative, lacking quantitative analyses such as chemical shift perturbation (CSP) and intensity ratio plots, which would offer deeper mechanistic insights. The NMR spectra detailing interactions among the IDR domains need to be quantified.

We thank the reviewer for the suggestion. We have now performed quantitative CSP analyses for the NMR data shown in Figure 4, and the corresponding CSP plots have been added to the revised manuscript (Figure S7).

As expected for interactions mediated by intrinsically disordered regions involved in phase separation, the observed CSPs are generally small. Notably, the CSP profile of IDRa closely matches that observed for the full-length IDR, whereas IDRb and IDRc show minimal perturbations. These results indicate that the interaction is primarily mediated by IDRa, with little contribution from the remaining regions.

Peak intensity analyses were also examined but did not reveal additional residue-specific trends. Together, the quantitative CSP data support our conclusion that the interaction is weak, dynamic, and region-specific, consistent with an IDR-driven, phase-separation-related mechanism. We add this statement in method: CSPs were calculated in Hz at 600 MHz using the following equation:

$$\Delta\nu = \left((\Delta^1\text{H} \times 600)^2 + (\Delta^{13}\text{C} \times 150)^2 \right)^{1/2}$$

Minor comments:

(1) Line 59-60: The Authors mention the HUSH-complex and then the MORC protein family, but do not discuss the relation between the two.

We thank the reviewer for this comment. We have revised the Introduction to explicitly state that MORC2 may serve as a component of the HUSH complex and to clarify the functional relationship between MORC family proteins and HUSH-mediated transcriptional repression.

(2) Line 74: 'Despite their structural similarities...', similarities between what all?

We agree that this statement was ambiguous. We have revised the text to explicitly specify that the comparison refers to structural similarities among MORC family members.

(3) Line 75: 'MORC-mediated repression remains...', this is the first time the word 'repression' is mentioned in the text and directly as an outstanding question.

We have revised the Introduction to introduce the concept of transcriptional repression earlier and to provide appropriate context before posing it as an outstanding question.

(4) The third paragraph does address issues in comments 1 and 3 to some extent, but the introduction needs some restructuring to provide a proper flow of information.

We agree that the Introduction required restructuring. We have revised this section to improve logical flow, better integrate prior studies, and more clearly articulate the motivation and scope of the present work.

(5) Line 83-85: How does the presence of IDRs suggest potential regulatory mechanisms?

We have revised this sentence to clarify that IDRs may contribute to regulatory mechanisms by enabling multivalent and dynamic interactions, rather than implying that IDRs inherently confer regulatory function or phase separation capability.

(6) Line 106-107: 'To determine whether MORC2 has N- and C-terminal dimerization interfaces similar to those...', reference 14 has already established that CC3 (denoted as

CC4 in ref 14) is responsible for dimerization. Consider acknowledging their work in this regard?

We thank the reviewer for this reminder. We have now explicitly acknowledged Ref. 14, which previously established the role of CC3 (denoted CC4 in that study) in MORC2 dimerization.

(7) Lines 117-122: Are the authors comparing morphology from negative stain EM with AlphaFold predicted structure (Figure S1a and S1b)? If so, providing a zoomed-in inset from Figure S1a would be helpful.

Yes, the comparison was intended to relate the negative-stain EM morphology to the AlphaFold-predicted architecture. We have added a zoomed-in inset in Figure S1a to facilitate clearer comparison.

(8) Line 152-153: '...even under varying physiological conditions', what are these varying conditions? Are the authors trying to point towards any of their specific results?

We have revised this phrase to explicitly refer to variations in salt concentration and protein concentration tested in our *in vitro* assays.

(9) Line 154-155: 'The dimeric assembly of CC3 is essential for maintaining the structural integrity of the protein', if it has been established, then please provide a reference.

We thank the reviewer for this suggestion. For MORC family proteins, C-terminal coiled-coil-mediated dimerization is necessary for correct homodimer formation and functional stability (Xie et al., 2019, Cell Commun Signal. 17:160, Ref 14 in the revised manuscript).

(10) Line 159-161: 'we noticed a long unstructured region at its C-terminus (Figure S1b), a characteristic often associated with proteins capable of phase separation²⁵.' again authors are generalizing a statement which is, in most cases, context-dependent. For example, ref 25 mentions that unstructured regions or IDRs serve as a scaffold for multivalent interactions.

We agree with the reviewer and have revised this sentence to avoid generalization. The revised text now emphasizes that IDRs may facilitate multivalent interactions in a context-dependent manner, rather than being intrinsically indicative of phase separation. Additionally, we have explicitly cited the mechanistic insight from Reference 25 that IDRs serve as scaffolds for multivalent interactions, to strengthen the logical link between the structural feature and its potential functional relevance.

(11) Methods section for NMR (Line 665-667) mentions that nucleotides were added to a final concentration of 10 mM. There is no figure or section for MORC2 NMR with added nucleotides/DNA.

We thank the reviewer for pointing this out. The nucleotide (ATP) addition was part of preliminary NMR trials and is not directly associated with the figures presented. We have deleted this in the Methods section to avoid confusion.

(12) Line 285-294: Authors compare the effect of DNA binding on the phase separation of both MORC2FL and MORC2 CTDdeltaCW and conclude that DNA-induced condensation is primarily mediated through interactions with the IDR-NLS region. This appears not to be backed by proper control experiments. The authors do not show whether DNA binding mediates any phase separation for the isolated NTD or not? Similarly, what is the effect of DNA binding on MORC2 deltaIDR?

We thank the reviewer for this insightful comment and agree that additional controls are essential for rigorously dissecting the contribution of DNA binding to MORC2 phase

separation. Our interpretation that DNA-enhanced condensation is primarily mediated through the IDR–NLS region was based on comparative analyses of MORC2FL and MORC2 CTDΔCW, together with EMSA results demonstrating that DNA binding activity is conferred by the IDR–NLS–containing region. We acknowledge, however, that DNA binding alone is not sufficient to infer phase separation behavior.

To address this point, we have performed additional analyses using the isolated NTD' (residues 1–536) and MORC2 ΔIDR–NLS mutants (Figure S6). The isolated NTD' exhibited detectable DNA binding [4] but did not undergo DNA-induced condensation under conditions while MORC2FL or MORC2 CTDΔCW (residues 537–1032) readily formed condensates, indicating that DNA binding by itself is insufficient to drive phase separation. In parallel, MORC2 ΔIDR–NLS mutants showed severely compromised solubility and stability *in vitro*, which limited their quantitative characterization in phase separation assays. Nevertheless, under the conditions tested, these mutants did not display DNA-enhanced condensation comparable to MORC2FL.

Taken together, these observations support a model in which the IDR–NLS region plays a critical role in coupling DNA binding to condensation, while additional domains are required to sustain robust phase separation. We have revised the manuscript text to clarify the experimental scope and to avoid overinterpreting the contribution of DNA binding in the absence of fully reconstituted control systems.

| (13) How did the authors assign the backbone amide NMR chemical shifts for MORC2?

Backbone assignments of MORC2 IBD (1004–1032) were obtained using SOFAST versions of standard triple-resonance experiments, including HNCACB and CBCACONH, recorded at 298 K. Residual assignment ambiguities were resolved using [15] N-edited HMQC–NOESY–HMQC spectra.

| (14) Line 256: 'The partial compaction of IDRa...', what does the author mean here with 'partial compaction'? How did they measure compaction here?

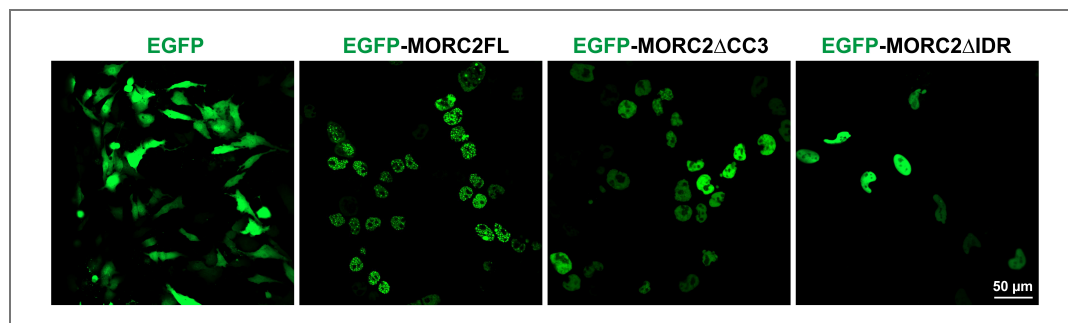
Regarding the term “partial compaction” mentioned previously, we apologize for the typographical error this phrase was erroneously used in place of “key component”.

| (15) Line 312–315: Why is there even a MORC2 readout for MORC2 KO cells with only EGFP? Also, the authors suggest that IDR deletion may impair mRNA stability or transcription; however, the expression levels of MORC2 deltaIDR and MORC2 deltaCC3 do not appear drastically different in Figure 3a.

We thank the reviewer for raising these points. The apparent MORC2 signal in MORC2 knockout cells transfected with EGFP alone is due to the presence of residual MORC2 mRNA. Although CRISPR–Cas9–mediated knockout introduces a frameshift that prevents MORC2 protein expression, the mRNA can still be detected by RNA-seq. This is because nonsense-mediated decay (NMD), which targets transcripts with premature stop codons for degradation, is not always 100% efficient. Therefore, some MORC2 transcripts remain and produce detectable RNA-seq reads, even though no functional protein is expressed.

Regarding the apparent discrepancy in expression levels, Figure 3a displays only EGFP-positive cells, within which the fluorescence intensity of MORC2ΔIDR and MORC2ΔCC3 appears comparable to that of WT MORC2. However, the overall fraction of EGFP-positive cells is markedly reduced for these mutants compared to WT. Thus, while expression levels among successfully transfected cells are similar, fewer cells express detectable levels of the ΔIDR or ΔCC3 constructs across the total population. We therefore interpret this reduction in EGFP-positive cell fraction as reflecting impaired expression efficiency of these mutants, potentially arising from altered transcriptional output, mRNA stability, or protein stability.

We have revised the manuscript text to clarify this distinction and to avoid overinterpreting the underlying mechanism in the absence of direct measurements.



Author response image 1.

EGFP, EGFP-MORC2 (FL), EGFP-MORC2 (Δ CC3), and EGFP-MORC2 (Δ IDR) were re-expressed in MORC2-knockout HeLa cells. Confocal imaging revealed that full-length MORC2 formed condensates in the nucleus, whereas mutants lacking either the CC3 or IDR domain failed to exhibit such behavior. Notably, under identical experimental conditions, we observed a marked reduction in the transfection efficiency of the EGFP-MORC2 (Δ IDR) construct. In contrast to the other variants, EGFP signals for Δ IDR were detectable in only a small fraction of the total cell population, despite consistent DNA loading and protocol synchronization. This observation suggests that the IDR might be required not only for biomolecular condensation but also for maintaining the steady-state levels of the MORC2 mRNA/protein or overall cellular fitness.

(16) Line 330: 'MORC2 deltaCC3 failed to repress any of the 18 downregulated targets...'. This does not appear to be entirely true as repression of some targets (LBH, TGFB2, GADD45A) are closer to MORC2 FL than the EGFP control.

We thank the reviewer for pointing out this inconsistency and for highlighting the need for precise wording. We have updated the dataset and revised the text to describe the results more accurately. We now describe that the mutants impair MORC2FL-mediated transcriptional regulation, consistent with the overall trend observed across these target genes.

(17) Line 347-350: Based on the percent of cells with condensates, the authors conclude that CMT2Z-linked E236G and SMA-linked T424R mutants promote MORC2 phase separation. Again, the effect of these mutations on MORC2 condensation in cells may be direct or indirect. This can be investigated by comparing the *in vitro* effect of these mutations on MORC2 phase separation.

We thank the reviewer for raising this important point and fully agree that the effects of disease-associated MORC2 mutations on condensate formation in cells may arise from either direct alteration in intrinsic phase separation propensity or indirect influences mediated by the cellular environment.

In our study, disease-associated MORC2 mutants were assessed for condensate formation in HEK293F cells. Attempts were made to characterize these mutants *in vitro*; however, the E236G mutant exhibited markedly reduced solubility and stability upon purification, which precluded reliable *in vitro* phase separation analysis. We therefore evaluated the impact of E236G in cells and found that this mutation significantly impaired the dynamics of nuclear MORC2 condensates. For the T424R mutant, we note that its intracellular condensates displayed FRAP recovery kinetics comparable to those of WT MORC2, suggesting broadly

similar dynamic properties of the assemblies formed in cells, but not necessarily implying a direct enhancement of intrinsic phase separation.

In light of these considerations, we have revised the text in Lines 347–350 to avoid attributing a direct causal role of these mutations in promoting MORC2 phase separation. Instead, we now describe the observed increase in the fraction of cells containing condensates as a descriptive cellular correlation. We further emphasize that systematic *in vitro* characterization of disease-associated MORC2 mutants will be required to distinguish direct from indirect effects and represents an important direction for future investigation.

(18) *The discussion section lacks referencing to individual figures in the results section as well as previous literature.*

We agree with the reviewer that the Discussion would benefit from clearer integration with both the Results figures and prior literature. In the revised manuscript, we have substantially restructured the Discussion to explicitly reference key figures when interpreting experimental findings and to more clearly distinguish conclusions drawn from specific datasets. In addition, we have expanded citations to previous studies where relevant, particularly in the context of MORC2 DNA binding, ATPase regulation, chromatin association, and disease-linked mutations. These revisions aim to better situate our findings within the existing literature and to guide readers more clearly between experimental observations and their interpretation.

Reviewer #3 (Public review):

Summary:

The manuscript by Zhang et al. demonstrates that MORC2 undergoes liquid-liquid phase separation (LLPS) to form nuclear condensates critical for transcriptional repression. Using a combination of in vitro LLPS assays, cellular studies, NMR spectroscopy, and crystallography, the authors show that a dimeric scaffold formed by CC3 drives phase separation, while multivalent interactions between an intrinsically disordered region (IDR) and a newly defined IDR-binding domain (IBD) further promote condensate formation. Notably, LLPS enhances MORC2 ATPase activity in a DNA-dependent manner and contributes to transcriptional regulation, establishing a functional link between phase separation, DNA binding, and transcriptional control. Overall, the manuscript is well-organized and logically structured, offering mechanistic insights into MORC2 function, and most conclusions are supported by the presented data. Nevertheless, some of the claims are not sufficiently supported by the current data and would benefit from additional evidence to strengthen the conclusions.

Thank you for your insightful review and constructive suggestions, which have been invaluable in refining our manuscript.

The following suggestions may help strengthen the manuscript:

Major comments:

(1) *The central model proposes that multivalent interactions between the IDR and IBD promote MORC2 LLPS. However, the characterization of these interactions is currently limited. It is recommended that the authors perform more systematic analyses to investigate the contribution of these interactions to LLPS, for example, by in vitro assays assessing how the IDR or IBD individually influence MORC2 phase separation.*

We appreciate the reviewer's insightful comment regarding the characterization of IDR–IBD interactions. In this study, we combined NMR spectroscopy, domain deletion analysis (*in vivo*), and *in vitro* phase separation assays to demonstrate that interactions between the IDR and

IBD contribute to MORC2 condensate formation. To systematically assess the individual contributions of the IDR and IBD to MORC2 phase separation, we performed *in vitro* reconstitution assays using purified domain constructs (Figure S6). Neither the isolated IDR nor the IBD alone exhibited phase separation under buffer conditions approximating the physiological environment, indicating that each domain is individually insufficient to drive condensation. Upon the addition of 10% PEG8000, phase separation was selectively observed for the IDR but not for the IBD, suggesting that the IDR possesses an intrinsic propensity for phase separation that can be enhanced by crowding molecular. Importantly, when the IDR and IBD were mixed, phase separation was robustly induced, supporting a model in which cooperative inter-domain interactions between the IDR and IBD promote MORC2 condensation. In the absence of PEG, no phase separation was observed for the IDR-IBD mixture. These observations imply that IDR-IBD interactions cannot drive phase separation on their own, but require cooperation with CC3-mediated dimerization to achieve this process, which is the central point we wish to emphasize.

(2) The authors mention that DNA binding can promote MORC2 LLPS. It is recommended that they generate a phase diagram to systematically assess how DNA influences phase separation.

We agree that constructing a full phase diagram would provide a more systematic evaluation of the effect of DNA on MORC2 phase separation. In the current study, we assessed DNA-dependent condensation across multiple protein and DNA concentrations, which consistently showed that DNA enhances MORC2 phase separation. At low protein concentration (0.5 μ M), phase separation requires sufficient DNA, whereas increasing either DNA or protein concentration promotes liquid droplet formation. At high DNA and protein concentrations, amorphous structures dominate, indicating a transition away from dynamic assemblies. We have clarified this point in the Results and Discussion sections and now note that a comprehensive phase diagram analysis represents an important direction for future work.

(3) The authors use the N39A mutant as a negative control to study the effect of DNA binding on ATP hydrolysis. Given that N39A is defective in DNA binding, it could also be employed to directly test whether DNA binding influences MORC2 phase separation.

We thank you for your constructive suggestions. The purified wild-type MORC2(1–603) exhibited weak but detectable ATPase activity, whereas the N39A mutant was completely inactive [5]. Based on this characteristic, the N39A mutant was used as a negative control for the ATP-binding-deficient mutant in this study [3]. However, no evidence has been provided to demonstrate that the N39A mutant is defective in DNA binding. Importantly, both our results and previous studies [5–6] indicate that MORC2 engages DNA via multiple domains, suggesting that a single-point mutation is unlikely to significantly compromise its overall DNA-binding capacity.

(4) Many of the cellular and *in vitro* LLPS experiments employ EGFP fusions. The authors should evaluate whether the EGFP tag influences MORC2 phase separation behavior.

We appreciate the reviewer's concern regarding the potential influence of the EGFP tag. The use of EGFP fusions in our study was primarily to maintain consistency with the in-cell experiments. Importantly, we confirmed that EGFP alone does not undergo phase separation in cells, and this observation is consistent with previous studies [7]. Additionally, *in vitro* phase separation of MORC2 was independently validated using Cy3-labeled CTD (Figure S5), which recapitulated the condensate formation seen with EGFP-fused protein. Together, these results indicate that the EGFP tag does not significantly influence MORC2 phase separation, supporting the validity of our conclusions.

Reviewer #3 (Recommendations for the authors):

(1) The authors claim to have obtained nucleic acid-free protein, but no data are provided to support this assertion. It is recommended that they include appropriate validation to confirm the absence of nucleic acids.

We thank the reviewer for highlighting this point. To validate that the purified MORC2 protein is indeed free of nucleic acid contamination, we have additional experimental evidence (e.g., A260/280 measurements, agarose gel analysis, or EMSA in Figure 5), which has been added to the Methods section and Table S2.

Note: Agarose gel analysis for MORC2 constructs to confirm the absence of nucleic acids. The pET32 vector as the positive control, the protein preparation for analysis is 0.05 mg. E means *E. coli* and H means HEK293F.

(2) The FRAP recovery curves are not normalized to 0, making comparison difficult. The authors should normalize the post-bleach intensity to 0 and re-plot the curves to allow a more standard interpretation of mobile fractions.

We agree with the reviewer and have now normalized the FRAP recovery curves by setting the post-bleach intensity to 0. The revised plots are presented in the Figures (2f, j, l; 6c, 7f), allowing for more direct comparison of mobile fractions across different conditions.

(3) The HSQC spectra for IBD appear inconsistent: the peak positions in Figure 4C do not align with those shown in panels D-F. The authors should verify the spectral assignments and ensure consistency across figures.

We thank the reviewer for pointing this out. The apparent inconsistency arose from the fact that different spectral regions were displayed in Figure 4c versus Figure 4d-f for visualization purposes, which may have given the impression of mismatched peak positions. The spectral assignments themselves are consistent across all panels.

To avoid confusion, we have now adjusted the spectral window shown in Figure 4c to match that used in Figure 4d-f. The revised figure ensures consistent presentation of the same spectral region across all panels.

Reference:

- (1) Zhang, Y., Stöppelkamp, I., Fernandez-Pernas, P. et al. Probing condensate microenvironments with a micropeptide killswitch. *Nature* 643, 1107–1116 (2025).
- (2) Fendler NL, Ly J, Welp L, et al. Identification and characterization of a human MORC2 DNA binding region that is required for gene silencing. *Nucleic Acids Res.*53(4):gkae1273 (2025).
- (3) Tchasovnikarova, I., Timms, R., Douse, C. et al. Hyperactivation of HUSH complex function by Charcot–Marie–Tooth disease mutation in MORC2. *Nat Genet* 49, 1035–1044 (2017).
- (4) Douse, C. H. et al. Neuropathic MORC2 mutations perturb GHKL ATPase dimerization dynamics and epigenetic silencing by multiple structural mechanisms. *Nat Commun* 9, 651 (2018).
- (5) Tan, W., Park, J., Venugopal, H. et al. MORC2 is a phosphorylation-dependent DNA compaction machine. *Nat Commun* 16, 5606 (2025).
- (6) Sánchez-Solana B, Li DQ, Kumar R. Cytosolic functions of MORC2 in lipogenesis and adipogenesis. *Biochim Biophys Acta.* 1843(2):316-326 (2014).
- (7) Li, C.H., Coffey, E.L., Dall’Agnese, A. et al. MeCP2 links heterochromatin condensates and neurodevelopmental disease. *Nature* 586, 440–444 (2020).

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